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Master Thesis

Comparing the Everi Labs AI Copilot and Paper Documentation: Task Time and Usability of a P&O Patient Assessment UI Prototype

Main area: *Product Development with specialization in Assistive Technology (MasTech)*

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This thesis is conducted at the *School of Healthcare and Welfare and the School of Engineering* at Jönköping University within Product Development with specialization in Assistive Technology as part of the two-year master's degree program. The author is fully responsible for the opinions, conclusions, and results presented herein.

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Abstract

Background: Prosthetics and orthotics (P&O) clinicians in low-resource settings rely heavily on paper-based documentation, which contributes to workflow inefficiencies, data loss, and a heavy documentation burden. Electronic medical records (EMRs) and AI-assisted tools offer potential solutions, but their usability and efficiency in specialized P&O workflows remain underexamined.

Objective: This study evaluated whether an iteratively designed, AI-assisted EMR prototype (Everi Labs AI Copilot) could improve task times and perceived usability compared to paper-based documentation for P&O clinicians in the Philippines.

Methods: A crossover mixed-methods design was used. Ten (10) ISPO-certified prosthetist-orthotists completed simulated patient assessments under three conditions: paper, prototype v1, and prototype v2. Task completion time and System Usability Scale (SUS) scores were collected. Qualitative feedback was gathered through focus group discussions and analyzed thematically to inform design decisions of the iterated prototype.

Results: The v1 prototype was 38.31% slower than paper (36.42 vs. 26.34 minutes; SUS = 60.00). After one iteration informed by user feedback, v2 achieved slightly slower total task time than the paper (+1.97%) and was faster in SOAP note creation (-21.30%). The SUS score improved 16 points to 76.00, indicating good usability. Welch's ANOVA showed no statistically significant differences ($p > 0.05$), but effect sizes were large ($\eta p^2 > 0.14$ for both time and SUS), suggesting meaningful differences that a larger sample would likely confirm. Three qualitative themes emerged: the burdens of paper-based documentation (data loss, fragmentation, storage demands), issues with v1 (AI transcription errors, connectivity problems, terminology mismatches), and desired features for v2 (visual prompts, structured inputs, data security, accessibility options), which were used to inform changes to the UI prototype.

Conclusion: Within the UI development cycle, a user-centered, iteratively designed AI-assisted EMR achieved good usability and total task time parity with paper. The study demonstrates that structured formative evaluation with limited resources can produce actionable design improvements, supporting the potential of purpose-built digital tools to reduce documentation burden in P&O practice.

Keywords – *AI-assisted documentation, assistive technology, clinical workflow, electronic medical records, EMR, EHR, human-computer interaction, interface design, mixed-methods, Philippines, prosthetics and orthotics, System Usability Scale, task time, thematic analysis, usability evaluation, user-centered design*

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List of Abbreviations

WHO	World Health Organization
AT	Assistive Technology
JU	Jönköping University
RQ	Research question
HQ	Hypothetical Question
P&O	Prosthetics and Orthotics
CPO	Certified Prosthetist-Orthotist
PWD	Persons with Disabilities
LMIC	Low- and Middle-income countries
DT	Digital Technology
AI	Artificial Intelligence
PD	Product Development
SUS	System Usability Score
UX	User Experience
UI	User Interface
HCI	Human Computer Interaction
HFE	Human Factors Engineering
V1	Version 1
V2	Version 2
PSPO	Philippines School of Prosthetics and Orthotics
ISPO	International Society for Prosthetics and Orthotics
UERMMMC	University of the East Ramon Magsaysay Memorial Medical Center
AFPO	Association of Filipino Prosthetists Orthotists
EMR	Electronic Medical Record
EHR	Electronic Health Record
IEC	International Electrotechnical Commission
ISO	International Organization of Standardization
SOAP	Subjective Objective Assessment Plan
IE	Initial Evaluation
ROM	Range of Motion
MMT	Manual Muscle Testing
QDAS	Qualitative Data Analysis System

1 Introduction

1.1 The Challenges of Prosthetic and Orthotic Service Delivery

The World Health Organization (WHO) and UNICEF estimate that more than 2.5 billion people worldwide need at least one assistive device, a figure representing one-third of the global population that is projected to reach 3.5 billion by 2050 (WHO & UNICEF, 2022). In low- and middle-income countries (LMICs), access to such devices can be as low as 3% of the demand, leaving hundreds of millions without the assistive products that could improve their health, independence, and quality of life (WHO & UNICEF, 2022). A major barrier to assistive technology (AT) access is the shortage of trained personnel, with 10% to 43% of countries reporting an inadequate pool of AT providers (WHO & UNICEF, 2022).

Among the different forms of AT, orthoses and prostheses are critical for individuals with mobility impairment. These devices provide support, assist with anatomical alignment, aid in deformity correction, and, in some cases, replace a missing limb. The field of prosthetics and orthotics (P&O) is dedicated to delivering these interventions, enabling individuals to regain mobility and participate more fully in society. However, P&O care delivery remains a complex, hands-on process, historically rooted in artisanal craftsmanship, and remains device-centric (Hovorka, 2023; Ngan et al., 2022). The conventional workflow begins with patient assessment, measurements, and limb-shape capture, typically through plaster casting. This negative mold is used to create a positive plaster model, which the clinician manually rectifies before fabricating the final device through thermoforming or lamination (Ngan et al., 2022). The entire sequence is labor-intensive, time-consuming, and highly dependent on the practitioner's subjective skill. It generates significant physical waste and requires dedicated workshop space and materials that are often scarce in low-resource settings.

An equally cumbersome documentation process compounds these workflow inefficiencies. In most LMIC clinics, patient assessments, clinical notes, and treatment plans are recorded on paper. This paper-based system is prone to illegibility, data loss, and fragmentation, making it difficult to track patient progress, ensure continuity of care, or aggregate data for quality improvement. The time clinicians spend on documentation detracts from direct patient care and adds to the administrative burden (Moy et al., 2021). This administrative burden contributes to clinician stress and threatens the sustainability of the healthcare workforce, including the P&O profession (Asgari et al., 2024; Li et al., 2022).

The combination of inefficient fabrication, burdensome documentation, and a shortage of skilled professionals creates a great barrier to scaling P&O services. As the global population ages and non-communicable diseases such as diabetes and vascular disease rise, the demand for P&O care is projected to increase dramatically (Hovorka, 2023). The current device-centric, artisanal model is ill-equipped to meet this growing need, necessitating a reassessment of the P&O workflow and the tools used to support it, especially in the context of emerging digital technologies and artificial intelligence (AI).

1.1.1 The Philippine Context and the Digital Shift

The Philippines, an archipelago of over 110 million people, exemplifies the challenges of delivering P&O services in a low-resource context. Its healthcare system is characterized by fragmentation, with a complex mix of public and private providers operating under a decentralized governance structure (Acacio-Claro et al., 2024; De

Mesa et al., 2024; Elepaño et al., 2025). While the Universal Health Care (UHC) Act of 2019 aims to integrate service delivery networks and strengthen primary care, significant logistical and infrastructural hurdles remain (Acacio-Claro et al., 2024; Britiller et al., 2023).

Disability prevalence in the Philippines increased markedly between the 2010 census (1.57%, or 1.44 million) and the 2020 census (8.7%, or 8.46 million)(Manlapaz, 2023). The 2016 National Disability Prevalence Survey reported that almost half of people with disabilities experienced moderate levels of disability, while 12% reported severe disability (Castillo et al., 2019). As of April 2026, the National Council on Disability Affairs reports 2,651,033 registered persons with disabilities (NCDA, 2026). With this growing population in need of assistive devices, a significant gap exists between workforce availability and demand. The Association of Filipino Prosthetists and Orthotists reported in 2025 that only 107 clinicians have graduated from the Philippines' only ISPO-accredited P&O program, with 95 holding ISPO Category 1 certification and only 80 actively practicing in the country (AFPO, 2025).

Clinical documentation in the Philippines is still predominantly paper-based, creating inefficiencies in record-keeping, storage, and data retrieval (Britiller et al., 2023; Derecho et al., 2024). Paper records are kept in filing cabinets that allow onsite access but present poor data security, high storage demands, and sustainability concerns (Derecho et al., 2024). The reliance on paper also makes it difficult to share patient information between facilities, a critical gap in a fragmented health system (De Mesa et al., 2024). The COVID-19 pandemic further exposed these limitations, disrupting supply chains and highlighting the need for resilient, digitally enabled models of care (Acacio-Claro et al., 2024).

P&O clinics in the Philippines still largely rely on a traditional paper-based assessment workflow shown in Figure 1-1. The process begins with the patient arriving at the clinic, followed by the receptionist handing out physical intake forms. The patient or guardian completes these forms by hand before the clinician proceeds with an initial assessment, interview, and objective tests. All information is manually recorded and stored as physical medical records in filing cabinets. Although the Department of Health planned d to implement eHealth in 2020, the pandemic delayed it. The Philippine Health and Services Act (House Bill No. 7422), which supports the integration of eHealth systems, remains pending, with no legislation in place yet. The Philippines currently has no national or centralized EMR system operating across healthcare institutions.

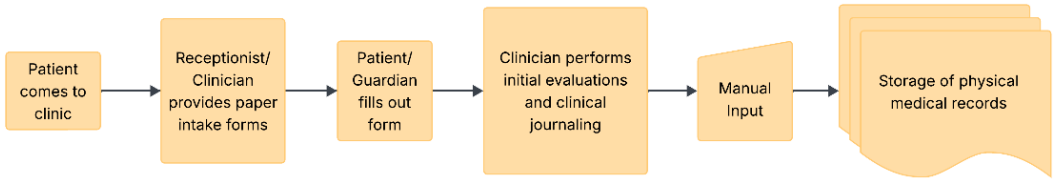


Figure 1-1: Paper-based workflow for patient assessment for a P&O clinic in the Philippines

1.1.2 The Pros and Cons of Electronic Medical Records (EMR)

In response to the limitations of paper-based systems, healthcare globally has been undergoing digital transformation through the adoption of Electronic Medical Record (EMR) systems. EMRs are digital systems used to store, access, and manage health-related data, playing a central role in how care is documented and delivered (Li et al., 2022). In the Philippine setting, EMRs are used for medical recording, reminders,

referrals, report viewing, and electronic claims filing, and they support transmission of mandatory reports to agencies such as PhilHealth and the Department of Health (Acacio-Claro et al., 2024). The terms EMR and EHR (Electronic Health Record) are used interchangeably in this study.

These systems promise to centralize patient information, improve data accessibility, enhance communication between providers, and increase administrative efficiency (Gopidasan et al., 2022). EMRs can improve patient safety by reducing medical errors, providing clinical decision support, and ensuring adherence to clinical guidelines (Hossain et al., 2025). However, widespread implementation has yielded mixed results, revealing complex trade-offs between benefits and drawbacks. A primary concern is the increased documentation burden and its negative impact on clinician workflow. Poorly designed interfaces require excessive navigation and increase cognitive load, fragmenting clinical tasks (Darejeh et al., 2026; Kremer et al., 2022). This information overload leads to clinicians spending more time interacting with the EMR than with patients, contributing to professional dissatisfaction, fatigue, and burnout (Asgari et al., 2024; Holmgren et al., 2024; Li et al., 2022). A scoping review identified several interlocking challenges: frequent task switching due to navigation dead ends, fragmentation of essential information across multiple screens, and cognitive strain caused by workflow misalignment (Olakotan et al., 2025).

Issues of interoperability persist, with different EMR systems unable to communicate, creating data silos that undermine seamless information exchange (Asgari et al., 2024; Derecho et al., 2024; Li et al., 2022). In low-resource settings like the Philippines, these challenges are exacerbated by unreliable internet connectivity, inconsistent power supply, and limited digital literacy among healthcare staff (Britiller et al., 2023; Derecho et al., 2024). Studies on EMR implementation in the Philippines highlight high user acceptance in principle, but also significant frustration with system bugs, slow performance, and lack of features aligned with local workflows (De Mesa et al., 2024). Self-efficacy, or the user's confidence in their ability to use the system, showed the most significant association with successful EMR use (De Mesa et al., 2024). For EMR adoption to succeed, systems must be designed as user-centered tools that are intuitive, efficient, and integrated into the specific context of clinical practice (Britiller et al., 2023; Busse et al., 2023; Elepaño et al., 2025).

1.1.3 Artificial Intelligence in EMR Systems

Artificial Intelligence (AI) has emerged as a powerful tool for addressing many challenges in healthcare practice through its capacity for data analysis and workflow streamlining (Sallam, 2023). AI can streamline clinical documentation, enhance decision support, and reduce cognitive burden on clinicians. One promising application is AI-powered speech recognition that converts clinician-patient conversations into structured clinical notes, reducing the time and effort required for manual data entry (Ng et al., 2025). AI can also assist in synthesizing patient data, identifying relevant patterns, and presenting information in an intuitive and actionable format, thereby reducing information overload and supporting clinical reasoning (Mohsen et al., 2022; Yang et al., 2022).

However, AI tools are not without drawbacks. Their effectiveness depends on accuracy, adaptability to different accents, alignment with clinical contexts, and seamless integration with existing workflows. Concerns about algorithmic bias, data privacy, and the unpredictability of some AI models require careful consideration to ensure ethical implementation(Chen et al., 2024; Sallam, 2023). The successful integration of AI into clinical practice hinges on a human-centered design approach, in which tools are developed in close collaboration with end users to meet specific needs

and enhance, rather than disrupt, clinical work. At present, AI adoption for national health is unevenly integrated and remains largely unstudied in southeast Asia (Wibowo et al., 2025).

1.1.4 Everi Labs AI Copilot UI Prototype

Everi Labs Inc., a startup based in Stockholm, Sweden, is developing an AI Copilot for P&O professionals designed to support clinical workflow, including documentation, component selection, and insurance compliance. The platform uses AI-powered intake, live transcription, and automated data structuring to convert clinical information into SOAP notes, prescriptions, and payer-ready authorization materials. The system aims to streamline administrative tasks and reduce the documentation burden on clinicians (Everi Labs, 2025).

This study is conducted in collaboration with Everi Labs Inc. and is situated within the product development process of the AI Copilot prototype. Prior to the start of the study, the company independently developed a high-fidelity web-based UI prototype. The initial UI prototype, version 1 (v1), features functional AI voice transcription and automatic data population into specific input form fields. The initial prototype UI includes a sign-in page, a dashboard (Figure 1-2), a referral intake page, a patient list page, and other form templates. For the patient assessment, there are face-to-face encounter notes (Figure 3-2) for initial evaluations (IE) and a SOAP note page (Figure 3-3) for clinical journaling. Also, the v1 UI prototype had limited data-saving capacity and allowed single-user accessibility. Although user-centered approaches such as interviews were used for initial concept and design, the v1 prototype remains unevaluated. The proposed evaluation focuses on the patient assessment module, specifically regarding usability, task efficiency, and workflow fit. (See Section 3.3)

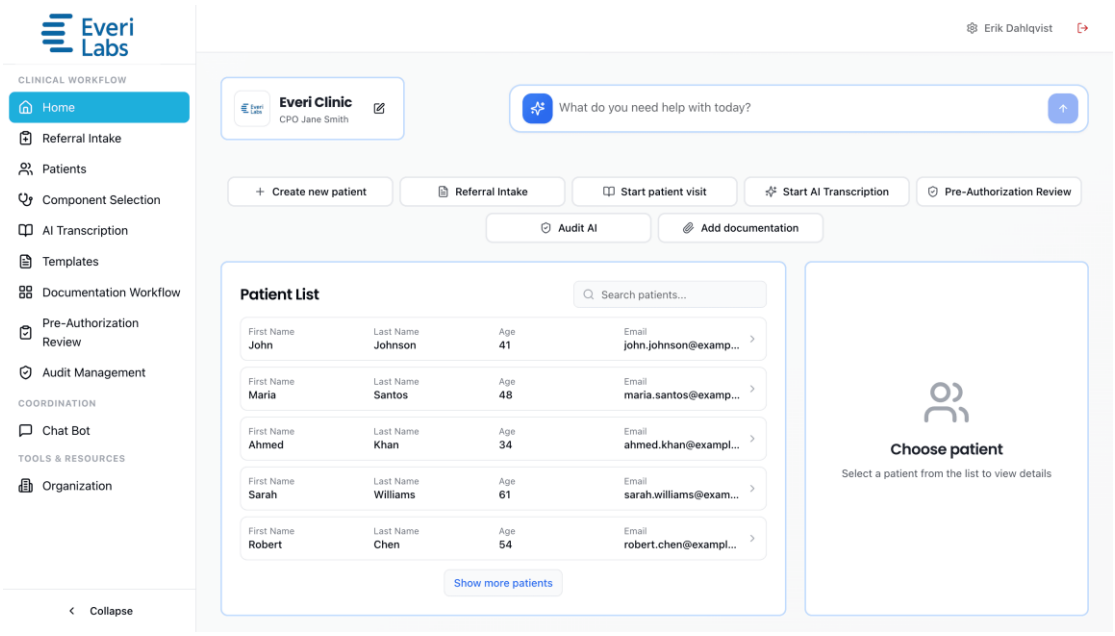


Figure 1-2: Everi Labs version 1 (v1) Dashboard

1.2 Problem Definition

The central problem addressed by this research is the significant burden that inefficient, paper-based patient assessment and documentation workflows place on P&O clinicians in low-resource settings, exemplified by clinics in the Philippines. This burden limits clinical efficiency, detracts from patient-facing time, and hinders the

scalability of essential P&O services. While EMR systems and digital technologies offer potential solutions, their adoption has been hindered by poor usability, high cognitive load, limited integration with specialized clinical workflows, and implementation barriers. There is limited evidence on the development and evaluation of user-centered, AI-assisted EMR systems tailored to the unique needs of P&O clinicians in these environments, particularly compared with existing paper-based workflows.

1.3 Research Questions

This study aims to fill that gap by evaluating a purpose-built AI-assisted UI prototype. In the context of prosthetic and orthotic clinicians in low-resource service delivery settings during patient initial assessment:

RQ1: Does the use of Everi Labs AI Copilot UI prototype v1 show a significant difference in task time duration and overall usability score compared to the paper-based system?

RQ2: Does the use of Everi Labs AI Copilot UI prototype v2 show a significant difference in task time duration and overall usability score compared to the paper-based system?

RQ3: What design features or improvements do clinicians desire to be integrated into the UI prototype v2 after testing v1?

RQ4: Does the UI prototype v2 show a significant difference in task time duration and overall usability score compared to v1?

1.4 Objectives

The research is structured around four (4) primary objectives:

1. To compare total task time and perceived usability (SUS) of the UI prototype v1 against the paper-based system using descriptive statistics, Welch's ANOVA, and effect size analysis.
2. To compare total task time and perceived usability (SUS) of the UI prototype v2 against the paper-based system using the same analytical approach.
3. To collect and describe clinician qualitative feedback through thematic analysis to inform design sketches for the UI prototype v2.
4. To compare total task time and perceived usability (SUS) between the UI prototype v1 and v2, testing whether iterative user-centered redesign produced measurable improvement.

1.5 Hypothesis

The study is structured around four research questions. RQ3 is exploratory and qualitative; no formal hypothesis is tested. For the quantitative comparisons (RQ1, RQ2, RQ4), the following hypotheses are evaluated using Welch's ANOVA and Games-Howell post hoc tests at $\alpha = 0.05$. Because this is a formative, small-sample evaluation, effect sizes (partial eta squared, ηp^2) are interpreted alongside p-values to assess practical significance.

H1 (RQ1: v1 vs paper): There is a significant difference in mean total task time or SUS score between the Everi Labs UI prototype v1 and the paper-based system.

H2 (RQ2: v2 vs paper): There is a significant difference in mean total task time or SUS score between the Everi Labs UI prototype v2 and the paper-based system.

H3 (RQ4: v2 vs v1): There is a significant difference in mean total task time or SUS score between the UI prototype v2 and v1. It is expected that v2 will show improved performance, reflecting the effect of iterative, user-centered redesign informed by focus group feedback.

1.6 Significance of the Study

This research contributes to the field of assistive technology, specifically the practice of prosthetics and orthotics, and the broader domain of human-computer interaction in healthcare. By focusing on improving the efficiency and usability of clinical tools, this study addresses a key barrier to expanding access to essential P&O services. If clinicians can perform their duties more efficiently, they can serve more patients and dedicate more attention to patient-centered care. This work aligns with global efforts to leverage digital technology to overcome barriers in P&O and move toward a client-centric, value-based model of care (Hovorka, 2023).

For the field of P&O, the findings will provide evidence-based insights into the benefits and challenges of adopting AI-assisted EMR systems into clinical practice. The results offer a comparison between a digital tool and the traditional paper-based standard, informing technology adoption decisions for clinic managers, developers, and policymakers. The thematic analysis yields a rich understanding of design features that support or hinder the work of P&O clinicians, offering a resource for future technology development.

From a Human-Computer Interaction (HCI) perspective, this study serves as a case study in applying user-centered design and mixed-methods evaluation in a specialized, low-resource healthcare environment. It contributes to the literature on EMR usability, task time, and the usability of AI-powered clinical tools. By combining quantitative metrics with qualitative insights from thematic analysis of focus group interviews, this research provides a dual perspective for evaluating health technology. This framework can be replicated in other domains.

This study is also relevant to product development because it follows an iterative, user-centered design process. The Everi Labs AI Copilot is treated as a product under development rather than a finished system. The UI prototype is co-designed, tested, and improved based on user feedback throughout the study. This approach helps identify what works well, what is confusing, and what needs to change. Product development is fundamental to this study because it enables developers to iteratively improve the design, supporting better decisions about layout, navigation, task flow, and overall usability.

1.7 Position of the Study in the Product Development Process

This study is situated in the testing and refinement phase of the Everi Labs AI Copilot product development process, following initial concept development and prototype creation. It serves as a proof-of-concept evaluation of the AI Copilot in simulated clinical workflow tasks, using early user feedback to assess whether the interface supports clinicians' work in practical, intuitive ways. This approach is consistent with healthcare usability research showing that early user-centered testing can identify workflow problems, interface confusion, and other issues before wider implementation. The focus on usability, workflow integration, and task support is especially important because poor interface design in clinical systems is associated with navigation burden, task switching, documentation inefficiency, and increased cognitive load (Olakotan et al., 2025). By evaluating the prototype at this stage, the study is expected to inform further design improvements before broader deployment or release.

1.8 Scope and Delimitations

The scope and delimitations of the study are summarized in Table 1-1. This study is focused on the user-centered evaluation of a UI prototype designed to reduce clinical documentation time and improve usability for P&O clinicians. Other allied healthcare professionals were not included in the study. The primary intent is to investigate whether the Everi Labs AI Copilot offers demonstrable advantages in usability and task efficiency over the current paper-based process. The study uses a mixed-methods approach that combines task time studies, the System Usability Scale (SUS), and thematic analysis of qualitative feedback for a multidimensional evaluation, but excludes other evaluation methods for cognitive load, such as NASA-TLX and eye-tracking. The research is limited to the patient assessment stage, including initial evaluation and clinical journaling (SOAP notes), and does not include evaluation of other processes in the P&O workflow, such as casting, fabrication, fitting and follow-up.

Due to geographical and resource constraints, the study is limited to remote online usability testing, thus precluding in-person observation. The study does not involve real patient interactions or protected health information; instead, simulated case scenarios are used. The scope does not extend to backend development, coding, or AI architecture, as the research focuses on the frontend UI presentation layer. Finally, the study is cross-sectional and does not track long-term clinical patient outcomes.

Table 1-1 Summary of study scope and delimitations

Scope (Included in the study)	Explicitly Excluded
Remote, online usability testing (no in-person observation)	In-person or in-clinic observation
Simulated patient scenarios (patient use cases)	Interaction with real patients or any handling of protected health information
Frontend UI development	Backend development, software coding, or technical architecture of the AI engine
Short-term evaluation for patient assessment workflow in P&O	Long-term clinical outcomes and other parts of the P&O process (i.e., casting, fabrication, fitting, delivery, and follow-ups)
Quantitative task-time measurement, System Usability Scale (SUS) scores, and qualitative thematic analysis	Other evaluation methods for usability, cognitive load, and cost-effectiveness (i.e., NASA-TLX, eye tracking, motion analysis).
P&O clinicians in the Philippines who consent to participate	Other health-professional groups in different settings (physiotherapists, physicians, etc.)

1.9 Thesis Structure

This thesis is organized into six chapters. **Chapter 1** establishes the global need for assistive technology and the workforce and documentation challenges facing P&O service delivery, with a specific focus on the Philippine context. It introduces electronic medical records and AI-assisted documentation as potential solutions, describes the Everi Labs AI Copilot prototype, and presents the research questions, objectives, hypotheses, study significance, scope, and delimitations.

Chapter 2 provides the theoretical foundation, framing EMR systems as assistive technology and drawing on the Double Diamond design framework, HCI principles,

and usability engineering standards (ISO 9241-11; IEC 62366). It reviews prior EMR or EHR usability research and identifies gaps in evidence on digital workflows for specialized P&O practice.

Chapter 3 describes the mixed-methods crossover design used to evaluate the UI prototype against paper-based documentation. It covers participant recruitment (10 ISPO-certified CPOs), randomization, the three testing conditions (paper, v1 prototype, v2 prototype), task time measurement, SUS administration procedures, focus group protocols with thematic analysis, and statistical analysis using Welch's ANOVA and Games-Howell post-hoc tests.

Chapter 4 reports the evaluation results, including participant demographics; descriptive and inferential statistics for task completion time and SUS scores across all three conditions with effect sizes; and the three themes emerging from thematic analysis, supported by participant quotations.

Chapter 5 interprets the findings by research question, comparing results to existing literature on EMR usability and efficiency. It discusses theoretical implications for design in low-resource contexts, offers practical redesign recommendations, examines unexpected findings, acknowledges study limitations, and outlines implications for the AT field and future research.

Chapter 6 summarizes the key findings and contributions, provides recommendations for clinical adoption and future research, and concludes with a discussion of the broader significance of user-centered, AI-assisted documentation for P&O practice.

2 Theoretical Framework

This chapter presents the theoretical foundation for the study, focusing only on the concepts that directly inform the research design, evaluation approach, and analytical lens. The framework draws on three interrelated domains: (a) the framing of EMR systems as assistive digital tools in prosthetic and orthotic (P&O) practice, (b) user-centered product development and human-computer interaction (HCI) design principles that guide the iterative prototyping approach, and (c) established usability evaluation methods following Human Factors Engineering (HFE) guidelines.

2.1 EMR Systems as Assistive Technology

The WHO defines assistive technology (AT) as an umbrella term for assistive products and the related systems and services, emphasizing that AT should be people-centered, accessible, and integrated into broader health and social systems (WHO & UNICEF, 2022). In the context of P&O clinical documentation, an EMR functions as an assistive digital tool by reducing barriers to data entry and retrieval, supporting clinicians with varying levels of digital literacy, and enabling more efficient completion of administrative tasks. This framing positions the EMR not merely as a data repository but as an access-enabling system that should meet the AT quality criteria of accessibility, adaptability, and acceptability (WHO & UNICEF, 2022). This study places EMRs as an assistive technology, as it directly supports the AT industry by improving the efficiency of service provision.

The European Medical Device Regulation (MDR) further reinforces the need for safety-critical usability in systems that support clinical decision-making (European Parliament & Council of the European Union, 2017), and relevant usability engineering guidance for medical devices requires manufacturers to analyze, specify, develop, and evaluate usability in relation to safety and use errors (BSN, 2015; IEC, 2016). These regulatory perspectives are directly relevant because the AI-assisted EMR prototype under evaluation involves voice transcription and automated data structuring, where poor usability could amplify clinical risks rather than reduce them. The combination of AT principles and medical device usability standards should be used as the evaluative lens for assessing whether the Everi Labs prototype is both accessible and safe for P&O documentation tasks and patient data management.

2.2 Product Development Frameworks

The study follows an iterative, user-centered design process grounded in two complementary frameworks: the Double Diamond design process (Design Council, 2019) and the interaction design model from Human Computer Interaction (HCI) theory(Rogers et al., 2023). Both frameworks converge on a four-stage structure: understanding user needs, defining the design problem, developing solutions through prototyping, and delivering a tested product. Like the double diamond framework, the interaction design process is non-linear and iterative in practice. This structure aligns directly with the study's position within the Everi Labs product development cycle, where the prototype has undergone initial development and is now being evaluated and refined (Section 1.1.4).

The Double Diamond framework provides a methodological foundation for why the study begins with initial usability evaluation rather than feature expansion: the prototype must first be tested to confirm that the problem has been correctly defined before investing in further development (Figure 2-1). The HCI interaction design model (Figure 2-2) operationalizes this logic through four specific activities: requirements discovery, design concept development, prototyping, and evaluation thus serving as

the guide for the UI design and development process (Rogers et al., 2023). This study mainly focuses on the development and evaluation stage of both frameworks by measuring whether the AI-assisted interface reduces the mismatch between system design and clinical workflow that prior EHR research has consistently identified as a source of frustration, inefficiency, and workarounds (Jansson et al., 2022; Lloyd et al., 2024; Olakotan et al., 2025).

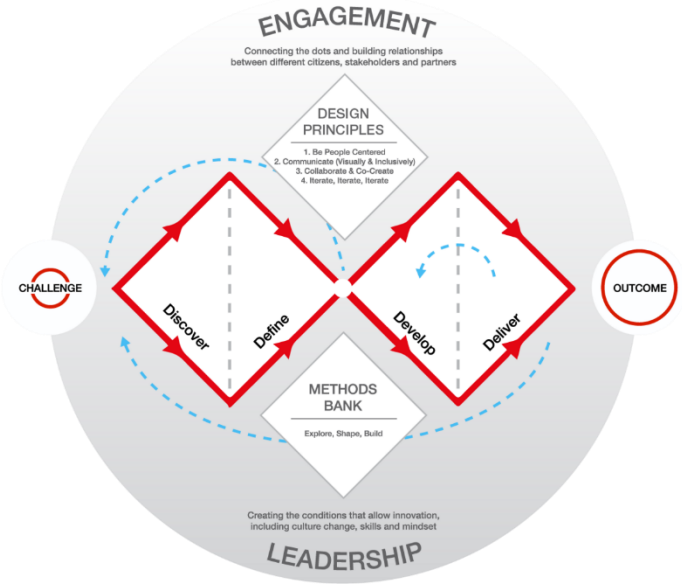


Figure 2-1: The Double Diamond Framework (Design Council, 2019)

Source: 2019, Design Council: https://www.designcouncil.org.uk/fileadmin/_processed_/4/9/csm_Framework_for_Innovation_transparent_14a10de530.png

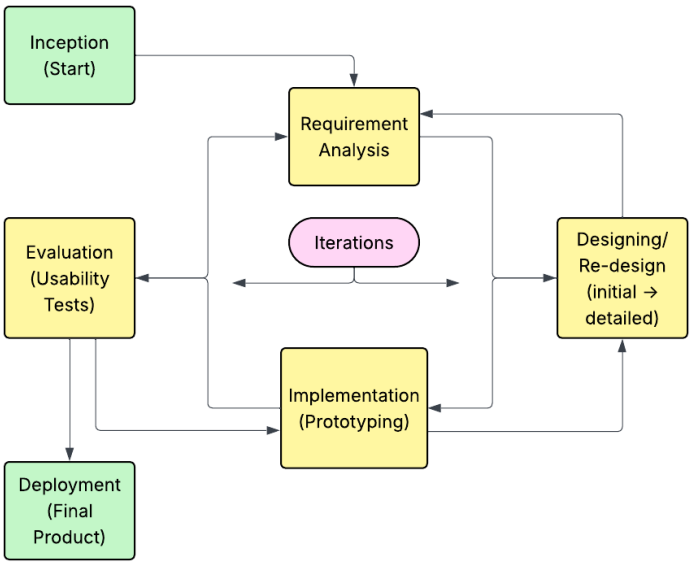


Figure 2-2: Interaction (HCI) Design Model

Because clinicians work with large volumes of information under time pressure, interface decisions such as layout density, navigation depth, and information organization directly affect cognitive load and task performance (Lazar et al., 2017; Mazur et al., 2019). The study therefore evaluates not only whether the prototype is

usable but also whether integrating user-centered design elements will help narrow the gap between the user and their goals, thereby addressing the primary aim of interaction design (Krishna, 2015).

2.3 Usability Engineering or Human Factors Engineering (HFE)

Usability engineering, also termed human factors engineering (HFE), provides the evidence-based foundation for assessing usability. IEC 62366-1:2015 defines usability engineering as applying knowledge about human behavior, abilities, and limitations to design so that adequate usability can be achieved, with usability itself defined by ISO 9241-11 as the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use (Ghasemifard et al., 2015; IEC, 2016).

The IEC standard describes an iterative process of user research, task or function analysis, user interface design, and evaluation through formative and summative testing (IEC, 2016). This study is situated in the formative evaluation phase: the system has undergone initial design, and the current evaluation aims to identify usability issues, measure baseline performance, and gather user feedback before further iteration. Formative evaluation typically uses smaller samples and focuses on identifying what to improve rather than confirming final safety (BSN, 2015; IEC, 2016), consistent with the study's sample and exploratory purpose.

The ISO 9241-11 definition directly guides the selection of outcome measures. Effectiveness is evaluated through task completion accuracy; efficiency through task completion time; and satisfaction through scales like the System Usability Scale (SUS) and qualitative feedback. To guide the evaluation phase of prototype development, usability tests are implemented in accordance with the usability engineering formative evaluation guidelines.

2.4 Usability Evaluation Measures

2.4.1 Task Time studies

Task time is one of the most direct indicators of efficiency in EMR systems. Time-and-motion studies, using structured observation to record duration on specific tasks, are considered a gold standard for quantifying clinical workflows (Moy et al., 2021). A scoping review of EHR usability challenges found time-motion studies among the most common quantitative methods for evaluating documentation burden (Olakotan et al., 2025), with task switching occurring as frequently as 1.4 times per minute in some hospital settings (Moy et al., 2021). In the same study, task completion time was measured during clinician work to provide an objective evaluation of the efficiency of the digital EMR interface in acute care, ambulatory, emergency, and intensive care units (Moy et al., 2021).

The usability engineering guidelines also recognize time-and-motion studies as an early and valuable technique for revealing what interface features and environmental characteristics facilitate or hinder task completion. They can determine whether users are able to complete tasks within the available time and which sub-tasks or steps consume the most time or are performed in a rush (IEC, 2016).

Task completion time is particularly relevant for comparing interface designs. In a systematic review, some studies have shown that completing tasks on a standard EMR interface can take twice as long as on a redesigned interface, and that the number of clicks and navigation steps correlates strongly with perceived workload and cognitive demand (Busse et al., 2023; Kremer et al., 2022). For the proposed AI-assisted EMR

system, measuring task completion time during simulated clinical scenarios will provide an objective indicator of whether the interface reduces documentation delays and supports faster workflow compared to conventional methods.

2.4.2 System Usability Scale

The System Usability Scale (SUS) is the most widely used standardized questionnaire for assessing perceived usability. It was developed by Brooke in 1996 as a quick but reliable tool for measuring how people perceive the usability of computer systems, and it has since been applied across thousands of evaluations in industry and research (Lewis, 2018; Schneiderman et al., 2018). It consists of 10 items on a 5-point Likert scale, producing a single score from 0 to 100, where scores above 70 are considered acceptable (Bangor et al., 2008). The SUS has been applied extensively in healthcare contexts, including EMR and EHR interface evaluation (Ghorayeb et al., 2023; Vlachogianni & Tselios, 2022), and poor SUS scores have been associated with higher clinician burnout and job dissatisfaction (Li et al., 2022; Olakotan et al., 2025). In this study, the SUS provides the primary measure of perceived usability, enabling comparison between the AI-assisted UI prototype and the paper-based system, and between v1 and v2 iterations. See actual SUS questionnaire used in Appendix 8.5.

For the SUS, Scores above 70 to upper 80s are generally considered acceptable, and scores above 90 indicate excellent usability (Bangor et al., 2008). See SUS scoring criteria illustrated in Figure 2-3.

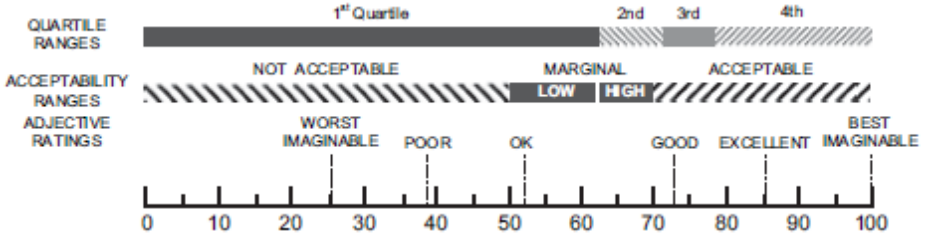


Figure 2-3: System Usability Scale scoring criteria with adjective ratings (Bangor et al., 2008)

Source: https://www.researchgate.net/profile/Joshua-Laurence/publication/351539733/figure/download/fig11/AS:1030115916013568@1622610181372/SUS-Score-assesmentBangor-et-al-2008.png

However, the SUS questionnaire has limitations that are important to acknowledge. It provides a global view of subjective usability but does not identify specific interface problems or provide detailed information on effectiveness or efficiency (Hägglund & Scandurra, 2021). For a complete evaluation, the SUS should be complemented by other methods such as task performance measures, cognitive workload assessments, and qualitative feedback. Some researchers have also noted that the SUS needs user feedback to have a deeper understanding of scores, which is why mixed-methods approaches are recommended (Ghorayeb et al., 2023).

2.4.3 Qualitative Feedback

Standardized scales capture perceived usability but do not identify specific interface problems or explain the reasons behind user ratings(Hägglund & Scandurra, 2021). In EMR usability research, qualitative methods such as focus groups and semi-structured interviews are recommended for eliciting detailed clinician perspectives on interface design, workflow fit, and contextual barriers that surveys alone cannot capture (Lloyd et al., 2024; Olakotan et al., 2025). Focus groups are particularly valuable in formative evaluation because group discussion can surface shared concerns, generate design ideas through participant interaction, and reveal the rationale behind usability

(Ghorayeb et al., 2023). Focus groups also engage participants in brainstorming and generating shared ideas that lead to better solutions (Coleman, 2018).

In this study, thematic analysis is used to systematically interpret open-ended user feedback from focus groups after they complete the assessment tasks. This will help identify what users liked, what frustrated them, what slowed them down, and what changes they recommend. Complementing the SUS and task timing measures, thematic analysis provides a fuller understanding of both the measurable and experiential aspects of usability, which is especially important in the development of the UI design prototype iteration.

The combination of task completion time, System Usability Scale scores, and qualitative user feedback provides a balanced, multidimensional assessment for this study. Together, these measures capture efficiency, perceived usability, and insights into the user experience, which are the three critical indicators of whether an EMR system is usable.

2.5 Previous Literature and Research Gaps

The EMR usability literature consistently demonstrates that poor interface design fragments clinical workflow, increases cognitive burden, and reduces clinician satisfaction. In Australian hospitals, clinicians reported problems with hybrid records, multiple logins, poor searchability, and lack of intuitiveness(Lloyd et al., 2024). ICU clinicians in Finland identified poor layout, inefficient navigation, and double documentation as major barriers(Jansson et al., 2022). Scoping reviews confirm that poorly designed interfaces increase documentation burden, encourage workarounds, and prolong task completion (Asgari et al., 2024; Olakotan et al., 2025), while enhanced systems with improved information tracking are associated with lower cognitive workload (Mazur et al., 2019). User involvement throughout development has been shown to improve adoption, efficiency, and satisfaction, and mixed methods approaches combining subjective and objective measures are recommended for reliable usability evaluation (Darejeh et al., 2026).

In Southeast Asia, AI adoption in health systems is shaped by governance, ethics, and infrastructure concerns, with stakeholders emphasizing the need for regulatory frameworks and contextual design (Wibowo et al., 2025). Participants in the same regional qualitative study emphasized the need for ethical and regulatory frameworks, legal protection, capacity building, multistakeholder collaboration and localized development to support responsible adoption (Wibowo et al., 2025). Studies specific to the Philippines show high EMR acceptance across urban and rural settings, with self-efficacy increased by 31% as the strongest predictor of intent to use (De Mesa et al., 2024). A mixed-methods study in Philippine primary care found that user involvement, training, iterative feedback, and contextual fit were essential for successful deployment, while navigation complexity and inadequate support reduced acceptability (Elepaño et al., 2025). Overall, the presenting literature suggests that a successful EMR system must do more than store information. It must reduce cognitive burden, support clinical workflow, remain accessible, and minimize the risk of error. When the system aligns with human limitations and user expectations, it is more likely to improve usability, acceptance, and safety. When it does not, users are more likely to experience frustration, inefficiency, and workarounds.

In the field of prosthetics and orthotics, digital technology adoption has focused primarily on fabrication workflows such as 3D scanning, CAD/CAM, and additive manufacturing, with practitioners reporting time savings and efficiency gains in device production (Hovorka, 2023; Ngan et al., 2022). However, the digital transformation of clinical documentation and assessment workflows in P&O has received less attention,

leaving a gap in understanding how digital systems can support the administrative and record-keeping aspects of P&O practice.

Despite this evidence base, several gaps remain that this study aims to address. First, there is limited literature on digital workflow in P&O practice in the Philippines; existing studies focus on digital fabrication rather than clinical documentation systems, mostly in developed countries.

Second, while Philippine studies show promising acceptance, detailed usability evaluation in specialized clinical practice remains limited, with most work focusing on general implementation conditions rather than interface-level performance (De Mesa et al., 2024; Elepaño et al., 2025).

Third, research on AI-assisted EMR systems specifically is sparse. A systematic review of AI-based speech recognition for clinical documentation found that word error rates vary from 0.087 in controlled dictation to over 50% in conversational settings, and that performance is heavily context-dependent, with challenges persisting for specialized terminology and accents (Ngan et al., 2022). Few studies have examined how AI tools affect navigation, attention, workflow, and trust in practice. Early tests indicate encouraging results for AI-based documentation tools such as ambient scribing (Olakotan et al., 2025). However, more evidence is needed on how AI-assisted EMR systems objectively affect the P&O clinical workflow.

Fourth, methodological limitations are common: most studies use small samples and fail to implement randomized designs, lack long-term follow-up, and cost-effectiveness analysis (Olakotan et al., 2025), and a study in Japan evaluating EMR efficacy across 3,111 patients acknowledged the absence of randomized assignment between EMR-use and no-use groups in available literature (Kodama et al., 2023). This study helps fill these gaps by evaluating the usability of an AI-assisted EMR prototype for P&O assessment documentation, using a combination of task time, SUS, and thematic analysis.

3 Methodology

This chapter describes the methodological approach used to evaluate the Everi Labs AI Copilot UI prototype against traditional paper-based documentation for P&O patient assessment. The chapter explains not only what was done but why each methodological choice was suitable, with explicit links to the research questions (Section 1.3), theoretical framework (Chapter 2), and established usability engineering standards. The study is positioned within the testing and refinement phase of product development, focusing on assessing a prototype under simulated patient assessment conditions.

3.1 Study Design

The study follows an explanatory sequential mixed-methods crossover design, where quantitative task performance data is collected first and followed up with qualitative user feedback to explain the quantitative results and inform iterative prototype development (Cresswell, 2022). This design was selected because usability is inherently multidimensional: the ISO 9241-11 standard defines usability through effectiveness, efficiency, and satisfaction, requiring distinct measurement approaches for each dimension (Ghasemifard et al., 2015; IEC, 2016). The literature on healthcare usability evaluation consistently recommends mixed methods designs because they capture both objective performance differences and the subjective contextual factors that explain why those differences occur (Darejeh et al., 2026; Kremer et al., 2022). No single measure is sufficient for a complete usability assessment; combining task performance metrics, standardized questionnaires, and qualitative feedback provides a holistic picture of how a system functions in practice.

The study is positioned within the evaluation stage of the Double Diamond design framework and the HCI interaction design model described in Chapter 2 (Design Council, 2019; Rogers et al., 2023). Within the Double Diamond, this work primarily spans the develop and deliver phases: the development phase involved translating user feedback and design principles into low-fidelity and high-fidelity prototypes, while the deliver phase evaluated those prototypes through structured usability testing. In parallel, the HCI model was applied in the “develop” stage as shown in Figure 3-1: Experimental cross-over study design following Double Diamond framework and HCI design process. To guide the formative evaluation phase of the HFE process, where usability testing with smaller samples identifies design shortcomings and generates improvement insights before summative validation (BSN, 2015; IEC, 2016).

The crossover design was chosen as it allows each participant to serve as their own control across documentation conditions and allows for within group comparisons that requires smaller sample sizes. This is favored due to the limited pool of available P&O clinicians in the Philippines (AFPO, 2025). Crossover designs are well-suited for comparing task-level interventions where individual differences in typing speed, clinical experience, and familiarity with digital tools would otherwise introduce substantial inter-individual variability. By exposing all participants to both the paper-based and digital conditions, the design directly addresses the comparative nature of the research questions. Random assignment of task order within and across conditions was used to control for learning and fatigue effects, consistent with best practices in experimental usability research (Lazar et al., 2017). Each participant completed the same assessment tasks for the paper-based system and smaller groups of five (5) tested each prototype version.

The research questions directly shaped the design. RQ1 and RQ2 (differences in task time and SUS score between paper-based and prototype v1, and between paper-

based and v2) required quantitative comparison across conditions. RQ3 (design features clinicians desire for v2) required qualitative data collection. RQ4 (differences between v1 and v2) required quantitative comparison across prototype iterations. The mixed-methods crossover design addresses all four questions within a single coherent framework, allowing both summative comparison across documentation methods and formative insight for iterative design.

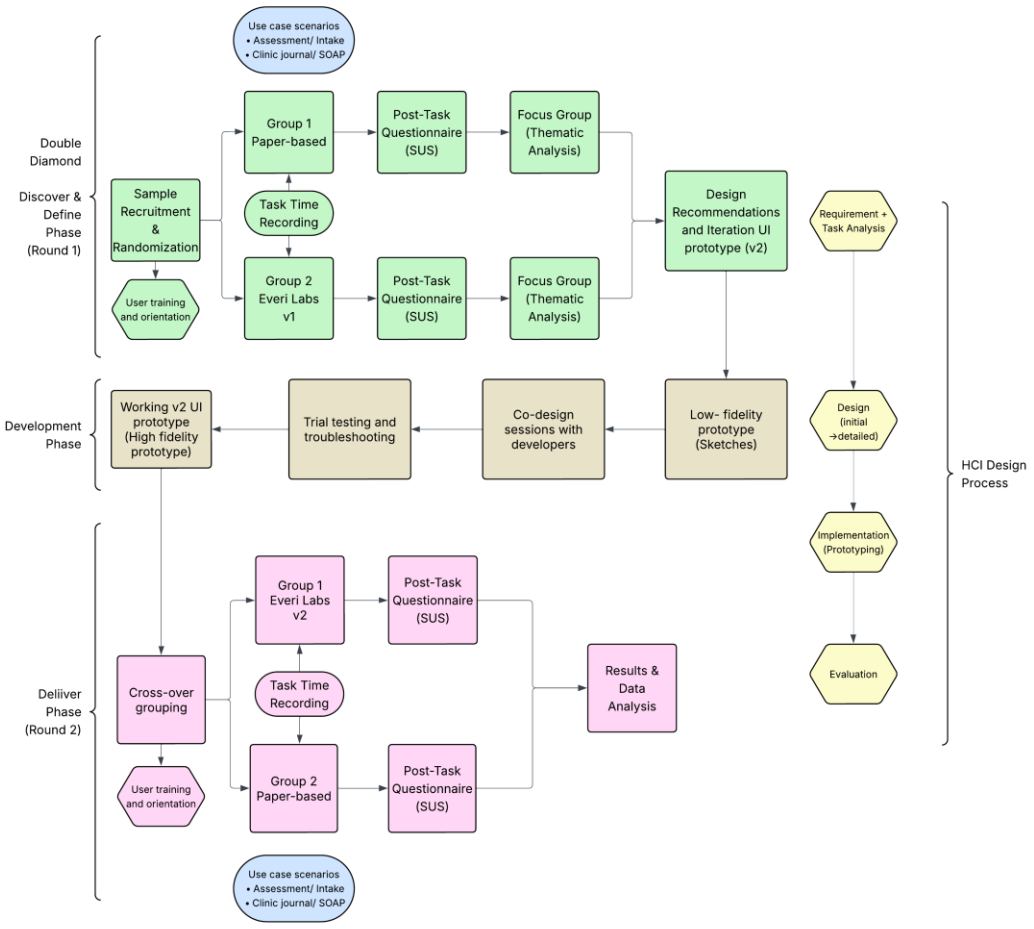


Figure 3-1: Experimental cross-over study design following Double Diamond framework and HCI design process

3.2 Participants

Ten (n=10) certified prosthetist-orthotist (CPO) clinicians were recruited through purposive sampling using a call-for-participants poster (Appendix 8.2) distributed through the Association of Filipino Prosthetists and Orthotists (AFPO), the local professional organization. Interested clinicians who met the inclusion criteria volunteered to join the study. Screening was conducted using an online information form (Appendix 8.4). Specific participant screening criteria are presented in Table 1-1.

Table 3-1: Inclusion and Exclusion criteria

Inclusion Criteria	**Exclusion Criteria**
Actively working as an ISPO certified prosthetist-orthotist clinician Category 1 - Professional	Non-active or non-practicing ISPO certified prosthetist-orthotist clinician Category 1 – Professional (CPO's not involved in direct patient care)

At least one (1) year clinical experience in the Philippines	Less than one (1) year of clinical experience
No prior exposure to the Everi Labs UI prototype	With prior exposure to the Everi Labs UI prototype
Can speak, read and write in the English language – Fluent Proficiency	Unable to speak, read or write in the English language – Non-Fluent English Proficiency
Age 18 years old and above	Below 18 years old
All genders or sex are accepted	Any condition inhibiting safe participation

The sample size of ten participants, with five per prototype group, is consistent with formative usability evaluation standards. The IEC established that 5 users per condition identify majority of usability issues, while another study demonstrated that sizes of 5-8 participants per condition are sufficient for formative testing (Lewis, 2018). The IEC 62366-1:2015 guidance specifically recommends smaller samples during the formative phase because the goal is to identify design shortcomings, not to achieve statistical generalization (IEC, 2016). This sample size is also comparable to prior usability studies in the previous literature.

3.2.1 Participant Randomization

Each participant was randomly assigned a numerical participant ID or code for the entire study duration using the Research Randomizer version 4.0 (Urbaniak & Plous, 2013). Two (2) groups were formed to test the v1 and v2 of the Everi Labs UI prototype. Each participant's numerical code (1-10) was randomly assigned to a test group before the start of usability testing. All participants also completed the paper-based condition in a randomly assigned sequence to control for order effects. Two simulated patient cases (Case 1 for prosthetics, Case 2 for orthotics) were randomly assigned to participants, and tests were conducted in random sequence for counterbalancing (Appendix 8.13).

3.3 Documentation Systems Evaluated

Three documentation conditions were compared. The paper-based system used standard assessment forms from the Philippine School of Prosthetics and Orthotics (PSPO): a transtibial prosthetics form (Appendix 8.8), a lower limb orthotics form (Appendix 8.9), and a clinical SOAP notes form (Appendix 8.10). Participants completed these using pen and paper, reflecting current practice in most Philippine P&O clinics. The design, structure, and content of these forms follow the standard assessment documentation used in Philippine P&O education and clinical practice.

The Everi Labs UI prototype v1 was a web-based, internet-dependent system developed prior to this study. It featured AI voice transcription and data auto-population that organizes voice transcripts into specific input fields. Three main functional components were identified: a dashboard for navigation, an assessment page for patient initial evaluation, and a clinical journal (SOAP) page for narrative documentation. As a prototype, it had specific limitations including internet-dependence, mono-lingual design (English only), and restricted data persistence. (See Figure 1-2, Figure 3-2 and Figure 3-3)

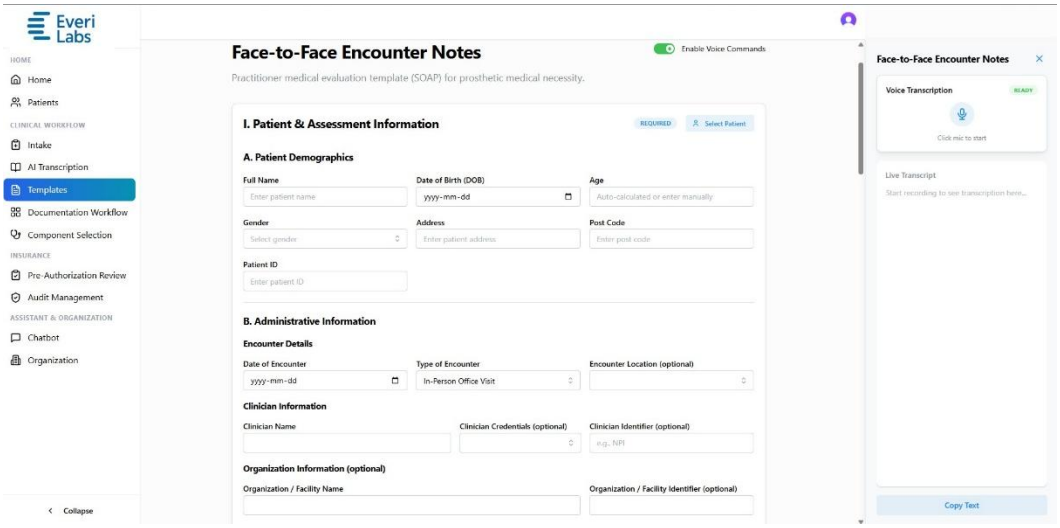


Figure 3-2: Assessment Form for Patient IE Everi Labs version 1 UI prototype

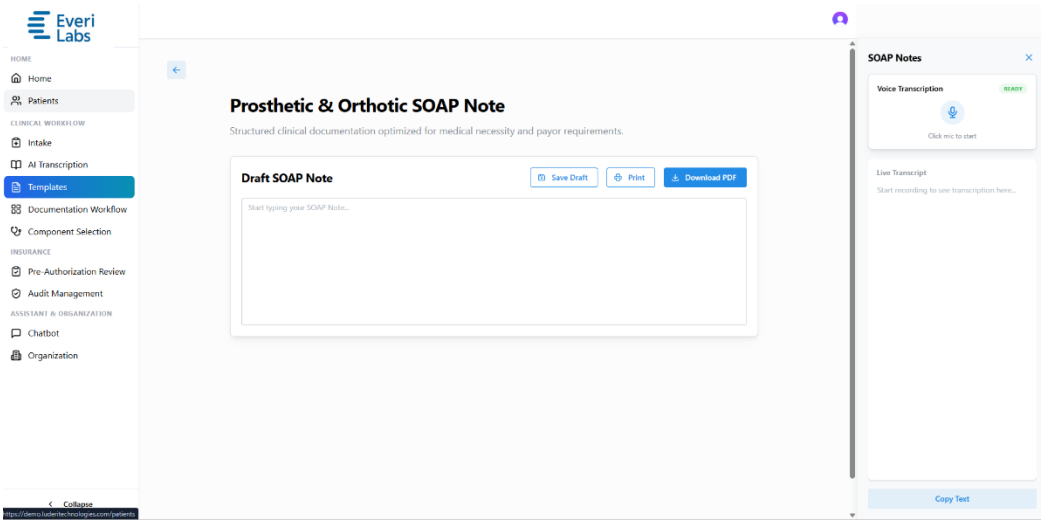


Figure 3-3: SOAP Notes for Clinical Journals Everi Labs version 1 UI prototype

The Everi Labs prototype v2 incorporated design changes based on focus group feedback and usability design principles, as described in detail in Section 3.7.3. All testing was conducted remotely via Zoom or Google Meet at the participant's preference, using their own personal laptop. All tests and interviews were recorded with consent.

3.4 Simulated Patient Cases

Two simulated patient cases were developed to mitigate the complexity of conditions. Case 1 described a 47-year-old male with a transtibial amputation requiring a prosthesis (Figure 3-4); Case 2 described a 57-year-old female with an ankle-foot orthosis prescription (Figure 3-5). These case types represent the most common device prescriptions in Philippine P&O practice, as analysis of patient data from clinics such as the University of the East Ramon Magsaysay Memorial Medical Center (UERMMMC) indicates high demand for Ankle-Foot Orthoses and Transtibial prostheses (PSPO, 2026).

Cases were carefully matched for complexity and information volume by ensuring that each case followed all components required to complete the standard assessment forms. This matching ensures that performance differences reflect the documentation

system rather than the scenario itself. Each case included sufficient clinical detail (patient social and medical history, key findings, subjective and objective data) to complete all sections of the assessment forms and SOAP note. The researcher role-played as the patient during the assessment interview.

Simulated cases are acceptable at this stage of product development because the study is a formative evaluation of a prototype to identify if the prototype is usable in terms of basic UI functionality. Using simulated cases eliminates patient safety risk, standardized the information clinicians receive across conditions, and is consistent with the IEC 62366-1 approach of using simulated use scenarios for formative usability testing (IEC, 2016).This approach also allows the researcher to control information delivery, ensuring that each participant works with identical clinical data regardless of which case they receive. See simulated case details in Appendix 8.11 and 8.12.

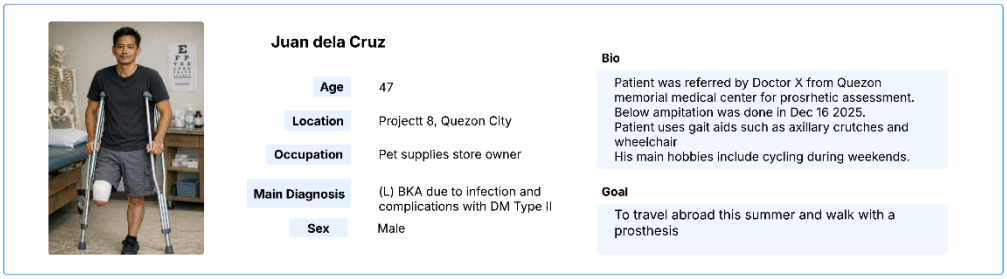


Figure 3-4: Simulated Patient Case 1 (Prosthetics)

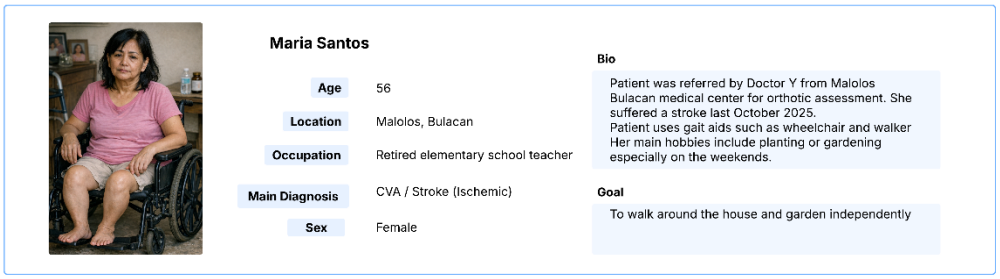


Figure 3-5: Simulated Patient Case 2 (Orthotics)

Source: Persona images generated by AI (Microsoft Copilot)

3.5 Data Collection Methods and Procedures

Three complementary methods were selected, each targeting a specific dimension of usability as defined by ISO 9241-11 and linked to specific research questions. The combination of task completion time (efficiency), System Usability Scale (satisfaction), and focus group discussions with thematic analysis (qualitative insight) provides a balanced, multidimensional assessment that captures both measurable performance and experiential aspects of usability.

The study was conducted in three sequential phases: pre-test preparation, Round 1 testing (paper-based and v1 prototype with focus group feedback), and Round 2 testing (paper-based and v2 prototype) as illustrated in Section 3.1.

3.5.1 Pretesting

Before testing, use-case scenarios were developed for each documentation system based on actual clinical workflows in accordance to HFE guidelines. Use cases are narrative descriptions of user behavior depicted as sequences of tasks and system

interactions (Rogers et al., 2023). They were developed to clarify and itemize the steps the user must follow to achieve functional goals (Coleman, 2018).

For the study, the use cases focus on two patient assessment tasks: patient initial evaluation (IE) as Task 1 and clinical journaling (SOAP) as Task 2. Task 1 includes gathering the patient's personal information, medical history, social history, treatment goals, and initial objective findings (ROM and MMT). Task 2 includes the clinician's narrative description of the consultation, overall key findings, and special notes. These two (2) documentation tasks were selected because they represent the main assessments completed by clinicians in practice. The use cases are further described in Table 3-2 and Table 3-3 below.

Table 3-2: Use case for Paper-based system

Step	User Action
1	User selects correct paper form depending on patient case
2	User conducts patient IE and writes in forms (Task 1)
3	User completes clinical journal (SOAP) using pen and paper (Task 2)
4	User stores physical files in a folder

Table 3-3: Use case for Everi Labs UI prototype (v1 and v2)

Step	User Action	System Response
1	User logs in using email and password	System validates login data
2	User navigates to patient assessment page	System opens dashboard
3	User conducts patient IE and inputs data (Task 1)	System accepts and stores data input AI accepts voice input encodes speech and repopulates data fields
4	User saves and downloads PDF	System saves and downloads completed form
5	User opens SOAP notes page	System opens SOAP interface
6	User composes clinical journal entry (Task 2)	System accepts and stores input
7	User saves and downloads completed SOAP note	System saves and downloads file

Prior to UI prototype testing, Participants were also given individual training demos with 30 minutes allocated for each. Trainings included a video tutorial and a live online demo focused on navigation and task overview.

3.5.2 *Task Time studies*

Task time was selected as the measure of efficiency, directly addressing RQ1, RQ2, and RQ4 by providing an objective, quantifiable indicator of whether the AI-assisted interface reduces documentation time compared to paper and across prototype versions. The practical rationale is that documentation burden in P&O clinics is fundamentally driven by the time required for manual recording; therefore, any measurable reduction in task time represents a meaningful improvement in clinical workflow efficiency that directly affects patient throughput and clinician workload.

Two task times were recorded per session: patient IE (completing the assessment form) and SOAP note creation. Timing started upon the participant's verbal signal ("start/go") and ended upon form completion or digital save ("end/stop") as illustrated in Figure 3-6. A digital stopwatch application in hh:mm:ss format was used. Recorded times were encoded in Microsoft Excel and converted to decimal minutes using the formula in Equation 3-1.

Equation 3-1: Equation for time decimal conversion

$$Decimal\ minutes = h \cdot 60 + m + \frac{s}{60}$$

All participants performed tasks at self-selected speeds. To ensure completion, all forms were submitted to the test moderator for inspection, and participants had screen sharing enabled during UI prototype testing.

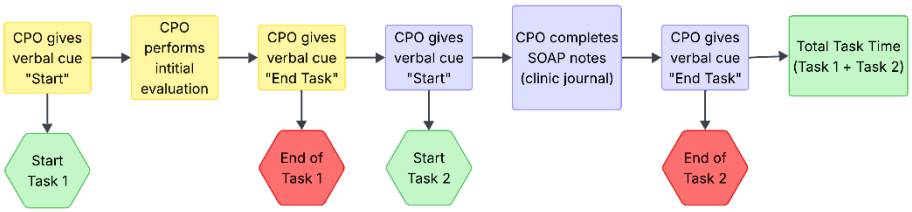


Figure 3-6: Task time recording procedure

The total time per participant per session was approximately 60 minutes, with the paper-based session and each prototype session conducted on separate days to reduce fatigue and order effects. Participants were encouraged to take breaks between tasks. The researcher was present throughout all sessions to monitor participants and provide technical assistance if needed.

3.5.3 *System Usability Scale (SUS)*

The SUS was chosen as the measure of perceived usability, directly addressing RQ1, RQ2, and RQ4. The practical rationale for selecting the SUS over other instruments is that it is technology-agnostic and can be administered across both paper-based and digital systems, enabling direct comparison across all three documentation conditions within a single framework. It also produces a single standardized score that can be benchmarked against established thresholds (above 70 acceptable, above 90 excellent), making results interpretable even with the small sample size typical of formative usability studies. Because this study is an early-stage formative evaluation, identifying whether the prototype reaches an acceptable level of perceived usability is more actionable than granular subscale analysis offered by longer instruments.

The SUS was administered via Google Forms using post-test questionnaire (Appendix 8.6) after each condition. Scores were computed using the standard formula (Table 3-4) and categorized using an adjective rating scale as summarized in Table 3-5. The SUS directly addresses RQ1, RQ2, and RQ4 by providing a comparable perceived usability benchmark across all three documentation conditions.

Table 3-4: SUS score computation (Jyoshita, 2025)

Step	**Formula**	**Applies to**
Adjust odd items	Response – 1	Q1, Q3, Q5 Q7, Q9
Adjust even items	5 − Response	Q2, Q4, Q6, Q8, Q10
Final SUS score	(Sum of all adjusted scores) × 2.5	Overall SUS

Table 3-5: SUS score and adjective criteria used (Bangor et al., 2008)

SUS Results	**Adjective Findings**
Above 80	Excellent usability
71 to 80	Good usability (above average)
70	Average usability (benchmark score)
51-69	Indicates usability issues
Below 50	Poor usability (significant improvements needed)

However, the SUS provides a global view of subjective usability but does not identify specific interface problems or explain the reasons behind user ratings (Hägglund & Scandurra, 2021). For this reason, focus group discussions were included as the third data collection method.

3.5.4 Focus Group Discussion

Focus group discussions were included to address RQ3: what design features clinicians desire for the v2 prototype. The practical rationale is that while the SUS captures perceived usability as a single global score, it does not explain why users rated the system as they did or what specific changes they would recommend. The group interaction format surfaces shared concerns, generates design ideas through participant dialogue, and reveals the rationale behind usability ratings in a way that individual interviews or surveys cannot, while also saving time (Ghorayeb et al., 2023).

On separate days, Two (2) focus groups of five (5) participants were conducted remotely using Google Meet after Round 1 testing. A focus group discussion guideline was prepared before sessions to ensure smooth facilitation and achieve objectives (Appendix 8.7). Each focus group discussion was allocated 2-3 hours per session. The sessions started with an introduction and orientation covering discussion rules. Focus group questions focused on open-ended feedback covering the main components of interface design: (1) visual and aesthetics, (2) navigation, (3) layout and (4) action (5) input (Tidwell et al., 2020). Additional questions were provided regarding task functionality, other comments, and follow-up inquiries to elicit deeper understanding and clarity of answers. Google Meet with Gemini AI was used simultaneously for video recording and audio transcription. Although English was the primary language, participants were allowed to use Tagalog for clarity and ease of expression. All focus

group transcripts were reviewed manually and corrected line by line against audio recordings in Microsoft Word. Multiple transcript reviews (3-5 times) were conducted prior to coding.

3.5.5 Thematic Analysis

Transcripts were analyzed using the six-phase thematic analysis: familiarization, initial coding, searching for themes, reviewing themes, defining themes, and reporting (Braun & Clarke, 2006). An inductive approach was used, allowing themes to be generated from the data rather than being imposed on existing frameworks.

Direct participant quotes were transferred to a Microsoft Excel spreadsheet, with each quote in a cell attached to the respondent's participant code. Coding was performed manually on the same data sheet. After codes were generated, subthemes and main themes were identified after multiple iterations. No specialized Qualitative Data Analysis Software (i.e. NVivo, Atlas.ti, MAXQDA) was used due to study funding limitations. Although the use of such systems provides better traceability through audit trails, thematic analysis can also be conducted without them if thorough code review and validation are performed (Guest et al., 2012). To address this concern, voluntary respondent validation was conducted. Each participant was given access to the focus group transcripts and coding spreadsheet, and their comments and suggestions for code changes were applied, aligning with the iterative approach suggested by thematic analysis guidelines (Braun & Clarke, 2006). The summary of the thematic analysis steps are detailed in Table 3-6.

Table 3-6: Applied Thematic analysis methodology (Braun & Clarke, 2006)

Thematic Analysis Stages	**Steps taken**
Review of Interview Transcripts	Used an AI transcription tool (Gemini) supplemented with manual encoding of transcripts. Transcripts were manually checked line by line with audio or video recordings.
Generation of initial codes	Inductive thematic analysis was used.
Searching for Themes	Thematic coding and analysis were performed.
Review and validation of codes and themes	Codes and theme spreadsheets were sent to participants for voluntary review. (Respondent validation)
Writing the qualitative findings report	Thematic analysis summary included in results chapter (Section 4.3)

3.6 Data Analysis

The independent variable was documentation method with three levels: paper-based, v1, and v2. Dependent variables were IE time, SOAP time, total task time, and SUS score.

3.6.1 Test of Normality

Normality was assessed using the Shapiro-Wilk test and visual inspection of Q-Q plots, with priority given to visual inspection given the small sample sizes. Before performing normality tests, outliers falling outside the whiskers (IQR x 1.5) were identified using boxplots in IBM SPSS Statistics version 32.0 (IBM, 2026). No extreme outliers

requiring removal were identified, and all values were retained as plausible observations.

3.6.2 Statistical Analysis

Because Levene's test indicated heterogeneity of variances across groups and group sizes were unequal, Welch's ANOVA was used to test overall differences for each dependent variable, with Games-Howell post hoc tests for pairwise comparisons. These tests are robust to unequal variances and sample sizes (Lazar et al., 2017). Effect sizes were calculated using partial eta squared based on the Welch-adjusted F statistic and degrees of freedom. Classical ANOVA, repeated-measures ANOVA, mixed ANOVA, and MANOVA were not used because their assumptions of sphericity, homogeneity of covariance, and equal variances were violated (Lazar et al., 2017).. This analytical approach provides a statistically robust evaluation of differences across documentation methods despite unequal variances and small, unbalanced group sizes. Statistical analysis was performed using IBM SPSS Statistics version 32.0 (IBM, 2026).

3.7 Prototype Development and Rationale

The v2 prototype was developed through an iterative process grounded in the HCI interaction design model described in Chapter 2, following the four basic activities of interaction design: discovering requirements, designing concepts, prototyping, and evaluating (Rogers et al., 2023). Within the Double Diamond framework, this work sits within the develop phase, where design proposals and sketches are refined into a high-fidelity version through iterations of prototyping, testing, and troubleshooting. This section presents the complete design rationale each grounded in UI principles and specific user feedback from focus group discussions.

3.7.1 User Needs and Prototype Requirements

User needs in this study were defined as the practical requirements the EMR system must satisfy to support clinical work effectively, safely, and efficiently. In healthcare information systems, these needs are shaped by documentation burden, workflow fit, digital literacy, and the ability of the interface to support clinical decision-making and patient assessment (Acacio-Claro et al., 2024; Moy et al., 2021). The identified user needs were:

1) reduce documentation time without compromising clinical accuracy, as time-motion studies show that data viewing and data entry account for a large share of task switches in clinical work (Moy et al., 2021);
2) support for clinicians with varying levels of digital literacy through system training (Acacio-Claro et al., 2024);
3) ensure that workflows align with existing clinical routines and local service realities (Acacio-Claro et al., 2024; Busse et al., 2023);
4) provide a clear and intuitive interface usable during target tasks in patient assessment (Jansson et al., 2022; Lloyd et al., 2024); and
5) include user feedback as part of prototype refinement (Braun & Clarke, 2006; Hägglund & Scandurra, 2021).

These user needs were translated into UI usability and functional requirements through collaboration with Everi Labs company representatives and developers (see Table 3-7 and Table 3-8). Each requirement was categorized as required (essential for Round 2 testing) or desired (targeted for future development). All “required” components for

usability and functionality were implemented and tested prior to user testing of the v2 prototype.

Table 3-7: UI prototype v2 Usability Requirements

Category	**Required**	**Desired**
Platform Compatibility	Web-based interface accessible via modern browsers	Cross-platform support (Web, iOS, Android)
User Roles	Single user role (clinician)	Multi-role support (clinician, admin, other allied rehab professionals)
Prototype Fidelity	High-fidelity clickable UI	Fully interactive workflow simulation with data recovery
Language	English	Multi-lingual options (English, Tagalog, etc.)

Table 3-8: UI prototype v2 Functional requirements

Category	**Required**	**Desired**
User Login	Basic login with email + password	Double authentication
Create Patient IE File	User can open and complete a patient initial evaluation form with required fields (demographics, complaint, history and key findings)	Dynamic form with auto-suggestions, conditional fields, and progress indicators
Create SOAP Notes	User can create a SOAP note with all four sections (Subjective, Objective, Assessment, Plan)	Pre-filled templates based on input data; smart suggestions
Prototype Stability	No critical system crashes	Fully stable with smooth animations and micro-interactions

3.7.2 Conceptual Designs and Personas

Conceptual models outline what people can do and how the product aids users towards specified use goals (Rogers et al., 2023). In HCI and HFE, conceptual models are derived from product requirements captured through case scenarios. The main areas of interface design: visual and aesthetics, navigation, motion, input, task functionality, and accessibility (Tidwell et al., 2020). These UI design principles were balanced with Everi Labs' brand identity and user feedback to create the improved v2 prototype.

A main user persona was developed from the mean demographic data of the participant sample group and goals and frustrations derived from related literature on common issues identified in Philippine healthcare settings (Section 4.1). Persona development is a design technique that specifies a user's personal, professional, and technical background as well as motivations (Coleman, 2018; Rogers et al., 2023). The main persona embodied the characteristics of a typical Filipino P&O clinician: experienced in manual documentation, time-constrained, moderately digitally literate, and working in a resource-limited clinical environment with variable internet connectivity (Figure 3-7). Two (2) simulated patient personas were also developed to ensure consistency across testing conditions, as described in Section 3.4.

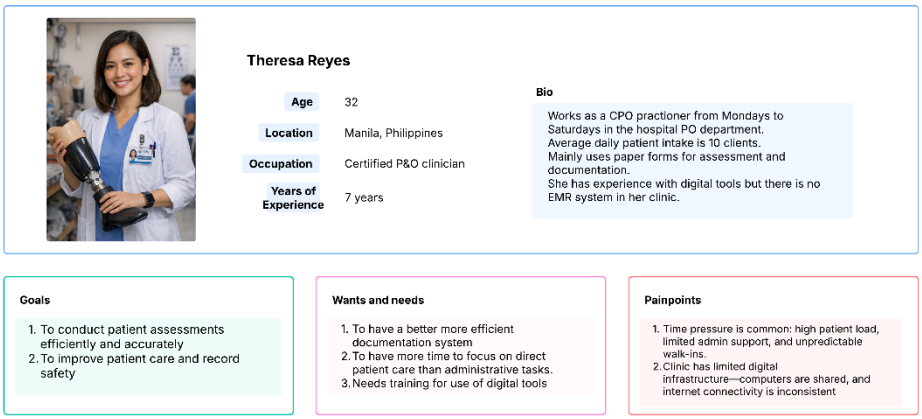


Figure 3-7: Main Persona Profile Summary

Source: Persona image generated by AI (Microsoft Copilot)

3.7.3 Prototyping Approach

Prototyping is an important stage from start to end of the UI design process (Coleman, 2018). Upon completion of the task time, SUS tests and thematic analysis of focus group discussions for the v1 prototype, Low-fidelity paper prototypes were first independently hand drawn by the researcher hand to explore layout and component placement for the dashboard (Figure 3-8), IE (Figure 3-9), and SOAP note page (Figure 3-10). These paper sketches allowed rapid exploration of design alternatives without engineering overhead (Schneiderman et al., 2018). The hand sketches were presented to the Everi Labs developer team, and continuous iterative design sessions were conducted over a 3-week timeline from sketches to functional v2 prototype. Each remote design session lasted 30 minutes to 1 hour, held biweekly. The researcher provided specific design specifications for UI elements (e.g., icons, images, text, placement). The development team granted the researcher full UI design flexibility for the v2 prototype, with restrictions on certain elements tied to AI features that may require complex backend configurations.

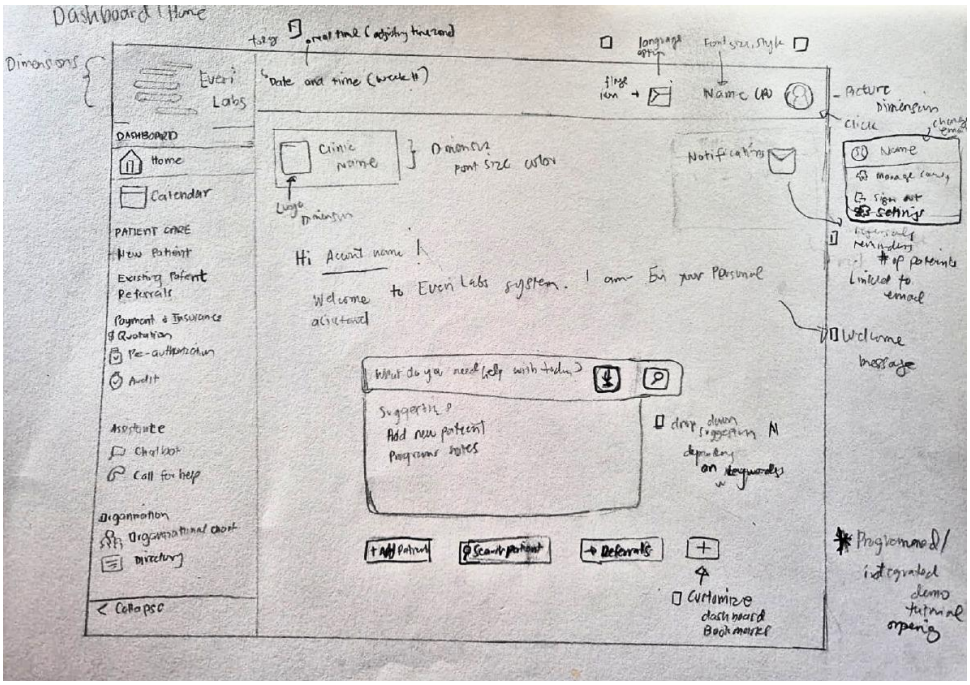


Figure 3-8: Low fidelity v2 UI prototype (Dashboard)

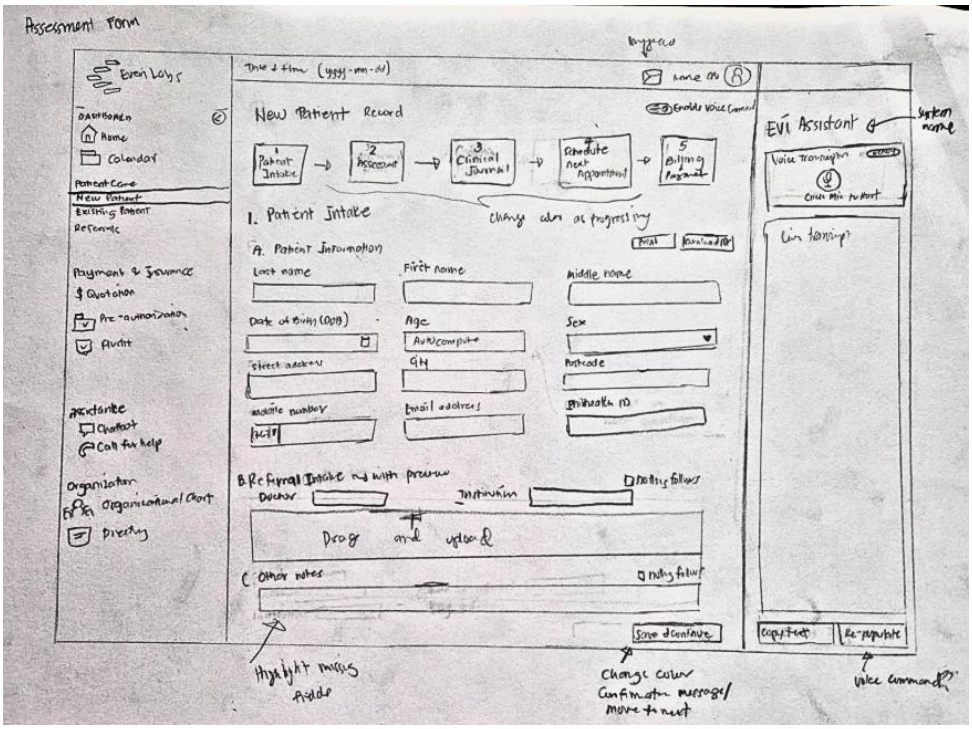


Figure 3-9: Low fidelity v2 UI prototype (Assessment)

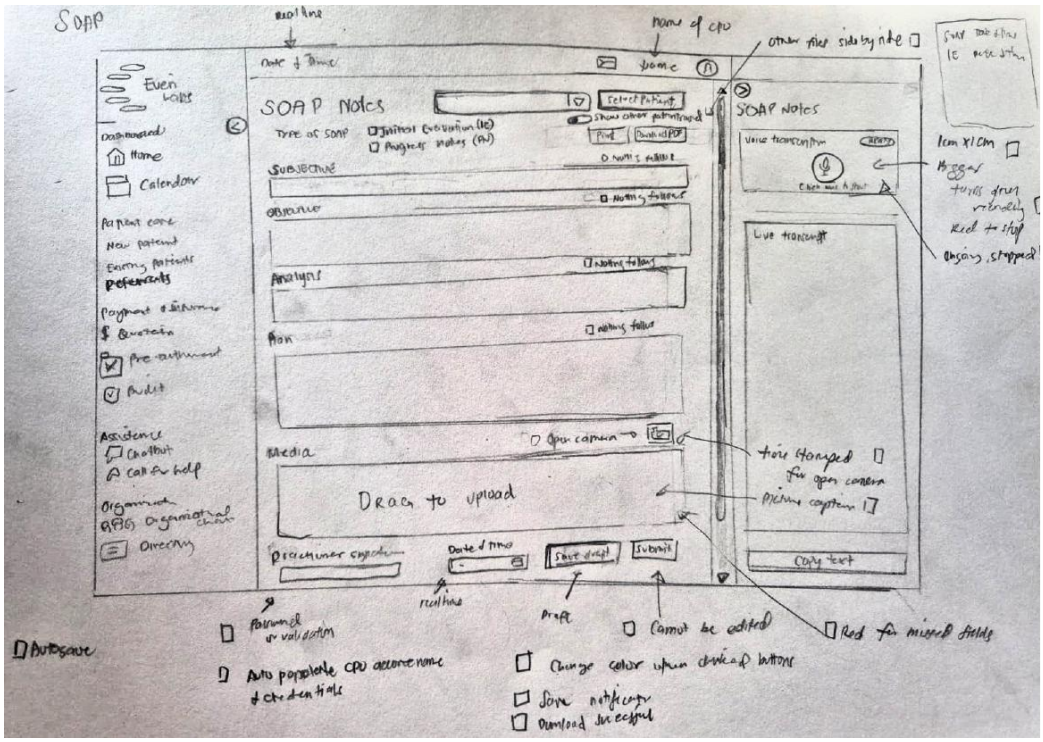


Figure 3-10: Low fidelity v2 UI prototype (SOAP)

These paper prototypes were refined into a co-designed high-fidelity prototype built using Mantine UI, React.js and Python, consistent with Everi Labs' existing technology stack. High-fidelity prototypes have the look, feel, and function of the final product with some limited features (Schneiderman et al., 2018). Mantine UI was chosen because it provides a comprehensive, accessible set of prebuilt components that allow designers to maintain visual consistency across the platform without having to build every element from scratch. It is highly customizable and integrates naturally with React.js, which is the UI library the Everi Labs AI Copilot uses. This combination supports rapid prototyping, high-fidelity UI functionality, and minimizes engineering overhead. Certain design aspects were predetermined by the Everi Labs developer team to preserve brand identity. This section describes the key elements of the v2 UI design such as visuals, layout, action, and data input and the theoretical foundations of each area of interaction design (Tidwell et al., 2020). These principles are combined with user feedback gathered from focus group discussions (Section 3.5.4). The design rationale and specific UI changes are described and demonstrated below.

3.7.4 Visual and Aesthetics

Typography. The template uses a San serif font to provide a more contemporary look and are better for readability at smaller sizes (Tidwell et al., 2020).

Color. The UI prototype follows Everi Labs pre-set color palette to maintain brand consistency. The main palette uses colors like blues and greens and a light white background to evoke an elegant, professional feel which are also common colors associated with the medical field. Color contrasts were also used to indicate response to clicks on certain buttons or icons by using reds for warnings or errors and greens as "okay" prompts (Tidwell et al., 2020). This design aspect was prioritized due to users' need for color prompts for completed and missed fields using red and green indicators (Figure 3-11).

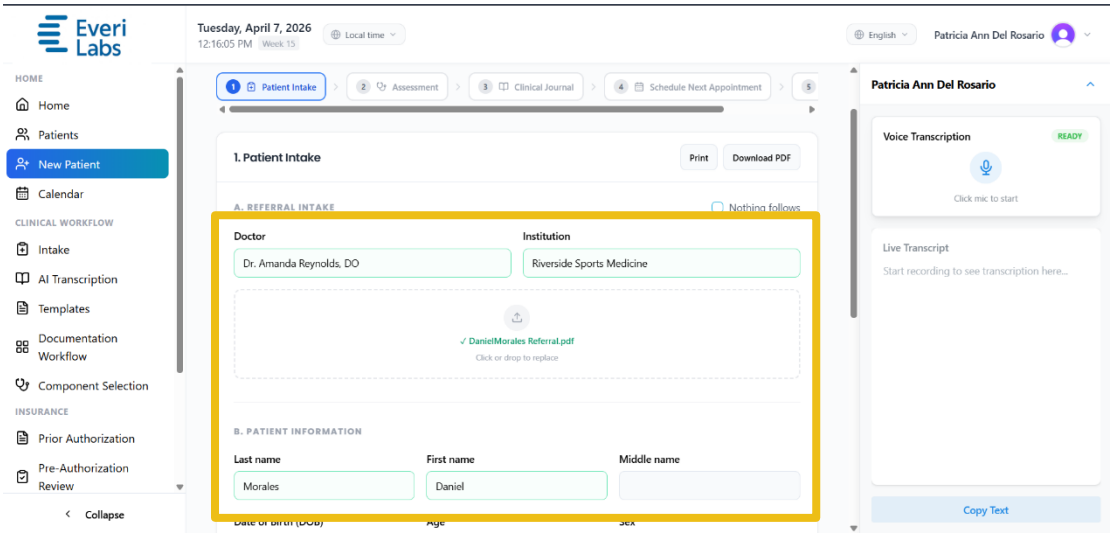


Figure 3-11: High fidelity v2 UI prototype visual color prompts

Images or icons. Everi Labs uses CSS pixels (px) and rem units as the primary measurements for design standardization across the platform. Since the system is a web-based application built in React.js and Mantine UI, all spacing, typography, icon sizes, and layout dimensions follow CSS conventions.

According to the Fitts Law of User Experience (UX), touch targets should be large enough for selection accuracy (Yablonski, 2024). During the design phase, Recommendations for images and icon sizes adjustments to the minimum recommended size by the Web Content Accessibility Guidelines (WCAG) at 44 x 44 css px were applied (Yablonski, 2024).

3.7.5 *Navigation*

Menu page. The dashboard or menu page is the first page that the users interact with upon logging in. This page shows the user the main functions of the systems. The main dashboard is intended to simplify the initial navigation of users. Following the Laws of UX, the Hick's Law is applied by eliminating steps and minimizing screen components (Yablonski, 2024). Breaking down complex tasks to simple ones like a simple search bar decreases mental effort. This design law is commonly applied in home pages like Google and AI search tools by putting a search bar on the center of the main page which is called the center stage visual framework used "hook" the users (Tidwell et al., 2020). Users mentioned the need for referral notifications, active patient count, and removal of the patient list and bookmark customization which were added to the v2 UI prototype dashboard. See menu page changes in Figure 3-12.

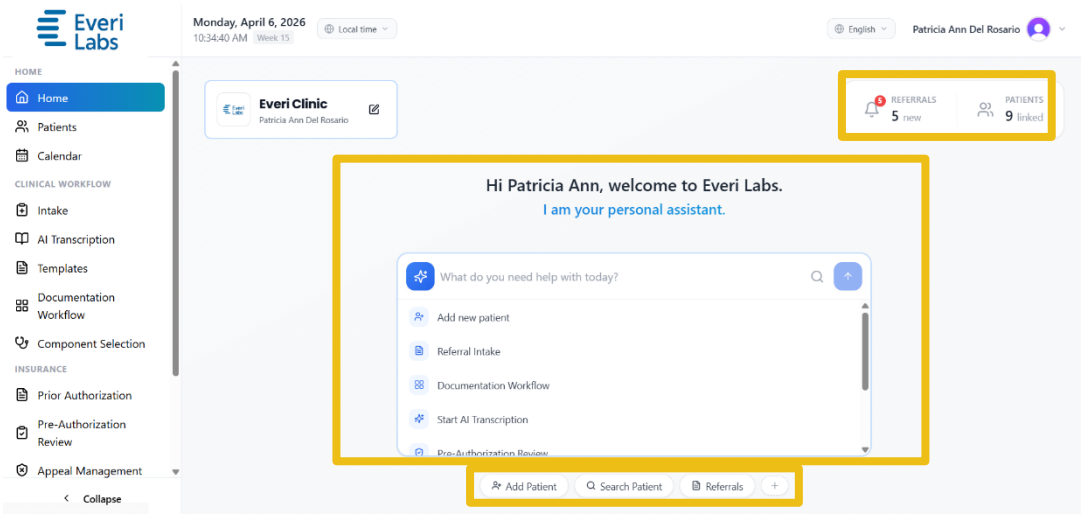


Figure 3-12: High fidelity v2 UI prototype (Dashboard)

Progress indicator. This is a navigation tool that lets users know where they are in the task at hand. Progress indicators help users decide whether to continue and estimate this allows users to go to a previously completed step and review input (Tidwell et al., 2020). These indicators were added for these reasons. Steps also follow the assessment workflow described by users in Section 4.3.1. Added progress indicators are shown in Figure 3-13.

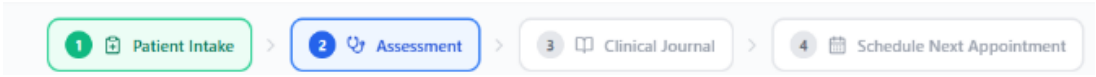


Figure 3-13: Progress indicators for v2 UI prototype

3.7.6 Layout

UI Regions. The regions of UI are the headers, menu panes, main content area, footers and panels. The v2 UI prototype follows the standard UI region positioning with headers at the topmost part of the page, menu panes are at the left-hand side of the screen, the main content area occupying the center of the screen, footers placed at the bottom and special panels at the right side (Tidwell et al., 2020). This pattern follows the common F-pattern of visual flow where users scan texts or screens following implied lines from left to right, top to bottom (Coleman, 2018).

Chunking. In UX design, Miller's law states that an average person can only keep 7±2 items in their working memory (Yablonski, 2024). By organizing the content or tasks into smaller chunks, users can reduce cognitive load. Chunking is a design strategy applied in the prototype through main task modularity and grouping input into smaller sections (Tidwell et al., 2020). For module tabs a specific "New Patient" module was created so users can focus on the main task of creating new patient assessments. The module tab and voice transcription panels were also collapsible to allow users to hide non-critical features and simplify the UI. An additional "Nothing follows" check box was also added to work as an accordion so users can customize the form with only relevant sections to the patient case. The "Nothing follows" was used to align with standard terminology used by clinicians. The checklist-based assessment format was also emphasized during feedback to provide user customizability and simply forms to relevant assessment fields only. Customization is also utilized in the objective evaluations specifically the range of motion (ROM) and manual muscle strength tests (MMT) to allow users to select specific joint segments relevant to evaluation. Chunking was also used in the SOAP page where separate input boxes for the Subjective, Objective, Assessment/ Analysis, and Plan components were created following user feedback. Samples of integrating chucking strategies are seen in Figure 3-14, Figure 3-15 and Figure 3-16.

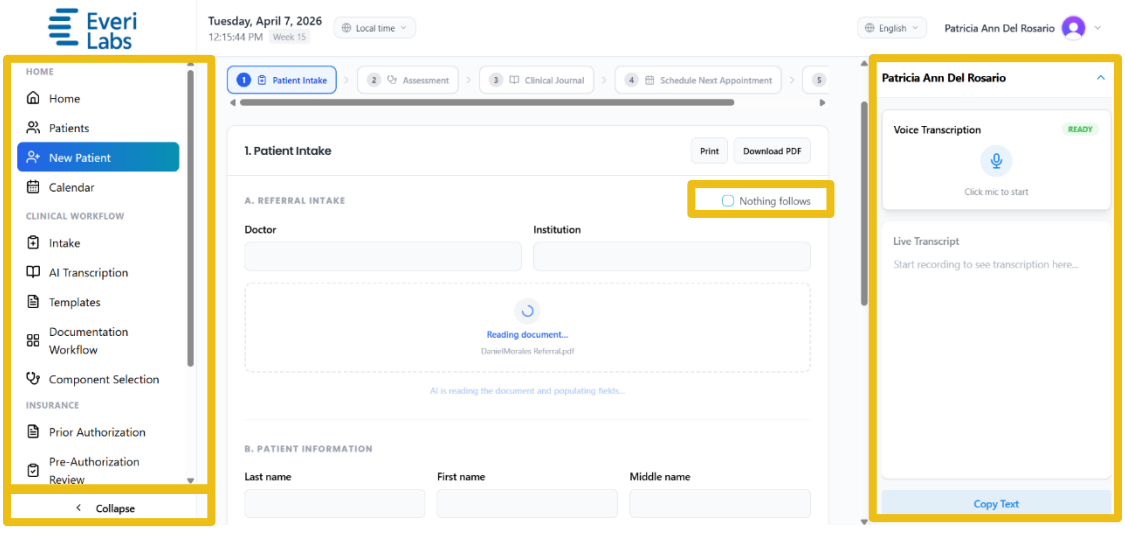


Figure 3-14: Chunking of information using module tabs, collapsible panels and accordions

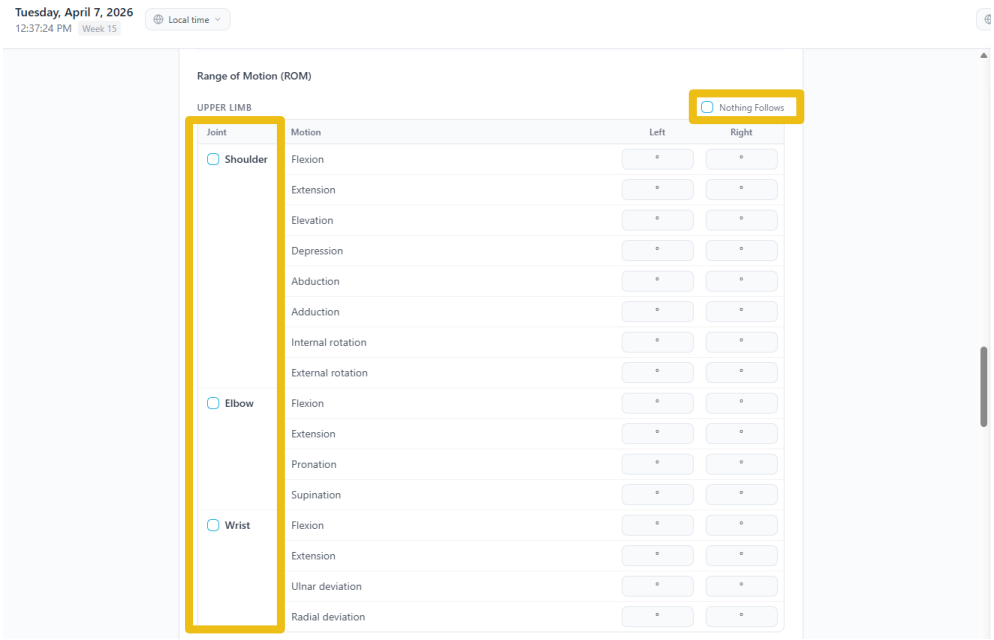


Figure 3-15: Collapsible, checklist-based assessment fields

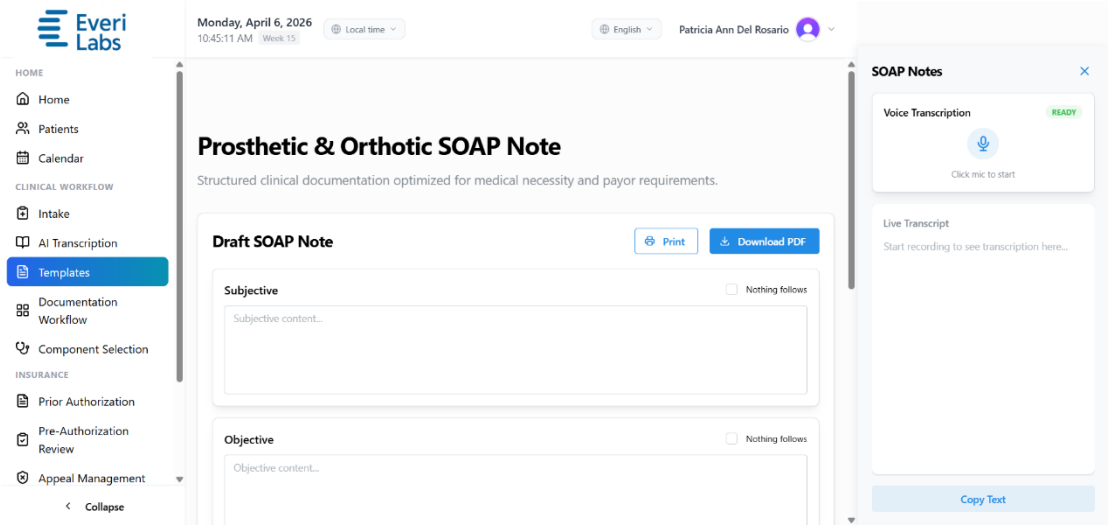


Figure 3-16: High fidelity v2 UI prototype (SOAP)

3.7.7 Actions and Commands

The Van Restorff Effect is a UX design principle used for action and command elements such as pop-ups and notifications, buttons, and drag and drop areas to distinguish and guide specific tasks. This law stipulates that when similar objects are present, the one that differs will be most remembered (Yablonski, 2024). To apply this, Contrasting colors (i.e. yellow and red) and larger input boxes were used to quickly signal to users any warning messages, reminders, notifications, critical buttons, special lists and input fields.

Pop-ups and notifications. Notification boxes (Figure 3-17) were positioned in the upper left corner to utilize the visual flow F-pattern and left-side placement for easier visual identification as it is the least crowded or busy section of the main content area.

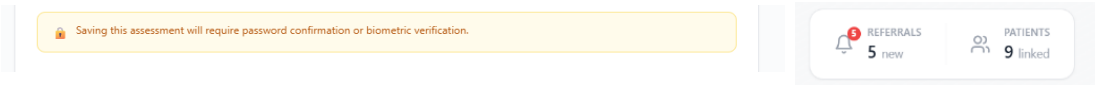


Figure 3-17: Warnings and notification boxes

Buttons. Critical buttons (Figure 3-18) were designed to have a prominent color and then they change color when selected so users remember to use the save or download buttons. Special buttons for data sharing were added as “Email” and “Print” as requested during user feedback sessions.

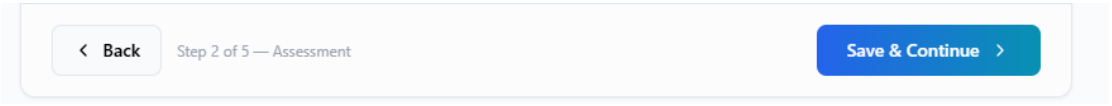


Figure 3-18: Save and continue button

Drop-down menus. Lists of common co-morbidities, amputation levels, residuum length and shape and skin condition were also integrated as per user comments.

Drag and drop. This feature was utilized for the open camera, media upload and referral section of the v2 UI prototype to meet the need for pictures and video attachments. The drop area borders were made bigger to stand out and help improve reachability (Figure 3-19).

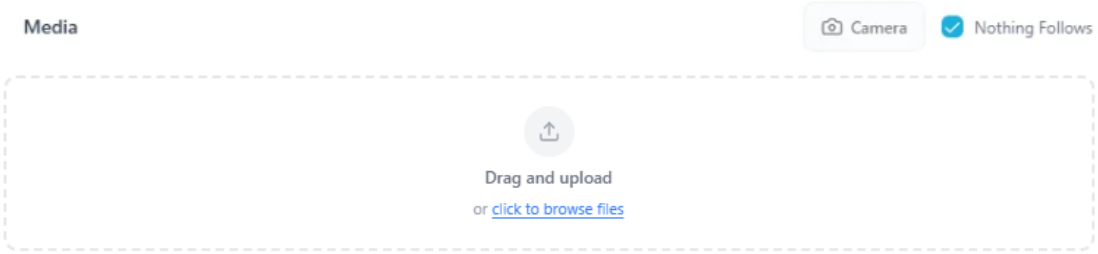


Figure 3-19: Media upload and open camera feature

3.7.8 Input and Controls

Data security features. Clinician signature verification was also included to improve user confidence in data security of the system as shown in Figure 3-20.



Figure 3-20: Signature verification and data saving

Autocompletion. Date and time stamping was also a feature users requested to verify and ensure SOAP notes' timely submission. Active user account details are also automatically filled in to reduce typing and identification errors (Figure 3-21).

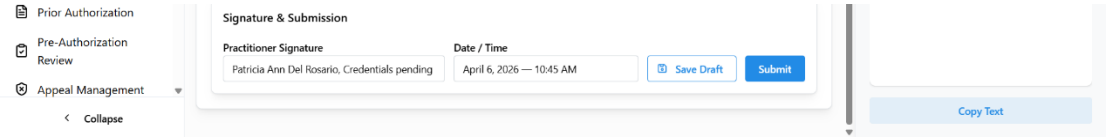


Figure 3-21: Auto filled clinician name with date and time stamp

Additional input fields. Missing input fields were added such as the comments (Figure 3-22) section, history of present illness (HPI), device history, functional goals, vital signs, main complaint, gait assessment findings, postural assessments, missing objective evaluation (ROM and MMT), pain scores, outcome measures, and fields for orthotic or non-amputee and congenital cases following user comments.

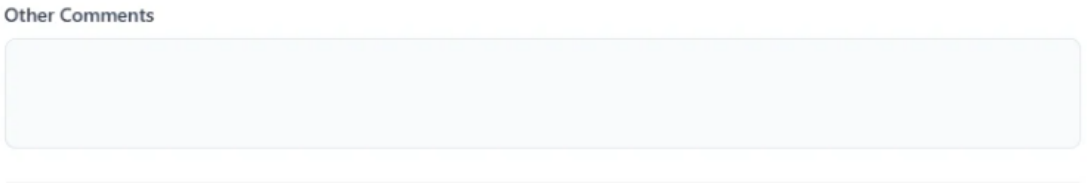


Figure 3-22: Other comments box

3.7.9 AI features and data retrieval

The AI features were reviewed, re-designed and troubleshooted by the Everi Labs team to address noted system errors in the v1 UI prototype tests. All AI related features were pre-tested by developers before actual user testing of the v2 UI prototype. AI transcription was integrated by the Everi Labs developers so that search can be activated with voice prompts. Auto-population of data upon referral upload or voice transcription was integrated by Everi Labs developers. This reduces the manual input and mental processing for clinicians when reviewing voice transcripts. Auto-computation of age and BMI were also included to reduce manual calculations for clinicians during patient assessment.

The combination of these design principles and user-driven changes produced the v2 prototype, which was subsequently evaluated through the task time and SUS, described earlier in this chapter. The results of that evaluation are presented in Chapter 4.

3.8 Ethical Considerations

Ethical review was conducted using the university's self-assessment process. The self-assessment of ethical issues (Appendix 8.1) was submitted and reviewed by the supervisors prior to implementation. All participants (n = 10) received an information letter explaining the study objective and signed informed consent forms (Appendix 8.3). The researcher ensured that all participants understood the scope of the study prior to participation during an initial orientation. Participation was voluntary, and participants were informed that they could withdraw from the study at any given point.

3.8.1 Data Storage and Protection

Personal information was handled in accordance with the General Data Protection Regulation (GDPR) to protect personal data and ensure privacy. Participants were assigned numerical codes to maintain confidentiality. As for the Everi Labs data storage and protection system, all patient data is encrypted with modern security standards, with TLS 1.3 in transit and AES-256 at rest. Data is protected by access control restrictions, ensuring it is only displayed to relevant parties with the necessary justification. All data saved and downloaded for the testing shall be deleted upon exiting the prototype post-tests. Backup copies of the testing results, video, audio recordings, and documents are stored securely in encrypted folders and shall be deleted after completion of the study or upon request of the participant.

3.8.2 Transparency and Funding

Another ethical priority in the study was transparency regarding how data were collected, interpreted, and how the findings were presented. All data were presented without manipulation or enhancement. The intention was not to prove the hypothesis but rather to contribute objectively to the development of the EMR system. Transparency about data, challenges, and uncertainties keeps the research reliable, respectful, and responsible, adhering to the International Electrotechnical Commission best practice guidelines (BSN, 2015; IEC, 2016). Although this study is a collaboration with Everi Labs Inc., no monetary funding was provided or offered by the company to the researcher or to any participants. This prevents performance bias and maintains the credibility of results. The aim of this study remains in providing a true representation of the EMR system’s test outcomes.

3.8.3 Risk management

Throughout the study, risk management was prioritized to protect the physical and mental well-being of participants. Several measures were taken to minimize potential risks associated with the experimental tasks and equipment. Before each testing session, all participants were screened based on the inclusion and exclusion criteria to ensure they were physically able to perform the tasks without increased health risks. Initial orientation meetings were given before implementation. Participants were informed that only simulated patient cases would be used during testing which eliminates any risk toward real patients.

For paper-based testing, all participants were provided with printed forms and encouraged to conduct the test in the ideal test setting in terms of ergonomic workplace setup, internet quality and adequate lighting. For UI prototype tests, individual 30-minute training sessions were given for both versions of the prototype. Participants used their personal laptops during the tests and were encouraged to perform assessment tasks at self-selected speeds. All user feedback was openly accepted, and each participant was given time to express their views during focus group

discussions and testing sessions. Although English was the primary language, participants were allowed to use Tagalog to express complex ideas and experiences. All participants were informed that their competencies were not being evaluated and that test results would not affect their professional standing. The researcher remained present during all sessions to monitor participants, to assist if needed, and to intervene immediately in case of discomfort, unexpected incidents, or errors during testing. The Everi Labs developer team was also on standby during testing sessions for any backend support needed. No adverse events occurred during the study. By implementing these risk mitigation strategies, the study aimed to create a safe and supportive environment that enabled participants to perform tasks confidently while preserving the reliability of the collected data.

4 Results

This chapter presents the outcomes of the usability tests performed. The results first describe participant characteristics, followed by quantitative findings for task completion time and SUS scores across all three documentation conditions (paper-based, v1, v2). Each set of quantitative findings is linked to its corresponding research question (RQ1, 2, and 4). Finally, the results of the thematic analysis from focus group discussions summarize user experiences and design suggestions for the v2 prototype, addressing RQ3.

4.1 Participant characteristics

All 10 participants were practicing certified prosthetist-orthotists (CPO) in the Philippines. Six participants were female (60%) and four were male (40%). The mean age was 32 years (SD = 7.65; range 23-49). Years of clinical experience varied: one participant (10%) worked for more than 10 years, five (50%) had 6-10 years, two (20%) had 3-5 years, and two (20%) had 1-2 years. Six participants (60%) practiced in city or urban areas (Metro Manila) and four (40%) in community or rural areas. Work settings included private P&O clinics (n = 4, 40%), hospitals (n = 4, 40%), and universities (n = 2, 20%). See participant summary in Table 4-1.

Table 4-1: Summary of participant characteristics

Characteristics	Category	N	%
Total participants	**~**	**10**	**100%**
Sex	**Female**	**6**	**60%**
	Male	4	40%
Age	20-29 years	5	50%
	30-39 years	3	30%
	40-49 years	2	20%
Mean age	-	**32 years**	-
Years of clinical practice	1-2 years	2	20%
	3-5 years	2	20%
	6-10 years	**5**	**50%**
	>10 years	1	10%
Work location	City or urban	6	60%
	Community or rural	4	40%
Work setting	Private PO clinic	4	40%
	Hospital	4	40%
	University	2	20%

4.2 Quantitative Findings

This section covers all the quantitative results gathered from time studies, SUS questionnaires, and statistical analysis.

4.2.1 Task Time Results

The descriptive task time results are summarized in Table 4-2.

4.2.1.1 v1 versus Paper-based

The v1 prototype showed longer mean task times than paper across all measures. Total mean time was 36.42 minutes (SD = 6.99) for v1, compared to 26.34 minutes (SD = 9.92) for paper, a difference of 10.09 minutes (+38.31%). Patient IE time was 27.61 minutes (SD = 9.67) for v1 vs. 19.93 minutes (SD = 7.96) for paper (+7.68 minutes, +38.56%). SOAP note time was 8.81 minutes (SD = 9.05) for v1 vs. 6.41 minutes (SD = 4.42) for paper (+2.41 minutes, +37.53%). The v1 prototype did not improve task time over the paper-based system.

4.2.1.2 v2 versus Paper-based

The v2 prototype showed comparable total task time to paper: 26.86 minutes (SD = 4.17) vs. 26.34 minutes (SD = 9.92), a difference of only 0.52 minutes (+1.97%). IE time was 21.80 minutes (SD = 2.09) for v2 vs. 19.93 minutes (SD = 7.96) for paper (+1.87 minutes, +9.38%). SOAP note time decreased for v2: 5.04 minutes (SD = 2.31) vs. 6.41 minutes (SD = 4.42) for paper, a reduction of 1.37 minutes (-21.30%).

4.2.1.3 Prototype comparison (v1 versus v2)

The v2 prototype was faster than v1 across all measures. Total time was 26.86 minutes (SD = 4.17) for v2 vs. 36.42 minutes (SD = 6.99) for v1, a reduction of 9.57 minutes (-26.27%). IE time decreased from 27.61 minutes (SD = 9.67) to 21.80 minutes (SD = 2.09) (-5.81 minutes, -21.06%). SOAP time decreased from 8.81 minutes (SD = 9.05) to 5.04 minutes (SD = 2.31) (-3.77 minutes, -42.78%).

Table 4-2: Descriptive Task Time data

Task	System	N	Mean	SD	Min	Max
IE	Paper	10	19.93	7.96	9.57	29.95
	v1	5	27.61	9.67	15.68	36.93
	v2	5	21.80	2.09	19.02	24.85
SOAP	Paper	10	6.41	4.42	2.07	15.28
	v1	5	8.81	9.05	2.65	24.52
	v2	5	5.04	2.31	2.07	7.75
Total	Paper	10	26.34	9.92	13.15	45.23
	v1	5	36.42	6.99	25.03	42.45
	v2	5	26.86	4.17	21.08	31.72

4.2.2 SUS Scores

For paper-based tests (n = 10), five participants (50%) scored below 70 (usability issues), one (10%) scored 70 (average), two (20%) scored 71-80 (good usability), and

two (20%) scored above 80 (excellent usability). For v1 (n = 5), one participant (20%) scored below 60 (poor usability), three (60%) scored below 70 (usability issues), and one (20%) scored 71-80 (good usability). For v2 (n = 5), one participant (20%) scored below 70 (usability issues), three (60%) scored 71-80 (good usability), and one (20%) scored above 80 (excellent usability). See full list of SUS adjective findings (Appendix 8.18) and summary of SUS descriptive data in Table 4-3.

4.2.2.1 v1 versus Paper-based

The v1 prototype mean SUS score was 60.00 (SD = 10.90), 8 points lower than paper at 68.00 (SD = 11.04) (-11.76%). Both scores fell below the acceptable threshold of 70, indicating usability issues, with v1 scoring lower than paper.

4.2.2.2 v2 versus Paper-based

The v2 prototype mean SUS score was 76.00 (SD = 7.20), 8 points higher than paper (+11.76%) and above the acceptable threshold of 70, indicating good usability.

4.2.2.3 Prototype comparison (v1 versus v2)

Comparing the prototypes, The v2 prototype mean SUS score was 16 points higher than v1 (+26.67%), moving from the "usability issues" range (60.00) to the "good usability" range (76.00), suggesting that iterative design improvements after v1 testing meaningfully improved perceived usability.

Table 4-3: Descriptives of SUS scores with adjective interpretations

System	N	Mean	SUS Score Interpretation	SD	Min	Max
Paper	10	68.00	Indicates usability issues	11.04	55.00	85.00
v1	5	60.00	Indicates usability issues	10.90	42.50	72.50
v2	5	76.00	Good usability	7.20	65.00	82.50

4.2.3 Test of Normality

All task time variables showed normal distribution on the Shapiro-Wilk test ($p > 0.05$), except SOAP time for v1 ($p = 0.033$). SUS scores showed normal distribution for all three systems ($p > 0.05$). Q-Q plots confirmed that data points followed the diagonal (Appendix 8.16 and 8.20)

4.2.4 Statistical Results

4.2.4.1 Task Completion Time

Welch's ANOVA showed no significant effect of documentation system on total task time, $F(2,9.888) = 3.589$, $p = 0.067$; patient IE time, $F(2,8.240) = 1.091$, $p = 0.380$; and SOAP note-taking time, $F(2,8.473) = 0.579$, $p = 0.581$. Although no statistically significant differences were observed on the task time variables, The effect size was large ($\eta p^2 = 0.243$), indicating that 24.3% of the variance in total completion time was explained by documentation system used: IE time ($\eta p^2 = 0.171$, 17.1%= large); and SOAP time ($\eta p^2 = 0.066$, 6.6%= medium).

Games–Howell tests indicated that for total task time for the v1 group was not significantly faster than the paper-based system ($p = 0.101$), likewise IE time ($p = 0.334$), and SOAP note-taking time ($p = 0.846$), showed no significant difference. Similar post hoc findings were observed between the v2 group compared to the paper-

based results: total task time (p = 0.989); IE time (p = 0.770); and SOAP note-taking time (p = 0.718) showing no significant difference the paper system and the new prototype. Between v1 and v2 task times, the task time results also did not differ significantly: total task time (p = 0.081); IE time (p = 0.455); and SOAP note-taking time (p = 0.665). Welch Anova and post hoc tests for task time are summarized in Table 4-4 and Table 4-5.

Table 4-4: Welch Anova for Task Time

Task	F	df1	df2	p-value
IE	1.091	2	8.240	0.380
SOAP	0.579	2	8.473	0.581
Total	3.589	2	9.888	0.067

Table 4-5: Games-Howell pairwise comparisons for Task Time

Task	(I) Group	(J) Group	Mean Difference (I-J)	Std. Error	p-value
IE	Paper	v1	-7.68400	5.00443	0.334
		v2	-1.87200	2.68519	0.770
	v1	v2	5.81200	4.42418	0.455
SOAP	Paper	v1	-2.40300	4.28321	0.846
		v2	1.36500	1.73895	0.718
	v1	v2	3.76800	4.17819	0.665
Total	Paper	v1	-10.08900	4.42734	0.101
		v2	-0.51900	3.64861	0.989
	v1	v2	9.57000	3.63718	0.081

4.2.4.2 SUS Score

Welch's ANOVA showed no significant effect of documentation system on SUS score, $F(2,9.302) = 3.736$, $p = 0.064$. The effect size was large ($\eta p^2 = 0.264$), indicating that 26.4% of the variance in total SUS score was explained by documentation system used.

Games–Howell tests indicated that for total SUS score for the v1 group was not significantly different than the paper-based system (p = 0.416). The pairwise comparison between the v2 group compared to the paper-based scores showed no significant increase in total SUS score (p = 0.252). For the UI prototype comparison of v1 and v2 groups, the SUS results also did not differ significantly (p = 0.067). Welch Anova and post hoc tests for SUS scores are summarized below (Table 4-6 and Table 4-7).

Table 4-6: Welch Anova for SUS scores

Variable	F	df1	df2	p-value
SUS score	3.736	2	9.302	0.064

Table 4-7: Games-Howell pairwise comparisons for SUS score

Variable	(I) Group	(J) Group	Mean Difference (I-J)	Std. Error	p-value
SUS score	Paper	v1	8.00000	5.99537	0.416
		2	-8.00000	4.75073	0.252
	v1	v2	-16.00000	5.84166	0.067

4.3 Qualitative Findings

Qualitative data was collected using focus group interviews after Round 1 usability testing (v1 prototype) to gather user insights for v2 design changes. Three major themes were identified: (1) current challenges with the paper-based system, (2) issues experienced with UI prototype v1, and (3) desired changes for UI prototype v2 with identified sub-themes listed in Table 4-8. A detailed thematic analysis code sheet can be seen in Appendix 0. For the results section, Redundant or filler words (i.e., um, uh, so, ehm, etc.) were removed, and some quotations were edited for grammar and conciseness with caution to preserve original meaning.

4.3.1 Theme 1: Current challenges with the paper-based system

The participants first described the current patient assessment process in the Philippines: 1) referral intake 2) initial assessment 3) SOAP note taking 4) appointment scheduling and 5) billing and payment using pen and paper for documentation. Although participants hope for improvements on efficiency, traceability and sustainability with digital EMR systems, they still emphasized that adequate training and testing of such systems are needed before full and effective clinical integration.

Documentation overload. Participants expressed that manual documentation takes time away from direct patient care, with SOAP note deadline policies adding further pressure. Some described preparing multiple forms as contributing to the overload. Documentation workarounds were common, leaving some assessment forms partially completed.

> "I'd also want to put in the thought about with the paper that we have different types of forms that we had to prepare." (CPO7)
>
> "So, there are times also that we don't get to do our SOAP notes ‘cuz when there are a lot of patients coming in on the same day, we don't get enough time to do the SOAP notes and the SOAP notes that we're used to are actually really long." (CPO4)
>
> "For example, in PSPO so we would actually do the SOAP notes on the same day, right? It should be within 24 hours." (CPO10)

Redundant record keeping. Participants described inefficiencies from duplicate patient records and redundant collection of patient information.

> "So, the main challenges that we've encountered while doing the paper-based system would be the redundancy of the general details of the patient." (CPO9)
>
> "There are cases where we have double folders of the same patient." (CPO6)

System fragmentation. Decentralized patient record keeping poses issues for interdisciplinary coordination and treatment efficiency like previous studies.

> “I think one of our challenges is that our clinic actually doesn’t have a proper filing for the patient's files so there's no like centralized folders or filings for our patients.” (CPO6)

> "If you have a lot of patients, it would be hard if it's not organized."(CPO2)

Data security and storage. A key concern was the vulnerability of paper records to natural disasters common in the Philippines such as floods and typhoons. Physical folders also required high storage space, which was problematic in small clinics. Participants envisioned digital storage as a solution.

> “During Karina typhoon, it (clinic) was unexpected to be flooded so the work for how many years’ worth of documents and charts were lost. It was flooded, everything.” (CPO9)

> "They sometimes also lose the assessment forms that they use even the folders get lost in the clinic, so I think that's the primary issue we have with forms with paper and pen in PSPO. " (CPO8)

> "I think it was just recently when I realized that our storage capacity is already full. So, it's difficult to store more folders" (CPO1)

> "Another problem is the storage of the folders because here at school, we're required to keep the physical copies of the folders for at least 5 years so that accumulates. So, it's mainly storage.” (CPO8)

Poor patient progress tracking. Participants noted that paper-based systems make it difficult to track patient progress because they lack visual baselines or references for gait and residuum condition.

> “It's a little bit hard to backtrack some of the important details so you have to like go through your notes to find other data.” (CPO2)

> “The lack of media in the paper-based system is really relevant in checking the patients’ improvements. It will serve as a baseline to check if the patient's gait improves or the ROM improve per day.” (CPO2)

Interdisciplinary misalignment. Participants reported inconsistent instructions across paper-based referrals and follow-ups, leading to confusion for patients and the healthcare team.

> "So sometimes, it is different in perspective of the rehab doctor. You will only know during follow-ups if you would ask the patient. Then sometimes you're going to realize that there's two different instructions made by the PO and the doctor as well. There's a different follow-up in doctor's point of view. So sometimes it's really not standardized for all rehab doctors. It’s really a challenge for the CPOs and patients as well." (CPO9)

Lack of connection to device inventory for prescription. Paper-based workflows lack visibility of component or material inventory and workshop timelines, creating a gap between assessment and fabrication.

> "So, we've all gone through this assessment objective and then I think the limitation with our paper-based is that we have no idea what's in stock, what's available, what can we prescribe to the patient. We have this but then component-wise so we assessed K3 then it would be really nice to suddenly like have a filter all of the things K3 down to K1 then we don't have to look at K4 or what's available in stock what can we prescribe from based on our assessment." (CPO7)

Poor sustainability. Paper forms generate waste and require labor and time to print and organize, which digital systems could reduce.

> "Okay, it's not that sustainable because you have to print it every now and then." (CPO5)
>
> "Then sustainability and because we used to print what do you call these patient files to insert it in the physical file or physical folder of the patient. So that may take also resources." (CPO5)

4.3.2 Theme 2: Issues experienced with UI prototype v1

This theme captures problems participants encountered during v1 testing.

Improper terminologies. Participants expressed confusion about some headings and input field labels, suggesting the use of universal medical terminology with descriptions or examples.

> "Some terminology really is very new to me that's why I've been asking you during the process what would what do you mean by the plan order is it the components being ordered or what would be my plan for the patient what would I order the patient to do like that." (CPO3)
>
> "For the reason for encounter, it would be I think it's just the choice of words in terms of what we're used to because we use the chief complaint or the patient's goal etc." (CPO6)

Data privacy. Concerns were raised about patient information displayed on the v1 dashboard, particularly on shared or in-clinic devices.

> "I don't want to see patient lists popping out on the first page for this page (dashboard). This is because of security issues or data privacy. So, when someone else sees, I'm on a homepage and I don't want anybody else to look at my records or my account and then just popping into my homepage, then someone else can actually try to look at it and then the patient list will just be open. So just more on the security side and then accessibility." (CPO5)

Connectivity. A participant reported being disconnected multiple times during v1 testing, highlighting a persistent issue in the Philippine context.

> "To be honest, I do have some problems. I think I was disconnected three or four times during the actual practice of the web-based app." (CPO6)

Data retrieval. Some participants also had issues with retrieving the data and downloading their IE forms post-test.

> "I think we do have a problem since the information was not saved when we try to download the PDF." (CPO6)
>
> "During the session, upon checking I think we save and download it twice but still the same the information is blank." (CPO6)

AI repopulation errors. A participant explained their issues with the AI data repopulation feature of the v1 UI prototype. The voice transcription repopulation feature did not work consistently. Some fields populated, others required manual copy-pasting.

> "I did the voice transcription. It was transcribed in words, but when I tried to repopulate it, it wouldn’t actually repopulate the other details." (CPO6)
>
> "I think the ones that were repopulated are the only details were demographics and then the rest I think I did have to manually copy and paste it on the text box that was provided." (CPO6)

AI transcription errors. One participant noted limitations in accuracy with accents and ambient noise, requiring manual correction.

> "During my experience, maybe the words that the mic would catch would not be the same word I'm trying to say. For example, if the area is, I'm in a clinical

> setting where many patients are being accommodated and even kids are present what are the chances that the words being received by the microphone are same as the same words I'm trying to say especially in an accent if the mic doesn't catch things like that." (CPO3)

Despite these issues, participants showed willingness to use the digital system and preferred typing or transcribing over handwriting. Additional backend support was provided by Everi Labs to address these issues before v2 testing. These findings clearly show the need for further system and interface refinement and extended training and long-term usability testing in real-life work conditions. Additional backend support was provided by the company to address system issues described in this section prior to testing the v2 UI prototype.

4.3.3 Theme 3: Desired changes for UI prototype v2

Visual prompts. Participants requested color indicators for missed input fields, visual feedback for selected buttons, and date/time stamps. These color or visual indicators are features participants feel will help identify and prevent critical errors.

> "I think it would be better if those boxes that need information will turn red if you miss them." (CPO1)

> "Maybe putting on this one is a progress note so that we can just actually indicate the date and time. I'm not sure if the practice is very specific in Sweden on the time stamp but on my previous practice the manual timestamping of your patient encounter is important." (CPO6)

Data security features. Participants requested password-protected signatures for attestation and clinician editor controls to prevent unauthorized access.

> “Yeah, password guarded signature. If you think you're already, you know, okay with your SOAP, then I don't know if how it can be translated into that easily. You like attach your signature, you just type in your password and then the signature will be there.” (CPO10)

Accessibility features. Difficulty with zooming and icon or text size was raised and participants recommend a size adjustment feature for those with vision concerns. Future addition of text-to-speech feature was also mentioned which can also help clinicians with visual impairments or for when they are away from the screen.

> “I'm just having a like a slight difficulty with zooming in zooming out, you know, making it bigger.” (CPO10)

> "I think if it can be read over by the AI so that if I missed something I would like it to be read to me especially if I’m using mobile. If I would access from mobile, I would like to listen to it instead because if I have problems with you know with viewing doing it especially if I'm using a mobile then I would just press read SOAP notes you know it's for accessibility anyway I think that's a good feature." (CPO10)

Multi-display compatibility. Currently, the Everi Labs UI prototype is designed for web-based interface compatibility, yet participants feel that a future proof system adapts with different types of interfaces and devices. Participants noted that mobile tablets are commonly used during clinic sessions and recommended mobile-friendly adaptation.

> "I think it's a good suggestion that it would be also friendly with mobile phones where they could easily zoom in or zoom out." (CPO10)

Display layout. Most participants complimented the minimalist, clean design of v1. Specific requests included customizable dashboards, a clinic calendar, form layouts resembling existing paper forms, and highlighting the AI transcription feature.

"If you could put or make a different interface, you know if you can customize what you can see on your own dashboard like what you what we used to do with other apps." (CPO10)

"It would be nice to have things that you could just hide or just not put in anything because you're not using them anyway." (CPO7)

"Maybe if you could just follow the form that we have at PSPO and that you know that will be more of a guided form translated into this." (CPO10)

“Yes. I think also a calendar would be nice so that you would have if you would like to see who you have seen yesterday or who you would like to schedule for tomorrow so that you would already balance your schedule if it's already full or not." (CPO10)

"I would like to see or improve is that to highlight the AI transcription because it is the I think the main difference of this website in comparison with other type of data documentation." (CPO1)

AI assistance features. Participants suggested auto-computation of BMI and age, and AI-generated prescription suggestions but with clinician final control.

"For the objective type, I think if there is an automatic computation for the BMI.” (CPO6)

"So, if it would already predict components, or you know that would fit to tailor to the case but still we (clinicians) have to be on top of what AI can do." (CPO10)

Multimedia upload and storage. Several participants also emphasized the need to have media storage and upload capability in the UI prototype to improve patient progress monitoring. An open camera feature was ideal for participants to ensure up-to-date images or recordings.

"I would like to recall the condition maybe if there would be a segment or a part where we could upload pictures that are relevant to the case then the images or the video of the gait if it's possible if we can record it as well in the assessment part." (CPO10)

"I was thinking if we can actually integrate an open camera option just to verify. For example, the actual condition during the time of the conversation or the consultation because the picture does not indicate how many weeks or months the image was taken." (CPO6)

Drop-down lists. Specific drop-down lists were suggested for fields with common classifications or findings particularly for list of co-morbidities, amputation levels, residuum length and shape and skin condition.

"I think on this part. It would be helpful if there were a drop-down option to click like I mean the most common diagnosis for the patient like diabetes, cerebral palsy, stroke etc." (CPO6)

"I think for common problems like that maybe you can already include there for the drop-down choices like if it's (skin condition) dry scaly or wound " (CPO10)

Pop-up messages and notifications. Due to issues during the first round of testing, Participants requested saving confirmation, transcription start/end cues, microphone detection messages, and referral notifications.

"I think just a warning if you have already saved it asking you if that would be yes like a pop-up command." (CPO6)

"So, it would also appear on my screen like when I arrive open it and then the dashboard will be showing me this thing so that I will see right away." (CPO10)

> "I think it's a good point where in the voice transcription has an indication if it's ready or already on." (CPO6)

Additional buttons. Special buttons for data sharing were mentioned by a participant such as quick printing for professionals outside the clinic and an express email button to send to patient or authorized professionals for referrals.

> "You can actually send it via email, something like that would be good." (CPO6)

> "For the doctors, I think the print option is good so that when you actually want to integrate a health professional from other clinics at least you have a printed format of a digital file." (CPO6)

Additional input fields. Missing assessment areas were identified: comments section, history of present illness, device history, functional goals, vital signs, main complaint, gait assessment findings, postural assessments, ROM and MMT fields, pain scores, outcome measures, and fields for orthotic, non-amputee, and congenital cases.

> "I just think that there are some missing areas where I can put the data based on the assessments." (CPO3)

> "I think for the subjective, we can put another box where we can add like other comments, something like that." (CPO1)

> "I think maybe we can put the patient stated functional goals at lower part of the subjective evaluation because I think you need to know." (CPO1)

User feedback from themes 2 and 3 were used to inform design sketches for the UI prototype v2 shown in Section 3.7.3. Not all comments were incorporated due to time constraints and technical complexity; however, most UI-related design comments were implemented, including visual prompts, data security features, display layout, multimedia upload, drop-down lists, pop-up messages, buttons, and additional input fields.

Table 4-8: Summary of main themes and sub-themes

Main Themes	Sub-themes
Current challenges with the Paper-based system (T1)	Documentation overload Redundant record keeping System fragmentation Data security and storage Poor patient progress tracking Interdisciplinary misalignment Lack of connection to device inventory for prescription Poor sustainability
Issues with UI prototype v1 (T2)	Improper terminologies Data privacy Connectivity Data retrieval AI repopulation errors AI transcription errors
Desired changes for UI prototype v2 (T3)	Visual prompts Data security features Accessibility features Multi-display compatibility Display layout AI assistance features Multimedia upload and storage Drop down lists Pop-up messages and notifications

Additional buttons
Additional input fields

5 Discussion

This chapter interprets the study findings in relation to the four research questions and four objectives that guided this investigation. The discussion is organized into seven sections: interpretation and comparison with literature, redesign recommendations, unexpected findings, study limitations, implications for the AT field, and future research directions.

5.1 Interpretation of Results

5.1.1 Task Time

Welch's ANOVA showed no statistically significant differences across the three documentation conditions for any task variable: total time ($p = 0.067$), IE time ($p = 0.380$), and SOAP time ($p = 0.581$). Nonetheless, the descriptive patterns and large effect sizes merit interpretation.

RQ1 addressed. The v1 prototype was consistently slower than paper across all measures: total time 36.42 versus 26.34 minutes (+38.31%), IE time 27.61 versus 19.93 minutes (+38.56%), and SOAP time 8.81 versus 6.41 minutes (+37.53%). These results align with a well-documented pattern in the EMR literature: task completion times often increase immediately after system implementation due to unfamiliarity and the learning curve. Olakotan et al. (2025) found that poorly designed EHR interfaces with deep menu hierarchies and excessive navigation demands force clinicians to switch tasks frequently, prolonging documentation time. Moy et al. (2021) reported that clinicians averaged 1.4 task switches per minute, with data viewing and entry accounting for 48.2% of all switches, indicating that interface complexity directly inflates task duration. Asgari et al. (2024) further linked EHR-related cognitive overload to excessive navigation and documentation burden, showing that clinicians spending more time on data entry experience higher cognitive strain. The v1 results extend these findings to prosthetics and orthotics, a domain where digital workflow adoption remains limited and largely unexamined (Ngan et al., 2022). Participants received only 30 minutes of training, and AI features were still under development, contributing to navigational delays and user uncertainty.

RQ2 addressed. The v2 prototype markedly improved over v1, with total time slightly slower than the paper: 26.86 versus 26.34 minutes (+1.97%). SOAP note time decreased below paper by 1.37 minutes (-21.30%), and the smaller standard deviation (v2 SD = 4.17 versus paper SD = 9.92) suggests structured digital forms produced more consistent task performance across clinicians. These improvements demonstrate the value of iterative, user-informed design: after focus group feedback, the v2 prototype incorporated visual prompts, display layout refinements, additional input fields, and drop-down lists, which appear to have driven the time reductions. Olakotan et al. (2025) reported that studies implementing usability-focused redesigns can substantially improve efficiency, citing an intervention in which an EMR redesign reduced documentation time by 45.2% and improved patient outcomes. The v2 result suggests the prototype is converging toward time parity with paper and, with continued refinement, may surpass it.

RQ4 addressed. The v2 prototype was 9.57 minutes (-26.27%) faster than v1 for total task time, with the largest improvement in SOAP note creation (-42.78%). This

improvement is practically meaningful: a reduction of nearly 10 minutes per patient assessment translates to substantial time savings across a full clinic day. These efficiency gains align well with similar studies finding a 45.2% decrease in documentation time after EMR redesign, and a 27% improvement after using AI scribes in EMR systems (Olakotan et al., 2025). The large effect size ($\eta p^2 = 0.243$) indicates that the documentation system explained 24.3% of the total variance in time. This is noteworthy for a 10-participant formative study, and the non-significant p-values may reflect limited statistical power; however, the absence of an effect should still be considered a legitimate cause, as elaborated in the limitations section.

5.1.2 Perceived Usability (SUS Scores)

SUS scores followed a clear ordinal pattern: v2 (76.00, good usability) greater than paper (68.00, usability issues) greater than v1 (60.00, usability issues). Welch's ANOVA approached significance ($p = 0.064$) with a large effect size ($\eta p^2 = 0.264$), indicating that 26.4% of the SUS variance was attributable to the documentation system

RQ1 addressed. The v1 SUS of 60.00 fell below the paper (68.00). This score is comparable to the mean SUS of 60.8 reported for in-patient EHR clinical notes documentation, rated as having usability concerns (Olakotan et al., 2025). It exceeds the U.S. physician median SUS of 45.9, which places EHRs in the bottom 9% of all software systems (Holmgren et al., 2024; Olakotan et al., 2025). The 60.00 SUS score of the v1 UI prototype is similar to the 60.80/100 SUS score reported by residents for inpatient EMR note-taking (Olakotan et al., 2025). That even v1 scored above this median is notable, though the paper-based system also scored 68.00, suggesting the documentation task itself contributes to perceived burden regardless of medium. V1's lower score confirms that the first prototype was perceived as inferior to paper, consistent with the task time results.

RQ2 addressed. The v2 SUS of 76.00 (good usability) was 8 points above paper (68.00) and crossed the acceptable usability threshold of 70. This indicates that after one iteration cycle, clinicians rated the prototype more favorably than the paper-based system they use daily. The v2 score places the prototype within the good usability range, whereas the paper remained in the usability issues range. This improvement is consistent with the task time results, where v2 approached and, in some measures, surpassed the paper. The result suggests that a purpose-built, user-centered design can achieve a level of perceived usability comparable to that of the paper system.

RQ4 addressed. The v1-to-v2 improvement of 16 points (+26.67%) was close to statistical significance ($p = 0.067$) and moved the prototype from below 70 to a score indicating good usability. This improvement can indicate the positive impacts of user-centered design changes (visual prompts, drop-down lists, multimedia upload, progress indicators, layout refinements) informed by focus group feedback, confirming that user involvement in the development cycle directly improves perceived usability (Busse et al., 2023; Ghorayeb et al., 2023) and highlights the value of iterative user-centered design.

5.1.3 Qualitative Findings and Design Implications

Three themes emerged from focus group discussions; each echoing patterns observed in prior EMR literature while revealing context-specific concerns.

Theme 1: Paper challenges. Participants described documentation overload, redundant recordkeeping, system fragmentation, poor patient progress tracking, data security and storage concerns (including typhoon flooding that destroyed years of

charts), interdisciplinary misalignment, lack of connection to device inventory, and sustainability issues. These findings mirror problems documented in Philippine primary care EMR research, where system fragmentation and poor workflow integration were identified as barriers to adoption (Elepaño et al., 2025). The documentation overload and workaround behaviors align with research showing that data viewing and data entry tasks account for 48.2% of all task switches in clinical workflows, with clinicians averaging 1.4 switches per minute (Moy et al., 2021). The sustainability concerns, particularly disaster-related data loss, represent a context-specific finding that is less emphasized in high-resource-country literature but is critically important for LMIC settings. As Ngan et al. (2022) observed in their study of digital workflow adoption in orthotics and prosthetics, practitioners value efficiency gains from digital tools but face structural barriers, including a lack of infrastructure and institutional support.

Theme 2: v1 issues. Participants identified improper terminologies, data privacy concerns (patient lists visible on dashboards), internet connectivity problems, data retrieval failures, and AI repopulation/transcription errors. The AI transcription errors align with Ng et al. (2025), who documented word error rates exceeding 50% in conversational clinical settings, and extend this finding to P&O-specific terminology and Filipino-accented English. The connectivity concerns are consistent with Elepaño et al. (2025), where unstable internet was a core barrier to Philippine EMR acceptability, and with broader challenges identified in technology adoption across developing economies (Britiller et al., 2023; Derecho et al., 2024).

RQ3 addressed; Theme 3: Desired v2 features. Participants specified visual prompts, data security features (password-protected signatures, role-based access), accessibility options (text size adjustment, text-to-speech), multi-display compatibility, customizable layouts, AI assistance (auto-computation, prescription filtering), multimedia upload, drop-down lists, pop-up confirmations, and additional input fields for orthotic and non-amputee cases. The importance of data security and signature verification is reflected in De Mesa et al. (2024), who found that self-efficacy (personal competence and confidence) was the strongest predictor of EHR intent-to-use in the Philippines, with each unit increase associated with a 31% rise in intent to use. The desire for form customizability and familiar PSPO-like layouts highlights the importance of user participation in interface development, consistent with findings that involving healthcare professionals throughout the design cycle produces more aligned and acceptable systems (Busse et al., 2023). These findings directly informed the v2 design sketches, completing the Double Diamond iteration cycle from discovery through delivery.

The convergence of quantitative trends (large effect sizes, non-significant p-values) and rich qualitative feedback supports the mixed-methods framework advocated by Darejeh et al. (2026), who argue that no single cognitive load or usability measure is sufficient and that combining subjective and objective methods increases reliability. The present study's combination of task time, SUS, and thematic analysis follows this recommended multi-method approach.

5.2 Unexpected Findings

Several findings were not anticipated. First, the combination of non-significant p-values and large effect sizes (ηp^2 greater than 0.14) across all variables was notable for a 10-participant study, indicating the system intervention's effect was larger than typical for formative evaluations.

Second, despite experiencing AI errors and data retrieval failures in v1, participants remained enthusiastic about AI features. Paper's severe limitations (flood vulnerability,

storage demands, documentation overload) may have motivated such strong interest that participants tolerated prototype imperfections more readily.

Third, the thematic alignment between Filipino P&O clinicians' concerns and the broader EMR usability literature (Asgari et al., 2024; Olakotan et al., 2025) suggests core usability principles may transfer across settings more readily than assumed, even when infrastructure differs substantially.

5.3 Recommendations for Redesign

5.3.1 Adoption Planning and User Training

Clinics adopting digital documentation should plan for an initial productivity dip of up to 38% (v1 pattern), similar to other studies identifying a 30-40% dip in efficiency after initial EMR integration (Moy et al., 2021; Olakotan et al., 2025). The 30-minute training used in this study proved insufficient. Practical recommendations include a phased, module-by-module rollout, workflow-based training (not just interface navigation), the designation of superuser clinicians for peer support, and layered training materials for varying levels of digital literacy. This aligns with Elepaño et al. (2025), who found that training, iterative feedback, and contextual fit were essential for successful Philippine EMR deployment, and with Derecho et al. (2024), who emphasized that infrastructure and user support are critical determinants of EMR adoption in developing economies. Standardized full-stack onboarding packages and interactive product walkthroughs are recommended, especially in remote training settings.

5.3.2 UI Design Improvements

Visual prompts and feedback. Addition of field-level indicators for required data, color-coded prompts (red for missing, green for complete), and date-time stamps address user requests from Theme 3 and reduce error risk yet indicators for user selection on click sensitive features should be considered as these issues were prevalent during AI transcription initiation delays causing participants to click buttons excessively which resulted in system connection errors.

Structured input controls. Although collapsible assessment sections and "Nothing follows" checkboxes allow clinicians to customize forms to each case, Toggle buttons for status fields can be implemented instead to improve selection speed and reachability by increasing target size over checkboxes and eliminate UI clutter from repeated words (Tidwell et al., 2020).

System feedback and notifications. Pop-up confirmations for saves, transcription status indicators (listening, processing, difficulty), and automatic data saving address the v1 data retrieval failures in Theme 2, however interactive coach marks can be added on the UI system to provide an interactive feedback chain specially for first-time users. This can be useful inclusion to reduce confusion and cognitive load in navigating the interface.

Language. Although participants are fluent in English, language options are an important aspect that reduces cognitive load of translating input. Multi-lingual support should be developed and tested, including English and Tagalog. Also, patients use their native languages more frequently than English for voice input in the Philippine context. The system is recommended to include Tagalog and other languages for universality.

5.3.3 Accessibility and Inclusivity

Text size adjustment, text-to-speech output, touchscreen compatibility and responsive layouts for mobile and tablet use should be adapted in future iterations. Participants noted that mobile tablets are more commonly used during clinic sessions due to portability, reflecting accessibility considerations that are especially relevant in low-resource settings where device availability varies. Color selection should also be considered for color-sensitive users with conditions that make common colors illegible through inverse or dark mode design strategies. Text-to-speech output also supports accessibility for users with hand impairments or typing difficulties. The text-to-speech feature requested by participants should be evaluated for usability and accuracy. Systematic accessibility compliance testing against international standards should be conducted, including screen reader compatibility, keyboard navigation, color contrast, and text resize functionality. Voice transcription should also be evaluated for accuracy across these languages and for Filipino-accented English, as the current prototype's transcription limitations represent a significant barrier to adoption.

5.3.4 Connectivity Infrastructure and Offline Compatibility

Internet connectivity was a persistent issue. Due to poor internet connections and the dependence of both prototypes on the internet, some users experienced system lags and crashes during testing, which affected some of the recorded task time results by adding to the total task time. Additionally, workarounds were done to save input data by taking screenshots of completed forms supplemented by video and audio recordings of test sessions. To mitigate these issues, an offline-capable mode, auto-save feature, or progressive web application was proposed to significantly improve viability for Philippine clinics. In disaster-prone regions, digital documentation should be framed as data continuity risk mitigation: participants reported losing years of clinical charts to typhoon flooding. These infrastructure barriers mirror challenges documented in Philippine EMR implementations, where unreliable connectivity and inconsistent power supply were critical obstacles (Britiller et al., 2023; Elepaño et al., 2025).

5.3.5 Mental Model and Standard Terminology

Forms should preserve familiar PSPO layout structures, assessment sequences, and terminology to reduce cognitive transition load, consistent with mental model principles in interaction design as established in this study's theoretical framework. This preserves and assists in ensuring that the UI system aligns with real clinical evaluations yet should also be balanced to reduce redundancies and simplify forms. The Tesler’s Law of UX should be balanced by the application of progressive disclosure technique where the UI display shows only important content by default but maintaining additional content accessible (Yablonski, 2024).

5.3.6 User Diversity and Safety

Design must accommodate clinicians with varying experience, work settings, and input preferences through customizable interfaces and multiple input modalities. It is recommended that diversity of sample groups should be expanded to different local and international contexts and physical conditions to ensure the system is accessible to a wider user population. Although having a P&O tailored EMR system is impactful to clinician work, It is also important to consider including other healthcare professionals in the system workflow design to improve interdisciplinary alignment and coordination which is important for patient continuity of care.

5.4 Study Limitations

5.4.1 Sample size and Statistical Power

The sample of 10 participants limits statistical power, particularly given the between-groups comparison (n = 5 per prototype version). Although randomization was used, the small group sizes greatly limit the power of the results. The non-significant ANOVA results ($p = 0.067$ for total time, $p = 0.064$ for SUS) should not be interpreted as evidence of no effect; they reflect insufficient statistical power within this sample to detect the observed effect sizes. The large effect sizes ($\eta p^2 > 0.14$ across both task time and SUS) suggest meaningful differences exist, but a larger sample is required for confirmatory inference. A post hoc power analysis indicates that studies with comparable effect sizes require 20-30 participants per group to achieve 80% power at alpha = 0.05. The risk of a Type II error is therefore higher, and all non-significant findings should be interpreted conservatively. Although the small sample size likely has an influence, it is equally important to interpret the results as evidence of no difference for the non-significant results.

5.4.2 Participant Homogeneity

All participants were ISPO-certified P&O professionals practicing in the Philippines with no prior Everi Labs exposure. This was appropriate for a formative evaluation targeting this specific user population, but it constrains generalizability to other clinician groups, practice settings, and geographical contexts. Filipino P&O clinicians work within a specific healthcare system infrastructure and regulatory environment that may not transfer directly to other LMICs or high-resource settings. Subgroup analyses by case type, gender, years of experience, or work setting were not feasible given the sample size, limiting the ability to identify differential usability patterns across clinician characteristics.

5.4.3 Prototype Constraints

The evaluation used a web-based, English-only prototype running on participants' personal laptops. The AI features, particularly voice transcription and auto-population, were still under development and produced errors during testing that affected user experience. Conclusions about a clinic-ready system are therefore preliminary. The prototype focused on the frontend interface only; backend architecture, data interoperability, and integration with existing clinic management systems were outside the scope. Additionally, participants used their own hardware, introducing variability in screen size, resolution, processing speed, and network connectivity that may have influenced task performance independently of the interface design.

5.4.4 Remote Test Environment

Remote testing introduced several uncontrolled variables that affect internal validity. Internet reliability varied across sessions, with participants reporting 3-4 disconnections during prototype use. Ambient noise affected AI transcription accuracy in unpredictable ways. Hardware variability (laptop specifications, screen resolution, input devices) could not be standardized. Task timing was conducted manually using stopwatches, introducing measurement imprecision of an estimated 1-3 seconds per task segment, which may affect the precision of task time comparisons. The absence of in-person observation meant the researcher could not control for environmental distractions or verify participant adherence to the testing protocol in real time. Also, motion analysis was not feasible to supplement task time studies due to the remote test setup.

5.4.5 Short Term Evaluation

Each participant completed a single session per condition, with only 30 minutes of training on the prototype before testing. This design cannot capture learning curve effects. Task time and SUS scores may improve substantially as clinicians gain proficiency, as suggested by the wider variance in v1 scores compared to v2. The single-session format also precludes assessment of system adoption over time, retention of workflow efficiency, or changes in user satisfaction after sustained use. Longitudinal evaluation across multiple sessions would provide a more accurate picture of the prototype's real-world performance trajectory (Coleman, 2018).

5.4.6 Task Complexity of Simulated Patient Cases

The study covered only initial patient evaluation and SOAP note creation, which are core but not exhaustive components of the P&O clinical workflow. Results may not generalize to more complex or time-sensitive tasks such as casting, measurement, fitting adjustments, or interdisciplinary team communication. The simulated case scenarios, while standardized, cannot replicate the cognitive demands of managing actual patients, including unexpected clinical findings, emotional interactions, or interruptions that characterize real clinic environments. It removed the variability, time pressure, and interpersonal demands of actual patient interactions. Participants' task performance and usability ratings in simulated conditions may differ from real clinical practice, where clinicians must manage patient communication, unexpected findings, and concurrent responsibilities. The use of only two case scenarios also limits the range of clinical presentations and documentation demands assessed.

5.4.7 Task Time Testing

Task time was recorded manually using a digital stopwatch rather than a validated time-motion analysis (TMA) system. Established TMA approaches, such as those used by Moy et al. (2021) to quantify task-switching frequencies and documentation time distributions, rely on structured observation protocols and specialized software to detect micro-level task transitions with greater precision. The absence of such a system in this study means that task switches, brief delays, and sub-task durations within each assessment session were not captured. This limits the granularity of the efficiency analysis: while overall task time differences were measurable, the specific activities driving those differences (i.e., navigation time, data entry time, and AI interaction delays) could not be isolated. Future evaluations should adopt validated TMA tools or automated log analysis to provide finer insight into where documentation time is spent, and which interface elements contribute most to delays.

5.4.8 SUS score Analysis

The System Usability Scale (SUS) was the only standardized usability questionnaire administered. While the SUS is widely used and validated across domains, it was designed as a general tool and may not capture usability dimensions specific to clinical documentation systems. Ghorayeb et al. (2023) noted that the standard SUS does not fully account for the clinical context of use, including workflow integration, patient safety implications, and decision support effectiveness, which are critical for healthcare systems. The SUS provides a global subjective score but does not identify specific interface problems or distinguish between efficiency, satisfaction, and learnability. Although the SUS items can be grouped by domain (i.e., efficiency with items Q1, Q5, Q6; efficacy with Q4, Q7, Q10; and satisfaction with Q2, Q6, Q9) to provide more granular insight, this subscale analysis was not applied in the present study. The use

of an adapted version, such as the Healthcare System Usability Scale (HSUS) questionnaire, should be explored in future studies.

5.4.9 Thematic Analysis Audit Trail

Manual coding of qualitative data in MS Excel, without qualitative data analysis software (QDAS), limits the formal reliability of the thematic analysis. Respondent validation was conducted to confirm theme accuracy, but the absence of independent coding or inter-rater reliability checks reduces the trustworthiness of the qualitative findings. Potential coder bias cannot be ruled out, and the themes identified may reflect the researcher's interpretive lens rather than fully objective patterns in the data.

5.4.10 Potential Order Effects and User Bias

The crossover design, while efficient for small samples, introduces the potential for order effects. Participants who completed the paper-based condition first may have become familiar with the assessment scenarios, potentially reducing task time in subsequent prototype conditions. Conversely, prototype-first participants may experience fatigue by the time they reach the paper tests; thus, testing was conducted on different days for each condition. Additionally, participants' awareness of testing a novel AI-assisted system may have introduced a Hawthorne effect or positive response bias, particularly in qualitative feedback, where enthusiasm for digital innovation may have inflated favorable responses (Yablonski, 2024).

5.5 Contributions to AT Product Development

At the clinician level, this study demonstrates that P&O practitioners in the Philippines are willing to adopt digital documentation tools designed for their specific workflows. At the methodological level, the study provides a replicable template that combines the Double Diamond framework with HCI design, links evaluation methods to HFE dimensions, and remains practical for small-scale formative evaluations in low-resource settings. This is significant given the limited evidence on structured usability evaluations in specialized clinical domains like prosthetics and orthotics (Hovorka, 2023; Ngan et al., 2022).

At the population level, this work aligns with the WHO and UNICEF (2022) call for people-centered, integrated AT service delivery. Even modest gains in documentation efficiency have a cumulative impact on service capacity in contexts where the P&O workforce is severely limited relative to population needs.

In the lens of product development, the v1-to-v2 iteration cycle demonstrates the value of early-stage, low-cost usability testing. The overall 26.27% task time improvement and 26.67% SUS score improvement show that formative evaluation with small samples yields actionable design direction. This supports the principle that user involvement throughout the development cycle produces more effective and acceptable systems (Busse et al., 2023; Ghorayeb et al., 2023), and that iterative redesign informed by user feedback can produce measurable improvements even within a single development cycle.

5.6 Ethical and Societal Implications

5.6.1 Data privacy and storage

The study adhered to GDPR protocols, including encrypted storage (TLS 1.3 in transit, AES-256 at rest) and numeric participant codes. However, participants' concerns about patient list visibility on shared devices highlight the need for role-based access

and minimized default data display, consistent with privacy considerations in Philippine EMR studies (De Mesa et al., 2024; Elepaño et al., 2025).

5.6.2 AI accuracy in medical record management

AI transcription errors (inaccurate word recognition in Filipino-accented English, failure to capture P&O-specific terminology) underscore the need for robust error checking, clinician oversight, and transparency about limitations in AI-assisted documentation. Ng et al. (2025) found that current AI speech recognition systems struggle with accented speech and specialty terminology, requiring substantial human review. Wibowo et al. (2025) identified ethical frameworks and localized development as prerequisites for responsible AI adoption in Southeast Asian health systems.

5.6.3 Patient consent to AI

Although current AI-implementation laws in the Philippines are still being defined, Informed consent procedures must address how patient data is transcribed, stored, and used by AI-assisted EMR systems. This was outside the present scope but warrants attention in clinical deployment.

5.6.4 Clinician decision-making support

Participants expressed cautious optimism about AI assistance for prescription suggestions (i.e., filtering components by K-level) but emphasized that clinicians must retain final control, consistent with findings by Asgari et al. (2024) on the importance of keeping clinicians in the decision loop when integrating AI into clinical workflows.

5.7 Future Research Recommendations

5.7.1 AI Systems Architecture Design

Future studies should systematically evaluate AI transcription accuracy under real clinic conditions, measuring word error rates across different P&O terminology domains, clinician accents (Filipino-accented English, Tagalog), and ambient noise levels. Ng et al. (2025) documented word error rates exceeding 50% in conversational clinical settings, and the present study's transcription errors suggest domain-specific terminology exacerbates this. Research should also evaluate the reliability of AI-powered prescription filtering and auto-computation features, assessing whether the accuracy of suggestions meets clinical safety standards. Clinician trust calibration over repeated use is another critical dimension: studies should examine how clinicians' reliance on AI suggestions evolves and whether appropriate skepticism is maintained. The prototype's architecture should also be evaluated for scalability, data synchronization across devices, and interoperability with existing clinic management and billing systems, particularly in low-resource settings where infrastructure constraints are most acute.

5.7.2 Other Parts of the PO Process

The present evaluation was limited to initial patient assessment. Future research should extend the prototype to cover the full P&O clinical workflow: casting and measurement documentation, component prescription and ordering, fabrication workflow tracking, fitting notes and adjustments, follow-up visits, and outcome measurement. Each workflow stage presents unique documentation demands and usability challenges. The casting and measurement phase, for example, involves spatial and anthropometric data that may require different input modalities than the text-heavy SOAP format. Evaluating the prototype across the full care cycle would

reveal where digital documentation provides the greatest efficiency gains and where interface redesign is still needed.

5.7.3 Advanced Testing Methods

More sophisticated usability testing methods should be employed in future evaluations. Eye tracking and gaze analysis could reveal which interface elements demand excessive visual attention and where clinicians experience search difficulty (Darejeh et al., 2026). Heart rate variability and electroencephalography (EEG) have been used to index cognitive load during EMR tasks (Darejeh et al., 2026)and could provide objective workload data beyond subjective SUS scores. The NASA Task Load Index (NASA-TLX) should be administered alongside the SUS to capture perceived cognitive load across mental demand, physical demand, temporal demand, performance, effort, and frustration dimensions. Think-aloud protocols during task completion would provide real-time insight into user confusion, navigation strategies, and decision-making processes that retrospective qualitative methods may miss. Automated log analysis (clickstream data, time per screen, error rates) should complement manual stopwatch timing to increase measurement precision. In time-and-motion analysis (TMA), using a validated system will mitigate the risk of time variance and tester delays. The use of such technology will help identify task switches that can affect true total task time duration.

5.7.4 Real Patient Cases

The system should be tested in real clinical settings with actual patients to evaluate performance under time pressure, with interdisciplinary team communication, and in the presence of clinical uncertainty. This would assess whether the efficiency gains observed in simulated testing translate to real workflow improvements and whether usability ratings remain stable when clinicians must divide attention between patient interaction and documentation. Real-world testing should also evaluate patient-facing implications: whether the prototype affects the quality of clinician-patient communication, patient wait times, and patients' perceptions of care quality. Safety outcomes, including documentation error rates and data integrity, should be systematically tracked.

5.7.5 Sustainability and Economic Impact

Cost-effectiveness analysis should model the total cost of digital documentation adoption for Philippine P&O clinics, including hardware, software licensing, training, internet connectivity, and technical support, against the value of time savings, reduced data loss, improved documentation quality, and increased patient throughput. Market research on willingness to pay among clinic owners and administrators in LMIC contexts would inform pricing and business model decisions. Environmental sustainability should also be assessed: digital documentation eliminates paper consumption, physical storage, and waste, but introduces electronic waste and energy consumption trade-offs that merit further study.

5.7.6 Longitudinal Implementation Studies

Future research should follow clinics through the full adoption lifecycle, from initial training to proficiency, using repeated measures of task time, SUS, error rates, and user satisfaction at multiple time points (i.e., 1 week, 1 month, 3 months, and 6 months post-implementation). This would capture learning curve trajectories, identify when the initial productivity dip resolves, and assess whether time parity or superiority over paper is sustained or improves. Longitudinal designs would also reveal whether the

16-point SUS improvement from v1 to v2 persists or grows with extended use, and whether the burnout risk reduction implied by the SUS improvement is realized in practice.

5.7.7 Practitioner Diversity

Future studies should include more varied experience levels (expand the >10 years group), and practice settings (rural community-based clinics, mobile outreach programs, hospital-based practices). The present sample was majority urban practitioners (60%) and experienced clinicians (50% with 6-10 years' experience). Purposive sampling should ensure adequate representation of novice and highly experienced clinicians, rural practitioners, and diverse work settings to assess whether usability patterns differ across these subgroups. Gender-balanced analyses should also be prioritized. The system should also include other allied health professionals (physical therapists, medical doctors, occupational therapists, etc.) workflows through integration of EMR module expansions.

5.7.8 Comparative Studies with Other EMR Systems

Head-to-head comparisons between the Everi Labs AI Copilot and other digital documentation systems (both P&O-specific and general EMR platforms) would establish the relative advantages in efficiency and usability of the purpose-built approach. Such comparisons would clarify whether the observed improvements are attributable to the prototype's specific design features or simply to the transition from paper to digital format. These studies should measure not only task time and SUS but also cognitive load, error rates, and user preference.

5.7.9 Multi-site or global Implementation

Replication studies in other LMICs (Southeast Asia, Sub-Saharan Africa, Latin America) and high-resource settings (Europe, USA) would clarify which findings are universal and which are context-dependent. The present findings suggest that core usability principles transfer across settings, but infrastructure constraints (internet reliability, device availability, power stability) vary dramatically. Multi-site studies would enable systematic comparison of how environmental factors moderate the relationship between interface design and usability outcomes, informing adaptation strategies for different deployment contexts.

6 Conclusion

This study evaluated whether an iteratively designed AI-assisted EMR could improve documentation efficiency and usability for P&O clinicians compared to paper-based methods. Ten (10) ISPO-certified CPOs in the Philippines evaluated three conditions (paper, Everi Labs v1, and v2) using task time, SUS, and qualitative feedback in a crossover mixed-methods design.

The v1 prototype was slower than paper (+38.31% total time, SUS = 60.00). After one user-informed iteration, v2 achieved slightly slower total task time than the paper (+1.97%) and was faster in SOAP note creation (-21.30%), while raising the SUS to 76.00 (good usability). While Welch's ANOVA did not reach the significance threshold ($p > 0.05$), large effect sizes ($\eta p^2 > 0.14$) indicate meaningful differences. Qualitative analysis revealed three themes: burdens of paper (data loss, fragmentation), v1 issues (AI errors, connectivity, terminology), and desired v2 features (visual prompts, structured inputs, security, accessibility). The convergence of quantitative and qualitative evidence strengthens the findings beyond what either method alone can provide.

This study fills a gap in P&O digital workflow research, validates design frameworks (Double Diamond, HCI) under resource constraints, extends findings on EMR performance in time and usability in the P&O context, and provides a replicable mixed-methods template for low-resource evaluations.

Clinics can anticipate an initial productivity dip (up to 38%) that can be managed through a phased rollout and full-stack workflow-based training. Developers should prioritize user-centered redesign features that directly improve efficiency and satisfaction in future iterations, supported by continuous qualitative studies using a mixed-methods approach. Future research requires a larger sample size, greater user diversity, longitudinal designs, real-world clinical testing, and multi-site replication.

Within the development cycle, an iterated user-centered AI-assisted EMR achieved good usability and similar task time to paper for P&O clinicians. Structured formative evaluation with modest resources produced actionable improvements. The study shows that incremental documentation efficiency gains have the potential to result in a cumulative impact in work-force-limited settings, where efficient PO workflows result in better quality care and ultimately lead to improved clinician and patient satisfaction. With continued refinement, the Everi Labs AI Copilot has the potential to reduce documentation burden and improve clinical efficiency by keeping users at the core and evaluating the usability of clinical work tools, supporting the global goal of accessible assistive technology.

Declaration of artificial intelligence use:

During the preparation of this work, the author used Microsoft/ Windows Copilot and Grammarly for proofreading, spelling correction, grammar check, conciseness, and clarity enhancement. AI was also used in the prototype development as a critical feature for transcription, auto-population, and auto-computations on certain input fields in the UI prototype. Embedded AI assistance was also used for search and review of related literature (i.e., Scopus AI). Google Gemini AI was also used for focus group transcription and initial thematic coding. Persona Images were also generated using Microsoft/ Windows CoPilot. After using the AI tools, the author reviewed and edited the content and takes full responsibility for the content.

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8 Appendix

8.1 Self-Assessment of Ethical Issues

JÖNKÖPING UNIVERSITY

Form for Self-Assessment of Ethical Issues in Degree Projects[1]

Date:	2026-02-24
Title of the degree project	Integrating Artificial Intelligence Into Prosthetics and Orthotics Clinical Workflow: Design and Usability Evaluation of the Everi Labs AI Copilot Prototype
Student(s):	Patricia Ann del Rosario
Student's/Students' e-mail address:	depa24bl@student.ju.se
Degree program:	Master in Product Development Specialization in Assistive Technology
Education cycle:	VT2026
Supervisor:	David Samvin, David Rusaw
Supervisor's e-mail address:	david.samvin@ju.se, david.rusaw@ju.se

This form is the basis for ethical reflections linked to your degree project. The intention is that students go through the form and then have a dialogue with their supervisor(s). If necessary or in doubt, the course examiner should be included in the discussion.

Ethical reflection is necessary in all degree projects, and it is especially important for degree projects that involve people in various ways. Issues that need to be considered include information about the study, consent, confidentiality, the type of data to be collected and how the study may affect the people involved.

Degree projects at Jönköping University must comply with the ethical principles expressed in the Swedish Law (2003:460) concerning the ethical review of research involving humans and biological material from humans, which process sensitive personal data or personal data about violations of law involving crimes, judgments in criminal cases, criminal procedural coercion or administrative deprivation of liberty. The law is also applied to research that involves physical intervention on a research subject or is carried out according to a method that aims to affect the research subject physically or psychologically or that involves an obvious risk of harming the research subject physically or psychologically[2]. Research that concerns humans, biological material, and sensitive personal data needs to go through an ethical review.

Degree projects are not normally considered research, and the intention is not for the results to be published in a scientific journal, which means that ethical review might not be necessary.

[1] The form also applies to quality improvement projects in health care and social welfare.
[2] https://www.riksdagen.se/sv/dokument-och-lagar/dokument/svensk-forfattningssamling/lag-2003460-om-etikprovning-av-forskning-som_sfs-2003-460/

8.2 Participant Recruitment Poster



8.3 Participation Letter and Informed Consent



JÖNKÖPING UNIVERSITY

Information Letter to Participants

You are hereby asked to participate in this degree project study conducted as part of a master's thesis at the School of Health and Welfare and the School of Engineering at Jönköping University.

About the study

Title: Integrating Artificial Intelligence Into Prosthetics and Orthotics Clinical Workflow: Design and Usability Evaluation of the Everi Labs AI Copilot Prototype

This study is a degree project that examines how an AI-assisted digital workflow system affects documentation efficiency and usability in prosthetic and orthotic (PO) clinical practice in low-resource settings. The purpose of the project is to compare traditional paper-based workflows with an AI-assisted prototype and to gather user feedback that can guide future design improvements.

The study is important because many PO clinics in low-resource settings still rely on paper-based documentation, which can be time-consuming and prone to errors. Understanding whether digital tools can support clinicians in their daily work may contribute to more efficient and user-friendly solutions in the future.

What participation involves

You are being asked to participate because you are an actively practicing prosthetist-orthotist clinician in the Philippines who meets the study's inclusion criteria.

If you choose to participate, you will be asked to:

- Complete a simulated patient assessment using **paper-based documentation** and the **AI-assisted digital workflow prototype** (Everi Labs Co-Pilot).
- Have your task time recorded through a time-motion study.
- Complete the **System Usability Scale (SUS)** questionnaire after using the prototype.
- Participate in a focus group discussion about some semi-structured interview questions about your experience.

Participation is expected to take approximately **60–90 minutes** in total per session. A total of 5 to 10 sessions is expected but subject to change depending on participants time availability. All sessions are to be conducted online.

- Session 1: Orientation
- Round 1
  - Session 2: Paper based task time studies
  - Session 3: Prototype user training
  - Session 4: Pilot Testing with Prototype and System usability scale questionnaire
  - Session 5: Focus Group Session
- Round 2
  - Session 6: Paper based task time studies
  - Session 7: Prototype user training)
  - Session 8: Pilot Testing with Iterated Prototype and System usability scale questionnaire
  - Session 9: Focus Group Session
- Session 10: Results sharing (Optional)

No physical risks are expected. Some participants may experience mild discomfort related to time pressure or unfamiliarity with digital tools. You may stop at any time.

Handling of data and confidentiality

All information collected in the study will be treated confidentially.

- Only the researcher, Everi Labs and academic supervisors will have access to the data.
- Data will be stored securely on password-protected and encrypted database.

- No names or identifying information will appear in the thesis or in any presentation of the results.
- Audio or video recordings (if interviews are recorded) will be transcribed and then deleted.
- Data will be stored only for the duration required to complete the degree project and will then be securely deleted.

You may request to read the completed thesis once it is published.

Processing of personal data (GDPR)

Jönköping University is responsible for the study and for the processing of your personal data in connection with the degree project. According to the General Data Protection Regulation (GDPR), you have the right to receive information about how your personal data are processed. You also have the right to request a register extract and to have any incorrect data corrected.
If you have questions about how your personal data are handled, you may contact the study's responsible researcher.

If you believe your personal data are being processed incorrectly, you may contact JU's Data Protection Officer at **dso@ju.se** or file a complaint with the Swedish Authority for Privacy Protection (IMY).

Voluntary participation

Participation in the degree project is completely voluntary. You may withdraw at any time without giving a reason. Your clinical work, teaching, or professional standing will not be affected by your decision to participate or not participate.

The degree project will be presented as a master thesis at Jönköping University.
If you have questions or would like more information, you are welcome to contact:

Student researcher:
Patricia Ann Del Rosario
Master of Product Development – Assistive Technology
Email: depa24bl@student.ju.se
Phone: +46 762 665 364

Academic supervisors:
David Rusaw
School of Health and Welfare (HHJ)
Jönköping University
Email: david.rusaw@ju.se

David Samvin
School of Engineering (JTH)
Jönköping University
Email: david.samvin@ju.se

Kind regards,

2026-02-19

Patricia Ann Del Rosario
Jönköping, Sweden



JÖNKÖPING UNIVERSITY

Participation consent form

Consent to participate in the degree project study:
"Integrating Artificial Intelligence into Prosthetics and Orthotics Clinical Workflow: Design and Usability Evaluation of the Everi Labs AI Copilot Prototype"

I have received oral and written information about the study and have had the opportunity to ask questions. I will keep a copy of the written information.

By signing below, I confirm that:

- I understand the purpose and procedures of the study.
- I understand that participation is voluntary and that I may withdraw at any time without giving a reason.
- I understand how my personal data will be handled according to GDPR.
- I consent to the collection of data related to my participation, including time study observations, SUS questionnaire responses, and focus group interview feedback.
- I consent to audio or video recording of the test sessions and interviews (if applicable). I have received oral and written information about the study and have had the opportunity to ask questions. I get to keep the written information.

☐ I agree to participate in the study Integrating Artificial Intelligence into Prosthetics and Orthotics Clinical Workflow: Design and Usability Evaluation of the Everi Labs AI Copilot Prototype.

Place and date	Printed Name of Participant & Signature

8.4 Participant Information Form

EVERI LABS Study Participant Information Form

Please fill up this form and upload all necessary documents required.
Thank you for your voluntary participation in our study.
For questions please email or contact:
Patricia Ann del Rosario | depa24bl@student.ju.se | +46 762 665 364

* Indicates required question

1. Email *

Recruitment poster with Inclusion criteria



2. First Name *

3. Last Name *

4. Age *

5. Sex *

Mark only one oval.

- Male
- Female
- Other:

6. Mobile number *

Please write your number starting with 09XXXXXXXXX unless using an area code outside the Philippines.

7. ISPO Certification Level *

Mark only one oval.

- Category I (Prosthetist/Orthotist): Requires university-level education (3–4 years) to provide comprehensive patient care, assessment, and device fitting.
- Category II (Orthopaedic Technologist): Requires 3 years of formal training below degree level to provide prosthetic/orthotic services.
- Category III (Prosthetic/Orthotic Technician): Focuses on on-the-job training for manufacturing and assembly of devices.
- Other:

8. Country of Practice *

Mark only one oval.

- ◯ Philippines
- ◯ Other: _____

9. Location of Practice (Region or Province) *

_____

10. Type of Work Setting (You can select more than 1) *

Tick all that apply.

- ☐ Hospital
- ☐ Clinic
- ☐ Community
- ☐ University
- ☐ Private Practice
- ☐ Other: _____

11. How many years have you been working as a CPO Clinician? (including clinical placements) *

Mark only one oval.

- ◯ 1-2 years
- ◯ 3-5 years
- ◯ 6-10 years
- ◯ More than 10 years
- ◯ Less than 1 year

12. How stable is your internet connection at work? *

Mark only one oval.

	1	2	3	4	5	
Poo	◯	◯	◯	◯	◯	Very Stable Connection

13. How often do you have access to stable internet connection? *

Mark only one oval.

	1	2	3	4	5	
Nev	◯	◯	◯	◯	◯	Always

14. Do you use a mobile device or computer for patient intake, and SOAP clinical journaling during initial assessment? *

Mark only one oval.

- ◯ Yes
- ◯ No

15. What record keeping devices or tools do you use for patient assessment? (You can select more than 1) *

Tick all that apply.

- ☐ Paper forms (hard copies)
- ☐ Smartphones
- ☐ Tablets
- ☐ Computer PC
- ☐ Laptop
- ☐ None
- ☐ Other: _____

16. How often do you use paper forms during patient intake and SOAP journaling during assessment? *

Mark only one oval.

	1	2	3	4	5	
Nev	◯	◯	◯	◯	◯	Always

17. How difficult do you think it is to store your patient assessment forms? *

Mark only one oval.

	1	2	3	4	5	
Extr	◯	◯	◯	◯	◯	Very Easy

18. How safe do you think is your current record keeping system? *

Mark only one oval.

	1	2	3	4	5	
Very	◯	◯	◯	◯	◯	Very safe

19. How satisfied are you with the current record keeping system at your workplace? *

Mark only one oval.

	1	2	3	4	5	
Very	◯	◯	◯	◯	◯	Very satisfied

20. Do you use digital electronic health/ medical record systems at work? *

Mark only one oval.

- Yes
- No
- Not sure

21. Is it your first time using an electronic medical record system for PO practice? *

Mark only one oval.

- Yes
- No
- Not sure

22. Are you willing to use an electronic health/ medical record system in your practice? *

Mark only one oval.

- Yes
- No
- Not sure

23. Do you use Artificial Intelligence (AI) tools in your practice? *

Mark only one oval.

- Yes
- No
- Not sure

24. What type of AI tools do you use for PO practice? *

Tick all that apply.

- [ ] ChatGPT
- [ ] CoPilot
- [ ] Gemini
- [ ] None
- [ ] Other: _____

25. Informed Consent *

Please upload your signed informed consent

Files submitted:

26. Data Privacy Consent *

Tick all that apply.

- [ ] I certify that the information provided are true and correct.
- [ ] I acknowledge and understand that the given information will be used and accessed by researcher, supervisors and Everi Labs for research purposes.
- [ ] I promise to keep all proceedings and information about the product and the study strictly confidential.
- [ ] I give my full consent to investigators and Everi Labs to record, collect, store and dispose of my personal information or test results as they see fit for this study.
- [ ] I fully understand my rights as a voluntary participant in this study by reading and signing the informed consent uploaded above.

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Google Forms

8.5 System Usability Scale (SUS) Form

System Usability Scale (SUS)

	Strongly Disagree				Strongly Agree
I think that I would like to use this system frequently.	1	2	3	4	5
I found the system unnecessarily complex.	1	2	3	4	5
I thought this system was easy to use.	1	2	3	4	5
I think that I would need the support of a technical person to be able to use this system.	1	2	3	4	5
I found the various functions in this system were well integrated.	1	2	3	4	5
I thought there was too much inconsistency in this system.	1	2	3	4	5
I would imagine that most people would learn to use this system very quickly.	1	2	3	4	5
I found this system very awkward to use.	1	2	3	4	5
I felt very confident using this system.	1	2	3	4	5
I needed to learn a lot of things before I could get going with this system.	1	2	3	4	5

8.6 Post-Test Questionnaire

Post-Test Questionnaire

Please answer this form after testing is completed.
Thank you for your voluntary participation in our study.
For questions please email or contact:
Patricia Ann del Rosario | depa24bl@student.ju.se | +46 762 665 364

* Indicates required question

1. Participant ID * Dropdown

Mark only one oval.

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10

2. Test Date *

Example: 7 January 2019

3. Testing Cycle *

Mark only one oval.

- ◯ Round 1
- ◯ Round 2

4. Type of System Tested *

Mark only one oval.

- ◯ Paper-based
- ◯ Everi Labs Prototype

5. Simulation Patient Case Tested *

Mark only one oval.

- ◯ Case 1 - Prosthetics
- ◯ Case 2 - Orthotics

6. What tasks were you able to perform during the test? (You may choose more than 1 if applicable)

Tick all that apply.

- ☐ Patient intake
- ☐ Clinical SOAP notes
- ☐ None

7. I think that I would like to use this system frequently. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

8. I found the system unnecessarily complex. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

9. I thought this system was easy to use. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

10. I think that I would need the support or assistance to be able to use the system. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

11. I found the various functions in this system were well integrated *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Disagree

12. I thought that there was too much inconsistency in this system. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

13. I would imagine that most people would learn to use this system very quickly. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

14. I found this system very cumbersome/ awkward to use. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

15. I felt very confident using this system. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

16. I needed to learn a lot of things before I could get going with this system. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

17. Did you finish your SOAP notes?

Mark only one oval.

- ◯ Yes
- ◯ No
- ◯ Not sure

18. Overall, how easy or difficult was the tasks to complete *

Mark only one oval.

	1	2	3	4	5	6	7	
Very	◯	◯	◯	◯	◯	◯	◯	Very easy

19. How satisfied are you with the system? *

Mark only one oval.

	1	2	3	4	5	
Very	◯	◯	◯	◯	◯	Very satisfied

20. Other notes or comments:

21. Upload Pictures or Scans of your forms

Files submitted:

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8.7 Focus group discussion guide

Focus Group Protocol for Evaluating Patient Assessment Documentation Systems

4. **Moderator Script (Opening)**

"Thank you for joining today's discussion. We're here to explore your experiences documenting patient assessments using two systems: the AI-assisted EMR tool and the traditional paper-based forms. Our goal is to understand what worked well, what created challenges, and what changes would make these systems more supportive of your clinical workflow.

There are no right or wrong answers. We're interested in your honest experiences and perspectives. Everything you share will be kept confidential and used only for research and product development purposes."

5. **Session Rules**
   - Speak one at a time so everyone can be heard.
   - All perspectives are valid—please feel free to disagree respectfully.
   - Share specific examples whenever possible.
   - The moderator may ask follow-up questions to clarify or deepen understanding.
   - You may choose not to answer any question.
   - Kindly use English language during sessions

6. **Warm-Up Question**

"Before we begin, could you briefly describe your typical workflow when documenting a patient assessment?"

Notes: This helps participants settle in and activates their mental model of documentation.

8. **Focus Group Questions: Paper-Based Documentation System**

A. Visual and aesthetics

- *"When using the paper forms, what did you notice about the spacing, font size, and overall appearance of the pages, and how did these elements influence how you worked through the documentation?"*

B. Layout and navigation

- *"How did the way the paper forms were arranged—such as the order of sections or how information was grouped—affect how you moved through the documentation?"*

C. Actions and commands

- *"How did the paper forms show you what step or section came next, and how did you understand what actions you needed to take as you worked?"*

D. Input forms and fields

- *"How did the wording of the labels and the structure of the fields on the paper forms shape how you entered information?"*

E. Task functionality

- *"How did the structure of the paper forms support the tasks you needed to complete during documentation?"*

9. **Additional Questions**
   - What do you wish to add to each system?
   - How much are you willing to pay for the service?

10. **Closing Reflections**

"Thinking about both systems, what is the single most important change that would improve your documentation experience?"

"Is there anything we haven't asked that you think is important for us to understand?"

11. Data Handling and Analysis Plan

A. Recording and Note-Taking

- Audio/ Video recording
- Moderator takes structured notes
- All data anonymized immediately after transcription
-

B. Thematic Analysis:

- Reviewing and familiarization with transcripts
- Initial coding + respondent's review
- Grouping codes into themes, thematic mapping
- Reviewing and refining themes
- Producing final theme descriptions and analysis with illustrative quotes

C. Linking to Product Development

Themes will be translated into:

- UI/UX requirements
- Interaction design improvements
- Workflow alignment recommendations
- AI suggestion behavior guidelines
- Prioritized feature suggestions

7. Focus Group Questions: AI-Assisted EMR (Everi Labs)

A. Visual and aesthetics

- *"When using the prototype, what did you notice about the spacing, font size, colors, and overall appearance of the screen, and how did these elements influence how you worked through the documentation?"*

B. Layout and navigation

- *"How did the way the prototype's screens were arranged—such as the order of sections or how information was grouped—affect how you moved through the documentation?"*

C. Actions and commands

- *"How did the prototype show you what step or action came next, and how did you understand what you needed to do as you worked?"*

D. Input forms and fields

- *"How did the wording of the labels and the structure of the input fields in the prototype shape how you entered information?"*

E. Task functionality

- *"How did the structure of the prototype support the tasks you needed to complete during documentation?"*

8.8 Prosthetic Paper Assessment Form

This form is for the research titled: "Integrating Artificial Intelligence into Prosthetics and Orthotics Clinical Workflow: Design and Usability Evaluation of the Everi Labs AI Co-Pilot"	**Transtibial Prosthetics Form**	Contact researcher: Patricia Ann del Rosario +46 762 665 364 depa24bl@student.ju.se

Patient Name:			Participant No.:
Age:	Sex: F ☐ M ☐	Weight (Kg):	Height (cm):
Address:		Date of Assessment:	
Subjective Assessment			
Affected Side: ☐ Left ☐ Right ☐ Bilateral		Cause of Amputation:	
Primary Prosthetic User: ☐ Yes ☐ No		Date of Amputation:	
K-Level: ☐ 1 ☐ 2 ☐ 3 ☐ 4		Revision: ☐ Yes ☐ No	
Occupation:		Date of Revision:	
Medical History:		Social History:	
Previous Prosthetic History: ☐ Yes ☐ No Detail:			
Patient Goals and Expectations:			
Objective Assessment			
General Condition of Residual Limb: (Wound Healing/Scars/Neuromas/Bony End/Bony Prominences/Pain/Edema)		General Conditions: Sound limb (Skin condition/Wound/Pain)	
Cutaneous Sensation: Phantom pain/sensation:		Upper Limb Condition: Eyesight: Hearing:	

Proprioception:	Standing Balance/Mobility:
Gait assessment with existing prosthesis (if applicable)	Gait aid……
Residual Limb Shape & Length: ☐ Conical ☐ Short ☐ Cylindrical ☐ Medium ☐ Irregular ☐ Long	**Residual Limb Position:** ☐ Flexion _____ degrees ☐ Abduction _____ degrees ☐ Adduction _____ degrees

Hip Joint	ROM		Muscle Strength	
	LEFT	RIGHT	LEFT	RIGHT
Flexion (120)/ Extension (30)				
ABD (45)/ ADD (30)				
IR (35)/ ER (45)				
Knee Joint	**ROM**		**Muscle Strength**	
	LEFT	RIGHT	LEFT	RIGHT
Flexion: (130)/ Ext (0-10)				
	LEFT		RIGHT	
M-L/ A-P Stability				
Ankle Joint	**ROM**		**Muscle Strength**	
	LEFT	RIGHT	LEFT	RIGHT
DF (30) / PF (45)				
Subtalar Joint	**ROM**		**Muscle Strength**	
	LEFT	RIGHT	LEFT	RIGHT
Inversion (30)/ Eversion (15)				

Plan			
Side:	☐ Left	☐ Right	☐ Bilateral
Socket Design:	☐ PTB	☐ PTB SC	☐ PTB SCSP
Polypropylene:	☐ 3mm	☐ 5mm	☐ Other, _____
Suspension:	☐ Cuff	☐ Self-Suspending	☐ Other, _____
Foot:	☐ SACH	☐ Single Axis	☐ Other, _____

Other Instructions:	P & O: (Name and Sign)
	Fitting Date:

8.9 Orthotic Paper Assessment Form

This form is for the research titled: "Integrating Artificial Intelligence into Prosthetics and Orthotics Clinical Workflow: Design and Usability Evaluation of the Everi Labs AI Co-Pilot"	**Lower Limb Orthotics Form**	Contact researcher: Patricia Ann del Rosario +46 762 665 364 depa24bl@student.ju.se

Patient Name:			Participant No.:
Age:	Sex: F ☐ M ☐	Weight (Kg):	Height (cm):
Address:		Date of Assessment:	
Subjective Assessment			
Primary Patient: ☐ Yes ☐ No		Affected Side: ☐ Left ☐ Right ☐ Bilateral	
Main Complaint:		Diagnosis:	
Medical History:		Social History:	
Objective Assessment			
Muscle Tone:		Joint Deformities:	
Sensation and Proprioception:		Leg Length Discrepancy	
Upper Limb Condition:		Spinal Deformity:	
Gait Deviations:			

Hip Joint	ROM		Muscle Strength	
	LEFT	RIGHT	LEFT	RIGHT
Flexion (120)/ Extension (30)				
ABD (45)/ ADD (30)				
IR (35)/ ER (45)				
Knee Joint	**ROM**		**Muscle Strength**	
	LEFT	RIGHT	LEFT	RIGHT
Flexion (130)/ Ext (0-10)				
Hamstrings Angle (140)				

	LEFT		RIGHT	
M-L/ A-P Stability				
Ankle Joint	**ROM**		**Muscle Strength**	
	LEFT	**RIGHT**	**LEFT**	**RIGHT**
Dorsiflexion Flex (20)/ Dorsiflexion Ext (10)				
Plantarflexion (45)				
Subtalar Joint	**ROM**		**Muscle Strength**	
	LEFT	**RIGHT**	**LEFT**	**RIGHT**
Inversion (30)/ Eversion (15)				

Assessment and Plan						
Prescribed Appliance:	☐ FO	☐PLS AFO ☐Rigid AFO ☐Articulated AFO ☐GRAFO	☐ KAFO	☐ HKAFO	☐ KO	☐ HpO
Side:	☐ Left	☐Right	☐ Bilateral			
Polypropylene:	☐ 3mm	☐5mm	☐ Other, _____			
Ankle Joints:	☐ None _____	☐Oklahoma	☐ Tamarac – Child ☐ Tamarac – Adult	☐ Other,		
Knee Joints:	☐ Paediatric ☐Adult	☐Swiss Lock	☐ Drop Lock	☐ Offset		
Hip Joint:	☐ Lock	☐Free	☐ Other, _____			
Straps:	☐ Calf	☐1 inch ☐1.5 inch	☐ Ankle	☐ Internal Pull ☐45 degrees		

Other Instructions:	P & O: (Name and Sign)
	Fitting Date:

8.10 Clinical SOAP Note Form

This form is for the research titled: "Integrating Artificial Intelligence into Prosthetics and Orthotics Clinical Workflow: Design and Usability Evaluation of the Everi Labs AI Co-Pilot"	**Clinical SOAP Notes**	Contact researcher: Patricia Ann del Rosario +46 762 665 364 depa24bl@student.ju.se

PATIENT NAME:	Participant No.:
DEVICE:	Dx or Co-morbidities:
LATERALITY:	Referring Doctor/ Clinic:

Date & Time	SOAP/ Progress Notes	CPO Name & Signature

8.11 Case 1 Prosthetics Details

This form is for the research titled: "Integrating Artificial Intelligence into Prosthetics and Orthotics Clinical Workflow: Design and Usability Evaluation of the Everi Labs AI Co-Pilot"	**Transtibial Prosthetics Form**	Contact researcher: Patricia Ann del Rosario +46 762 665 364 depa24bl@student.ju.se

Patient Name: Juan dela Cruz			Participant No.:
Age: 47	Sex: F ☐ M ☑	Weight (Kg): 56	Height (cm): 165
Address: Project 8, Quezon city		Date of Assessment:	
Subjective Assessment			
Affected Side: ☑ Left ☐ Right ☐ Bilateral		Cause of Amputation: non-healing wound at big toe Oct 2025 DM Type 2	
Primary Prosthetic User: ☑ Yes ☐ No		Date of Amputation: Dec 16 2025	
K-Level: ☐ 1 ☐ 2 ☑ 3 ☐ 4		Revision: ☐ Yes ☑ No	
Occupation: pet supplies store owner		Date of Revision:	
Medical History: Dx with DM Feb 2025 Dengue 2023 No other conditions Metformin BP 120/90		Social History: Hobby: cycling Lives in a town house with 2nd floor With wife and 2 children No flood, cement	
Previous Prosthetic History: ☐ Yes ☑ No Detail: First time user			
Patient Goals and Expectations: Travel abroad and walk with prosthesis			
Objective Assessment			
General Condition of Residual Limb: (Wound Healing/Scars/Neuromas/Bony End/Bony Prominences/Pain/Edema) Open wound at scar or seam Non- healing 6/10 VAS		General Conditions: Sound limb (Skin condition/Wound/Pain) Sound limb good condition Upper Limb Condition: Good	
Cutaneous Sensation: Less at distal end Phantom pain/sensation: Tingling at distal end		Eyesight: Prescription glasses, near-sighted Hearing: Normal	

Proprioception: Intact	Standing Balance/Mobility: can stand on one foot with aid
Gait assessment with existing prosthesis (if applicable)	Gait aid..... Wheel chair and axillary crutches
Residual Limb Shape & Length: ☐ Conical ☐ Short ☑ Cylindrical ☑ Medium ☐ Irregular ☐ Long	**Residual Limb Position:** ☑ Flexion 5_____ degrees ☑ Abduction 5_____ degrees ☐ Adduction _____ degrees

Hip Joint	ROM		Muscle Strength	
	LEFT	RIGHT	LEFT	RIGHT
Flexion (120)/ Extension (30)	WNL	WNL	5	5
ABD (45)/ ADD (30)	WNL	WNL	5	5
IR (35)/ ER (45)	WNL	WNL	5	5
Knee Joint	**ROM**		**Muscle Strength**	
	LEFT	**RIGHT**	**LEFT**	**RIGHT**
Flexion: (130)/ Ext (0-10)	WNL	WNL	5	5
	LEFT		**RIGHT**	
M-L/ A-P Stability	Stable		Stable	
Ankle Joint	**ROM**		**Muscle Strength**	
	LEFT	**RIGHT**	**LEFT**	**RIGHT**
DF (30) / PF (45)	WNL	WNL	NA	4
Subtalar Joint	**ROM**		**Muscle Strength**	
	LEFT	**RIGHT**	**LEFT**	**RIGHT**
Inversion (30)/ Eversion (15)	WNL	WNL	NA	4

Plan			
Side:	☑ Left	☐ Right	☐ Bilateral
Socket Design:	☐ PTB	☐ PTB SC	☐ PTB SCSP
Polypropylene:	☐ 3mm	☐ 5mm	☐ Other, _____
Suspension:	☐ Cuff	☐ Self-Suspending	☐ Other, _____
Foot:	☐ SACH	☐ Single Axis	☐ Other, _____

Other Instructions: Referred for wound healing and consultation with Doctor	P & O: (Name and Sign)
	Fitting Date:

8.12 Case 2 Orthotics Details

This form is for the research titled: "Integrating Artificial Intelligence into Prosthetics and Orthotics Clinical Workflow: Design and Usability Evaluation of the Everi Labs AI Co-Pilot"	**Lower Limb Orthotics Form**	Contact researcher: Patricia Ann del Rosario +46 762 665 364 depa24bl@student.ju.se

Patient Name: Maria Santos			Participant No.:
Age: 56	Sex: F ☑ M ☐	Weight (Kg): 50	Height (cm): 158
Address: Malolos Bulacan 4400		Date of Assessment:	
Subjective Assessment			
Primary Patient: ☑ Yes ☐ No		Affected Side: ☐ Left ☑ Right ☐ Bilateral	
Main Complaint: Unable to walk without wheelchair, hard to transfer to garden stool Afraid to slip and fall		Diagnosis: CVA Oct 16 2025- Ischemic	
Medical History: HBP - 140/90 Gait training 6/6 PT		Social History: Lives in house 1 story Planting or gardening on weekends Mall with wheelchair > 30 mins	
Objective Assessment			
Muscle Tone: Normal		Joint Deformities: None	
Sensation and Proprioception: Intact		Leg Length Discrepancy None	
Upper Limb Condition: Normal		Spinal Deformity: None	
Gait Deviations: Foot drop at swing			

Hip Joint	ROM		Muscle Strength	
	LEFT	RIGHT	LEFT	RIGHT
Flexion (120)/ Extension (30)	WNL		5	5
ABD (45)/ ADD (30)	WNL		5	5
IR (35)/ ER (45)	WNL		4	4
Knee Joint	**ROM**		**Muscle Strength**	
	LEFT	RIGHT	LEFT	RIGHT
Flexion (130)/ Ext (0-10)	WNL		5	5
Hamstrings Angle (140)	WNL		5	5

	LEFT		RIGHT	
M-L/ A-P Stability	Stable		Stable	
Ankle Joint	**ROM**		**Muscle Strength**	
	LEFT	**RIGHT**	**LEFT**	**RIGHT**
Dorsiflexion Flex (20)/ Dorsiflexion Ext (10)	WNL		4	3
Plantarflexion (45)	WNL		4	4
Subtalar Joint	**ROM**		**Muscle Strength**	
	LEFT	**RIGHT**	**LEFT**	**RIGHT**
Inversion (30)/ Eversion (15)	WNL		4	3

Assessment and Plan						
Prescribed Appliance:	☐ FO	☐PLS AFO ☐Rigid AFO ☐Articulated AFO ☐GRAFO	☐ KAFO	☐ HKAFO	☐ KO	☐ HpO
Side:	☐ Left	☑Right	☐ Bilateral			
Polypropylene:	☐ 3mm	☐5mm	☐ Other, _____			
Ankle Joints:	☐ None _____	☐Oklahoma	☐ Tamarac – Child ☐ Tamarac – Adult	☐ Other,		
Knee Joints:	☐ Paediatric ☐Adult	☐Swiss Lock	☐ Drop Lock	☐ Offset		
Hip Joint:	☐ Lock	☐Free	☐ Other, _____			
Straps:	☐ Calf	☐1 inch ☐1.5 inch	☐ Ankle	☐ Internal Pull ☐45 degrees		

Other Instructions:	P & O: (Name and Sign)
Referred for follow up with Md and possible gait training K2	Fitting Date:

8.13 Randomized Test Sequence and Patient Cases

System	Participant Code (CPO)	Case Type (1=Prosthetics; 2= Orthotics)
Paper	2	1
	4	2
	7	2
	8	1
	9	2
	5	2
	3	3
	1	1
	6	2
	10	1
v1	1	2
	5	1
	10	2
	6	1
	3	1
v2	9	1
	8	2
	4	1
	2	2
	7	1

8.14 Individual Task Time Results

System	Participant Code (CPO)	Assessment Time (min)	SOAP Time (min)	Total Time (min)
Paper	1	29.95	15.28	45.23
	2	20.03	2.07	22.05
	3	10.18	2.97	13.15
	4	9.57	11.53	21.10
	5	26.60	4.10	30.70
	6	27.98	2.72	30.70
	7	14.00	3.37	17.37
	8	15.78	4.98	20.77
	9	15.80	7.75	23.55
	10	29.37	9.32	38.68
v1	1	15.68	24.52	40.20
	3	18.85	6.18	25.03
	5	31.97	2.65	34.62
	6	36.93	2.88	39.82
	10	34.62	7.83	42.45
v2	2	21.30	3.70	25.07
	4	22.12	7.75	29.87
	7	21.70	4.83	26.53
	8	19.02	2.07	21.08
	9	24.85	6.87	31.72

8.15 Descriptives Task Time

System				Statistic	Std. Error
IE	Paper	Mean		19.9260	2.51805
		95% Confidence Interval for Mean	Lower Bound	14.2298	
			Upper Bound	25.6222	
		5% Trimmed Mean		19.9444	
		Median		17.9150	
		Variance		63.406	
		Std. Deviation		7.96277	
		Minimum		9.57	
		Maximum		29.95	
		Range		20.38	
		Interquartile Range		15.28	
		Skewness		0.078	0.687
		Kurtosis		-1.810	1.334
	v1	Mean		27.6100	4.32478
		95% Confidence Interval for Mean	Lower Bound	15.6025	
			Upper Bound	39.6175	
		5% Trimmed Mean		27.7550	
		Median		31.9700	
		Variance		93.519	
		Std. Deviation		9.67050	
		Minimum		15.68	
		Maximum		36.93	
		Range		21.25	
		Interquartile Range		18.51	
		Skewness		-0.522	0.913
		Kurtosis		-2.788	2.000
	v2	Mean		21.7980	0.93256
		95% Confidence Interval for Mean	Lower Bound	19.2088	
			Upper Bound	24.3872	
		5% Trimmed Mean		21.7828	
		Median		21.7000	
		Variance		4.348	

		Std. Deviation		2.08526	
		Minimum		19.02	
		Maximum		24.85	
		Range		5.83	
		Interquartile Range		3.33	
		Skewness		0.317	0.913
		Kurtosis		1.678	2.000
SOAP	Paper	Mean		6.4090	1.39866
		95% Confidence Interval for Mean	Lower Bound	3.2450	
			Upper Bound	9.5730	
		5% Trimmed Mean		6.1572	
		Median		4.5400	
		Variance		19.563	
		Std. Deviation		4.42296	
		Minimum		2.07	
		Maximum		15.28	
		Range		13.21	
		Interquartile Range		6.97	
		Skewness		1.033	0.687
		Kurtosis		0.082	1.334
	v1	Mean		8.8120	4.04841
		95% Confidence Interval for Mean	Lower Bound	-2.4282	
			Upper Bound	20.0522	
		5% Trimmed Mean		8.2817	
		Median		6.1800	
		Variance		81.948	
		Std. Deviation		9.05251	
		Minimum		2.65	
		Maximum		24.52	
		Range		21.87	
		Interquartile Range		13.41	
		Skewness		1.917	0.913
		Kurtosis		3.840	2.000
	v2	Mean		5.0440	1.03328

		95% Confidence Interval for Mean	Lower Bound	2.1751	
			Upper Bound	7.9129	
		5% Trimmed Mean		5.0589	
		Median		4.8300	
		Variance		5.338	
		Std. Deviation		2.31049	
		Minimum		2.07	
		Maximum		7.75	
		Range		5.68	
		Interquartile Range		4.43	
		Skewness		-0.096	0.913
		Kurtosis		-1.586	2.000
Total	Paper	Mean		26.3350	3.13725
		95% Confidence Interval for Mean	Lower Bound	19.2381	
			Upper Bound	33.4319	
		5% Trimmed Mean		26.0178	
		Median		22.8250	
		Variance		98.423	
		Std. Deviation		9.92084	
		Minimum		13.15	
		Maximum		45.23	
		Range		32.08	
		Interquartile Range		12.78	
		Skewness		0.763	0.687
		Kurtosis		-0.054	1.334
	v1	Mean		36.4240	3.12395
		95% Confidence Interval for Mean	Lower Bound	27.7505	
			Upper Bound	45.0975	
		5% Trimmed Mean		36.7222	
		Median		39.8200	
		Variance		48.795	
		Std. Deviation		6.98537	
		Minimum		25.03	
		Maximum		42.45	

		Range		17.42	
		Interquartile Range		11.50	
		Skewness		-1.434	0.913
		Kurtosis		1.723	2.000
	v2	Mean		26.8540	1.86281
		95% Confidence Interval for Mean	Lower Bound	21.6820	
			Upper Bound	32.0260	
		5% Trimmed Mean		26.9044	
		Median		26.5300	
		Variance		17.350	
		Std. Deviation		4.16536	
		Minimum		21.08	
		Maximum		31.72	
		Range		10.64	
		Interquartile Range		7.72	
		Skewness		-0.320	0.913
		Kurtosis		-0.671	2.000

8.16 Test of Normality Task Time

		Kolmogorov-Smirnov[a]			Shapiro-Wilk		
System		Statistic	df	Sig.	Statistic	df	Sig.
IE	Paper	0.199	10	.200*	0.887	10	0.157
	v1	0.274	5	.200*	0.859	5	0.226
	v2	0.239	5	.200*	0.954	5	0.763
SOAP	Paper	0.227	10	0.156	0.875	10	0.114
	v1	0.343	5	0.055	0.755	5	0.033
	v2	0.185	5	.200*	0.965	5	0.845
Total	Paper	0.211	10	.200*	0.935	10	0.500
	v1	0.287	5	.200*	0.856	5	0.214
	v2	0.165	5	.200*	0.976	5	0.912

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

8.17 Effect Size Calculations Task Time

Source	Dependent Variable	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	IE	198.595[a]	2	99.297	1.755	0.203	0.171
	SOAP	36.841[b]	2	18.421	0.596	0.562	0.066
	Total	369.624[c]	2	184.812	2.731	0.094	0.243
Intercept	IE	9614.407	1	9614.407	169.880	0.000	0.909
	SOAP	821.340	1	821.340	26.585	0.000	0.610
	Total	16060.980	1	16060.980	237.343	0.000	0.933
System	IE	198.595	2	99.297	1.755	0.203	0.171
	SOAP	36.841	2	18.421	0.596	0.562	0.066
	Total	369.624	2	184.812	2.731	0.094	0.243
Error	IE	962.120	17	56.595			
	SOAP	525.208	17	30.895			
	Total	1150.390	17	67.670			
Total	IE	11119.899	20				
	SOAP	1451.428	20				
	Total	18324.938	20				
Corrected Total	IE	1160.714	19				
	SOAP	562.050	19				
	Total	1520.014	19				

a. R Squared = .171 (Adjusted R Squared = .074)

b. R Squared = .066 (Adjusted R Squared = -.044)

c. R Squared = .243 (Adjusted R Squared = .154)

8.18 Individual SUS Scores

System	Participant Code (SUS)	Individual SUS Scores	SUS Adjective Findings
Paper	1	60.00	Indicates usability issues
	2	75.00	Good usability
	3	70.00	Average usability
	4	55.00	Indicates usability issues
	5	75.00	Good usability
	6	85.00	Excellent usability
	7	82.50	Excellent usability
	8	60.00	Indicates usability issues
	9	62.50	Indicates usability issues
	10	55.00	Indicates usability issues
v1	1	72.50	Good usability
	3	62.50	Indicates usability issues
	5	62.50	Indicates usability issues
	6	60.00	Indicates usability issues
	10	42.50	Poor usability
v2	1	72.50	Good usability
	3	62.50	Indicates usability issues
	5	62.50	Indicates usability issues
	6	60.00	Indicates usability issues
	10	42.50	Poor usability

8.19 Descriptives SUS Score

System				Statistic	Std. Error
SUS	Paper	Mean		68.0000	3.49205
		95% Confidence Interval for Mean	Lower Bound	60.1004	
			Upper Bound	75.8996	
		5% Trimmed Mean		67.7778	
		Median		66.2500	
		Variance		121.944	
		Std. Deviation		11.04285	
		Minimum		55.00	
		Maximum		85.00	
		Range		30.00	
		Interquartile Range		18.13	
		Skewness		0.317	0.687
		Kurtosis		-1.422	1.334
	v1	Mean		60.0000	4.87340
		95% Confidence Interval for Mean	Lower Bound	46.4693	
			Upper Bound	73.5307	
		5% Trimmed Mean		60.2778	
		Median		62.5000	
		Variance		118.750	
		Std. Deviation		10.89725	
		Minimum		42.50	
		Maximum		72.50	
		Range		30.00	
		Interquartile Range		16.25	
		Skewness		-1.087	0.913
		Kurtosis		2.485	2.000
	v2	Mean		76.0000	3.22102
		95% Confidence Interval for Mean	Lower Bound	67.0570	
			Upper Bound	84.9430	
		5% Trimmed Mean		76.2500	
		Median		80.0000	

	Variance	51.875	
	Std. Deviation	7.20243	
	Minimum	65.00	
	Maximum	82.50	
	Range	17.50	
	Interquartile Range	12.50	
	Skewness	-1.083	0.913
	Kurtosis	-0.062	2.000

8.20 Test of Normality SUS

		Kolmogorov-Smirnov[a]			Shapiro-Wilk		
System		Statistic	df	Sig.	Statistic	df	Sig.
SUS	Paper	0.191	10	.200*	0.911	10	0.286
	v1	0.300	5	0.161	0.888	5	0.349
	v2	0.311	5	0.129	0.871	5	0.269

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

8.21 Effect Size Calculations SUS

Dependent Variable: SUS

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	640.000[a]	2	320.000	3.056	0.073	0.264
Intercept	83232.000	1	83232.000	794.912	0.000	0.979
System	640.000	2	320.000	3.056	0.073	0.264
Error	1780.000	17	104.706			
Total	94900.000	20				
Corrected Total	2420.000	19				

a. R Squared = ,264 (Adjusted R Squared = ,178)

8.22 Thematic Analysis Code Sheet

Direct Quote	CPO	Code	Sub-Theme	Main Theme
"Sometimes during the casting assessment then that's the only time that we need to we need to endorse uh with PO to PO and there's really no paper base because we don't have time to do everything at once"	9	Time management of documentation with clinical tasks	Documentation overload	Current challenges with Paper Based System
"Also, if there's a lot of if it's the day that's our busiest, there's a lot of referrals. So, we have to do a quick assessment for everybody."	9			
"For students, it takes too much time especially because they're basically writing their assessments and performing the assessments at the same time.	8			
"So there are times also that we don't get to do our soap notes cuz when there are a lot of patients um coming in at the same day, we don't get enough time to do the soap notes and the soap notes that we're used to are actually really long."	4	SOAP notes too long		
“Sometimes we overlook the um as well like updating the soap notes. So um sometimes during the casting assessment then that's the only time that we need to we need to endorse uh with PO to PO and there's really no paper base because we don't have time to do everything at once.”	9	SOAP notes overlooked		
“We don't know what patient is coming in so we have to look into files of what type of form we should be getting because each and every one would have a different type of assessment and different type of measurement.”	7	Choosing the right forms		

"I'd also want put in the thought about with the paper is that um the different types of forms that we had to prepare"	7	Preparing different forms		
"If you have a lot of patients, it would be hard if it's not organized."	2	Difficulty with disorganized data		
"I'm a little bit not really used to this type of pattern um of just how the data are or are since um during my internship the soap note is um a little bit different on the formatting"	2	Unfamiliarity with different form layouts		
“In the paper based, I guess you're forced to go through each and every one of the entries. So for example in the um assessment of the limb um then if it's not applicable”	7	Forced to fill irrelevant sections		
"So we don't have to fill up it's unnecessary to put in that we received this patient with subjective objective just my thought for at least since we have comments anyway at the bottom"	7			
“For the interns, they really had to go through the whole section because they are graded to that.”	9	Interns required to complete full forms		
"So it really needs to be documented as soon as possible so that there will be no mix up of um information among those uh patients."	6	Need for quick documentation to avoid documentation mix-up		
"We do it all manually"	1	Manual documentation		
"So the main challenges that we've encountered while doing the paper-based system would be the redundancy of the details the general um details of the patient "	9	Redundant patient information	Redundant record keeping	
"So um redundancy really is one of the basic challenges that um we've encountered."	9			

"So we don't have redundant or like duplicate things."	7			
"Um uh but let's say if ever the same patient comes back after five years and wala folder then we will create another one. We'll start the assessment from the uh from the start"	8			
"then the third would be the redundancies mentioned by Krisha. So um if for example at entry of the patient by initial assessors or the preassessment or by the hospital already "	7			
"It becomes a problem um especially if you want to target PhilHealth. Um then you would have to link up to a hospital which then has its own patient management system and then, so we'd have duplicate databases for the patient as well as for the PO center"	7	Duplicate databases		
“I think um one of our challenges is that uh we our clinic actually don't have a proper filing for the patient's files. So there's no like centralized folders or um filings for our patients”	4	Lack of centralized filing	System fragmentation	
"I usually have problem. I take a picture in the phone and then I upload it into the drive so that the soap notes as well as whatever I put in are in the same file and folder. "	7	No centralized media storage		
"it's a bother because there are times where we forgot is the patient already due for a follow-up or possibly next month"	2	Scheduling issues		
"we really have a problem in terms of setting the deadline for the fitting but most of the time the device it is not done yet in fabrication. So we had to um cancel the appointment and then reschedule again"	9			

“during Karina typhoon um it was unexpected to be flooded. So the work for how many years’ worth of documents charts it was flooded everything.”	9	Catastrophic data loss or damage	Data security and storage	
"Ah for the disadvantages first po I experienced trying to retrieve a paper-based type of documents when I was in PMR and it so happened that there is a returning patient but the PMR experienced flooding prior to my transition so there were no uh backups of the file so technically it's all flashed out during the flood so it was a big problem for us to track the history of the patient and will start again from assessing its uh history. So it was a big problem during our transition"	3			
"they sometimes also lose the assessment forms that they use even the folders get lost in the clinic, so I think that's the primary um issue we have with uh with forms with paper and pen in PSPO "	8	Data retrieval issues		
“It's a little bit hard to backtrack some of the important details. So you have to like to go through your notes to find other data.”	2			
"We couldn't really retrieve any um I think we only have like active cases because we put the charts in a workshop "	9			
"and the of course data privacy because it once the folder gets lost um anyone can look through the patient um information"	8	Data privacy issues		
"It's a matter of like who's keeping who uh who's keeping what of the patient data and whose ownership would the data be?"	7	Unclear Data ownership and responsibility		

"So it's like um you get to be responsible for the data of your patient."	4			
"But mostly we lost the papers like after how many how after usage and not being able to if it's already done if you're already delivered it "	3	Missing records		
"So and also if we misplace the folder, it will because we experience that when we cannot find the folder of the patient. So it will cause it causes a lot of trouble because the file was missing."	1			
"There are cases that we have double folders of the same patient."	6	Duplicate patient folders		
"In TYK, um since we have the PT and the PO program um the patient folders are dedicated for PO services and then for the PT services they have a separate folder but again ideally it's supposed to be uh one folder so that we can see uh any progress in terms of the referring to certain um exercises or if patients have um adjustments at least it can be easily communicated even though that the PO or the student is not directly present they can just read the uh notes directly from the last um patient um contact. So I think that would be advantageous but again there is a separate folder, so I think that is also uh a problem"	6			
"Let's just say the patient is using a microprocessor me so all of the data such as when was the knee first connected to the internet or software to identify when is the maintenance something like that so if we don't have the specific date in which the Germany will request us will request us then we cannot provide	1	Stress with missing component or maintenance records of device		

them the information that they need so it will really be um stressful something like that also uh the for example um the patient is using a liner or knee sleeve where it has different sizes. If we cannot look the or if we cannot um retrieve the folder in which all the stickers or reference number are stored, then it would be a hassle to order a new or a spare component just because we lack the record. So it will be stressful also."				
"We the way we compile or save the data are manually."	1	Manual data storage		
"Sometimes it gets wet when it's being uh compiled by the other papers "	3	Risk of water damage		
"And another thing is when we when we're traveling and we have forms with us; the best thing we also think about is to protect the papers that we have. Like for example, if we do missions and we casted around 24 patients, we had to make sure that the papers are fully safe on our travel and not be wet during any extra activities that we do. "	3			
"we tried to have pictures before it was lost so we can just try to uh place it in a digital copy but that's how really the reality of our clinic right now like using papers that's all."	3	Data retrieval with pictures		
"I took photos of those um soap notes or uh measurement forms just forum safekeeping just in case the hard copy will be lost uh somewhere."	6			
"Then the storage is usually limited"	5	Limited storage capacity		

"So we need to um we need to discard most of the patients um especially um those patients who are patient files who are really like old already usually 5 years um from the present that's what we usually keep um and then the rest I believe we have to um destroy it or um discard most of us that patient's um files."	5			
"I think it was just recently when I realized that our storage capacity is already full. So, it's difficult to store more folders"	1			
"The biggest issue would be, um, yeah, mainly storage and then use of so much paper when printing the quotation or other important records “	1			
"I think we need to upgrade the cabinet storage so we can uh properly store the file of the patient because it's not enough to just store it. You need to store it properly "	1	Need for upgrade and proper storage system		
"I think it matters that um on the perspective of actually organizing those uh files and then um disorganization regardless of if it's five or 10 years "	6			
"So we need to um we need to discard most of the patients um especially um those patients who are patient files who are really like old already usually 5 years um from the present that's what we usually keep um and then the rest I believe we have to um destroy it or um discard most of us that patient's um files."	5	Discarding and archiving files		
"Another problem is the storage of the folders because uh here at school we're required to keep the physical copies of the folders for at least 5 years.	8			

So that accumulates so it's mainly storage"				
"So that we can so that it will be efficient if in case we need to look for a specific folder otherwise if it's not properly stored it will cause us so much trouble"	1	Inefficiencies due to improper storage		
"Obviously other problems were trying to print on areas. For example, we were run out of forms, and we need more forms for additional patients on a mission, and we don't have printer on this certain community. "	3	No access to printers in community settings		
“The lack of media in the paper-based system is really relevant in checking the patients’ improvements. it will serve as a baseline to check if the patient's gate improve or the angulations um improve per day.”	2	No visual baseline for patient progress	Poor patient progress tracking	
"I think if you're also working in a um in a like department in one department with physical therapist OT speech and rehab doctors it's also important to have a centralized patient data in terms of uh progress and um for example we have endorsed patients to undergo gait training for example so it's really good to have a progress notes as well"	9	Inconsistent instructions for endorsements or referrals	Interdisciplinary misalignment	
"So sometimes, it is different in perspective of the rehab doctor. You will only know if during follow-ups if you would ask the patient. Then sometimes you have you're going to realize that there's two different made by the PO and the doctor as well and also follow-ups actually. There's a different follow-up in doctor's point of view. So sometimes it's really not standardized to all rehab	9	Inconsistent instructions for follow up		

doctors. It's really a challenge for the CPOs and patients as well."				
"So, we've all gone through this assessment objective and then I think the limitation with our paper-based is that we have no idea what's in stock, what's available, what can we prescribe to the patient. We have this but then component-wise so we assessed K3 then it would be really nice to suddenly like have a filter all of the things K3 down to K1 then we don't have to look at K4 or what's available in stock what can we prescribe from based on our assessment"	7	No inventory link or visibility	Lack of connection to device inventory for prescription	
"Okay, it's not that sustainable because you have to print it every now and then"	5	Not sustainable or eco-friendly	Poor Sustainability	
"Then sustainability and because we used to print um uh what do you call this um patient files to insert it in the physical file or physical um folder of the patient. So that may take also resources."	5			
"I think for the first part of subjective, we can change the reason for encounter into chief complaint or concern."	1	Change reason for encounter to chief complaint	Improper terminologies	Issues with UI prototype v1
"For the reason for encounter, it would be I think it's just the choice of words in terms of what we're used to because of the chief complaint or the patient's goal etc."	6			
"I think for subjective; we can put a dedicated uh box or maybe just change the living situation into a K level or activity level"	1	Change living situation to activity (K) level		

"The order, I don't know if this one is something to do with the CPO giving the order, or um, that would be the doctor's, um, orders."	6	Specify plan description		
"Some terminology really is very new to me po that's why I've been asking you during the process what would what do you mean by the plan order is it the components being ordered or what would be my plan for the patient what would I order the patient to do like that"	3			
"But the plan is if you're going to continue on with uh fabricating the device or still defer it because the wound is there's still a wound like that or too risky to continue with the treatment because of um co-morbidities?"	10			
"I don't want to see patient lists popping out in the first page for this page (dashboard). This is because of security issues or data privacy. So when someone else sees, I'm in a homepage and I don't want anybody else looking at my records or my account and then just popping in my homepage, then someone else can actually try to look at it and then the patient list will just be open. So just more on the security side and then accessibility."	5	Security concerns with displayed patient list	Data privacy	
"To be honest I do have some problems. I think I was disconnected three or four times during the actual practice of the web-based app."	6	Internet dependent	Connectivity	
"I think we do have a problem since the information was not saved when we try to download the PDF"	6	Failed saving input	Data retrieval	
"During the session, upon checking I think we save and download it twice but	6			

still the same the information is blank"				
"I don't know if a problem would be and uh in terms of refreshing there are items that were deleted just in case you accidentally refresh the web base app."	6	Losing data after page reload		
"I don't know if there would be an um retrieval something like that. Um yes, because it would be a problem if you already completed the details and then suddenly it was not saved. So you'll still have to go back from uh from the start"	6			
"I did the voice transcription. It was transcribed in words, but when I tried to repopulate, it wouldn't actually repopulate the other details."	6	Failure to repopulate	AI repopulation errors	
"I think the ones that were repopulated are the only details was demographics and then the rest I think I did have to manually copy and paste it on the text box that was provided."	6	Manual input due to repopulation errors		
"During my experience, maybe the words that the mic would catch would not be the same word I'm trying to say. For example, if the area is I'm in a clinical setting where many patients are being accommodated and even kids are present what are the chances that the words being received by the microphone is same as the same words I'm trying to say especially in accent if the mic doesn't catch things like that."	3	Transcription errors	AI transcription issues	
"I think a contrast on the color something like that just to know if the user is able to click it properly, I think uh that's just something that is useful most especially if an older	6	Changing color when clicking buttons	Visual prompts	Desired changes for UI prototype v2

note, but an older user would be um using this kind of layout. I think it would be helpful for them'"				
"Um the search bar for uh patient uh search patient. I think that's um also good"	6	Having a search bar for patients		
"Um I don't know what part in the web base app would be the actual measurements where we're going to input it. Um since this one is just the interview if we're going to retrieve it and then we are on the session of uh measurement and casting the patient. I don't know if this would be the same um page that we'll be doing it. "	6	Smooth transition to measurement charts		
"I would like to ask if this specific uh SOAP note portion is connected to the record of the patient or it's just a page dedicated to create a note and then you just save it."	1	Connection of SOAP note to patient assessment form		
"I think it would be better if those boxes that needs information will turn red if you miss them."	1	Prompt for missed fields		
"So if there's a prompt on which area you would like to um to add. For example, in fitting um what happened at the end of this fitting. Did it get delivered like um they finished the fitting and proceed to deliver something like that]. m, it's actually it's more like I know PN more like it's just a specific section for section usually use a still right but it's less information. I think just a section reviewing it may prompt on where to write and then what sections to look for."	10			
"Maybe putting on this one is a progress note so that we can just um actually indicate the date the time. I'm not sure if the practice is very specific in Sweden on the time stamp but on	6	Time and date stamping for SOAP notes		

my previous practice the manual timestamping of your patient encounter is important. "				
"Like if there was a there was a problem with uh if there would be a problem with a device if it's not adjusted during the last time has it been adjusted this time? It should for the for the day soap note it should be addressed already. I think that will be easier to check if you address all your problems because you also have your problem list, right?"	10			
"if you weren't there were no uh SOAP notes made during the time that the patient uh had a visit with the clinician so I think there would be a I'm not sure how it should be done but you know it there will be a note there that that patient or the clinician failed to do a SOAP notes for that day I because in the hospital they are also checking that especially if they would be it would be seen by another clinician especially for the for the doctor that uh who referred the patient then it is very important for him or her to see what was done during the time that patient was uh consulted by the clinician."	10			
"I'm not sure if how you're going to do it like if you would like to see the IE so should there be a tab for that IE and then uh should there be included the last SOAP notes made or maybe it should correspond to the last date when the patient was seen "	10			

"Um yes. Um also uh a good thing will be uh adding just a clearance for those like we cannot actually uh mandate to only have those patient lists to a certain CPO. We can actually endorse these patients to for example Jonacel a patient of Alice can be endorsed to Jonacel maybe a clearance level wherein he or she can actually only access this pertinent um information and just have the progress notes so that once this CPO would come back um he or she can actually uh read uh the treatment that was done during the time he was or she was not uh present."	6	Clearance transition CPO to CPO endorsements		
"Uh I think since we are actually used to using the date format, I think that is important."	6	Change date format		
"Very uh concerned about the privacy of not seeing the patient first especially if it's a shared device. Yeah, it's a shared device where so that no other person can access the same the same information. So if you could put password "	10	Password protection in shared devices	Data security features	
"I just wondering because this is AI you know we are connected for the AI application if there's a command that you would like to attach your uh signature can you just you know click uh attach signature like that because that makes it legal"	10	Signature verification		
“Yeah, password guarded signature. If you think you're already, you know, okay with your SOAP, then I don't know if how it can be translated into that easily. You like to attach your signature; you just type in your password and then the signature will be there.”	10			

"So I think you know you know what I mean that we can still control uh"	10	Clinician editor control		
"I think it's good naman po but we can still quite big make it bigger for the other sub areas because if we're going to look at this um parang dashboard po there are still areas on the this side parang we can still maximize it to especially if the clinician has eye, poor eyesight and doesn't have his or her eyeglass at the moment might be a problem to view it."	3	Bigger viewing size of text and icons for visually impaired	Accessibility features	
"I'm just having a like a slight difficulty with zooming in zooming out, you know, making it bigger"	10	Size adjustments		
"I think if it can be read over by the AI so that if I missed something I would like it to be read to me especially if I'm using mobile. If I would access from mobile, I would like to listen to it instead because if I have problems with you know with viewing doing it especially if I'm using a mobile then I would just press read SOAP notes you know it's for accessibility anyway I think that's a good feature."	10	Text reader		
"I think it's a good suggestion that it would be also friendly with mobile phones where they could easily zoom in or zoom out."	10	Mobile phone compatibility	Multi-display compatibility	
"if you could put or make up a different interface you know if you can customize what you can see on your own dashboard like what you what we used to do with other apps."	10	Customizable dashboard	Display layout	
"Yes. I think also a calendar would be nice so that you would have if you would like to see who you have seen yesterday or who would you like to	10	Calendar schedule		

schedule for tomorrow so that you would already balance your schedule if it's already full or not."				
"Maybe if you could just follow the form that we have at PSPO and that you know that will be more of a guided form translated into this."	10	Mental Model		
"I would like to see or improve is that to highlight the AI transcription because it is the I think the main difference of this website in comparison with other type oof data documentation."	1	Highlight AI transcription	AI Assistance features	
"So I think the start AI transcription we can they can put it somewhere where it is easy to it is highlighted. "	1			
"I think skin condition and the living situation uh I think it was helpful to uh have the patient if they have a specific need to be addressed like for example don't know if it would be helpful if there would be a drop-down selection so that it would be a less um typing for the CPO. "	6	AI suggestions		
"For the objective type I think this one um there is an automatic computation for the BMI.”	6	Auto computations		
"The voice transcription um like for example if we're going to ask the question from the patient I think what I did was um for it to be translated in the live transcript I need to mirror the answer of the patient yes because I don't know if the web page app or they have the capability of actually recognizing the answer of the patient so that it would also be populate"	6	Patient voice detection during online consult		
"Uh noise uh cancellation something like that. So that it could actually filter	6	Noise cancellation		

unnecessary noise and just know um the voice of the um clinician and the patient. "				
"Um, I didn't have the chance to actually test it in an environment wherein there would be additional noise. Um, what we did is in a control environment wherein um I need to use an earphone and a quiet place."	6			
"So if there would be a link from the uh from the previous page and it would be seen already here then you could just compose your impression so that you would customize also or you would justify your prescription for the device and then for the daily soap notes or for the um routine notes for like for the times that you see the patient you would just focus on those details."	10	Patient file integration		
"Like if there was a there was a problem with uh if there would be a problem with a device if it's not adjusted during the last time has it been adjusted this time? It should for the for the day soap note it should be addressed already. I think that will be easier to check if you address all your problems because you also have your problem list, right?"	10			
"If there would be um a way that it can create an algorithm that it could actually uh suggest uh components suggest for this type of patient. You know it would and the price depends on which classification price"	10	Clinical prescription support		
"So, if it would already predict components, or you know that would fit to tailor to the to the case but still we have to be on top of what AI can do."	10			

"You know, so with you have you have suppliers, you have you have their you have their uh specifications like specifications for weight or for which activity it would already suggest according to the to the data that is has been uh what you call this been um included in your assessment or in your SOAP notes or in in you know the information they put in there it could already predict oh this one you can use this one but it will be easier for you could you could already discuss with the patient I think for your case would also help your decision making it's not I think AI is uh it creates an algorithm that it improves every time as long as you have add more information to it. I think it will be a nice feature at least."	10			
"I think it would be helpful if we can have the uh photo"	6	Add videos and images	Multimedia upload and storage	
"Um I think it would also be helpful if again uh videos can be uploaded. "	6			
"Then the images to be or the video of the gait if it's possible if we can record it as well in the assessment part"	3			
"I would like to recall the condition maybe if there would be a segment or a part where we could upload pictures that is relevant to the case then the images e or the video of the gait if it's possible if we can record it as well in the assessment part"	10			
"They can just like put pictures compared to it "	5			
"Uh more on uploading media section especially for example skin, stump conditions for postural assessment "	5			

"I was thinking if we can actually integrate an open camera option just to verify. For example, the actual condition during the time of the conversation the consultation because the picture does it indicate how many weeks or months the image was taken."	6	Open camera feature with date and time stamp		
"I also thought of po if this Everi Lab's logo can be placed beside in institution logo where the people would want to subscribe also to use their own logo if the I mean if it's possible po for the formality of the usage of the institution for the subscription on the matter um I think that's all po"	3	Icons and Logos		
"I think on this part it would be helpful if there would be a drop-down option to click like I mean the most common diagnosis for the patient like diabetes, cerebral palsy, stroke etc."	6	List of common diagnosis	Drop down lists	
"I think it's a good note in terms of what um Alex said earlier uh if the patient is uh for example for follow-up maybe uh, we can have uh choosing like a drop down I don't know if it's a drop down "	6	Type of SOAP note		
"Doon sa residual limb. Nasabi ko na siya before. I think um that's a two separate um for moderate or then the short or extra short. Um for this shape you can just do cylindrical um just a drop down "	10	Residuum length		
"I think for common problems like that maybe you can already include there for drop down the drop-down choices it like it's if it's dry scaly wound "	10	Skin condition		
"Again I don't know if what if the details were actually saved by clicking save"	6	Add saving confirmation	Pop-up messages and notifications	

"I think just a warning if you have already saved it asking you if that would be yes like a pop-up command."	6			
"So it would also appear in my screen like when I arrive open it and then the dashboard will be showing me this thing so that I would see right away. "	10	Referral notifications		
"I think it's a good point where in the voice transcription has an indication if it's ready or already on."	6	Add notification that transcription started or ended		
"Um and then there would be a description if there would be a problem in terms of um the microphone that was used like for example earphones or wireless connection is um not detected"	6	Add confirmation for microphone detection		
"Then save para it's easier nasa taas na siya."	3	Placement of the save and download buttons	Additional buttons	
"For the doctors, I think the print option is good so that when you actually want to integrate a health professional from other clinics at least you have a printed format of a digital file "	6	Print function		
"You can actually send it via email, something like that would be good"	6	Send referrals via email		
"I just think that there are some missing areas where I can put the data based on the assessments "	3	Missing assessment areas	Additional input fields	
"I think for the subjective we can put another box where we can add like other comments something like that"	1	Add other comments section		
"I would just like to add another box where it's dedicated for other comments or other findings. So, it's general. So you can just put uh put other findings that is that is not um that doesn't belong	1			

to other boxes in the upper part. So that's I think is that's my comment"				
"But I think you want this part should be I think an additional information just in case there are areas that was missed during the assessment."	6			
"Maybe that adds confusion but um I think overall this one is helpful in terms of additional uh information uh something like that."	6			
"I need to input the history of the presenting illness. "	1	Add history of present illness		
"The device history if the patient is already using a device something like that and also um I also uh observed	1	Add device history section		
"I think something like that. Um I it would be helpful if there would be a previous picture of it, but I think that would be unnecessary. But if that can be provided, yes, at least you'll have an idea of the previous um illustration or at least the background of the old devices that he has. Again, serial numbers can be added for that."	6	Add previous device description and media		
"I think Alex uh mentioned about the code for example liners or code for um the specific um knee joints so that it can be retrieved"	6			
"I think this specific form the face to face was more leaning towards prosthetic cases that's why there was already a box dedicated for the level of amputation so it will not be used if it's an orthotic case, so I think"	1	More tailored to prosthetic case		
"Um it will it is better to change I think maybe that the title instead of level of amputation it will be maybe diagnosis or uh case. So you can just uh choose if it's diabetes or cerebral	1	Add selection for diagnosis of non-amputees		

palsy or if it's an amputation you can just check if it's transtibial something like that."				
"I think maybe we can put the patient stated functional goals at lower part of the subjective evaluation because I think you need to know"	1	Add stated functional goals		
"What you want to put um siguro necessarily vital signs um during those visits. Usually the vital sign."	10	Add vital signs		
"I ask the patient's concern or main complaint "	1	Add main complaint section		
"Maybe for the objective assessment we can put another box for special tests for maybe for prosthetic cases such as yung ligament test something like that."	1	Add special test section		
"And then if it's and then if it's orthotic case, I think it's also important to indicate uh if the patient has spasticity"	1			
"Um I would just like to ask lang po if there we can add some special test "	3			
"I know other assessments like for example Thomas test, special tests."	5			
"Maybe special forms. Oh, if this is for if may special, do you have you have special forms before I'll go my first assessment."	5			
"I don't know how will that be integrated but I think that would be helpful I guess um in terms of I think um outcome measures uh I think from the other options on the right side there is a specific outcome measure but if the patient would have I think orthosis it would be helpful if there would be a specific outcome measure like	6	Add outcome measure section		

GMFCS for the patient or specific um outcome."				
"Um which movements are we like talking about even the hip extensor there's no knee extensor strength, no strength regarding especially if it uses a cause or what hip abductor."	5	Add ROM and MMT fields		
"Since we are used to um giving or testing the uh patients overall range of motion muscle testing special test. Uh I think it would be helpful if those would be um integrated and then just manually input by the CPO "	6			
"Um so it's tricky like um which one which one should be like um is it the real measurement of or within normal range"	5	Add key objective findings		
"So that you fill up you would just fill up the your we have in in PT we have the impression we have the PT impression or like it's like a mini diagnosis of why is it why does the limb has uh no uh decrease ROM because you know there could be some uh contracture "Um so it's tricky like um which one which one should be like um is it the real measurement of or within normal range"	10			
"Um so it's tricky like um which one which one should be like um is it the real measurement of or within normal range"	5	Add ligament laxity		
"Why pain, uh, five or 10 over 10, maybe because there's still a fresh wound."	10	Add pain assessment score		
"Um, next is um for pain um well I think I noted in terms of just a way of clarifying it to the patient "	6			
"And if there are cases like for example um congenital,	6			

we can just have to manually trace it."		Add description box for congenital cases		
"I think it's this way guess you need to describe where or you can just put others and then they can do um description if it is like for example irregular. "	10			
"If we have patients for pedia on the assessment part and if there is an option wherein because in reality we have pedia patients but we interview their parents like would it matter if they will also sign in terms of data privacy because in our experience making SOAP aside from having a signature of the clinician on the latter part we also as signature for the patient as courtesy that we had their information with us. "	6	Add patient consent specially for pediatric patients		
"If I'm going to have an ortho patient and for the spinal or orthosis type of assessment where other features are actually being considered if it's the dextroscoliosis the apex if we have an apex part on the um boxes if there is an option for it to be placed"	3	Add postural assessment section		
"They can describe general gait assessment"	5	Add gait assessment findings		
"Um about the contralateral limb assessment maybe we can focus also MMT and ROM and skin condition. Um not necessarily a drop down. Um but yeah, it's a maybe you can just um it's a free it's similar to this one, but you just have to like um categorize it content and then categorize it into um MMT and ROM limitations everything and um skin sensation I would say"	5	Add contralateral limb assessment		

"Before I forget, um since the practice here in the Philippines is we do have a professional fee when we um actually consult a patient. Does this web um application would require to have an online um payment integration? something like yes that would be basing on the consultation like for example uh you need to have your consultation for the patient and you did um adjustments etc. is the patient able to pay directly via I don't know QR code bank transfer etc. if that would be something that should be also integrated"	6	Add payment management		
"I think the um classification of the patient if it's charity pay or uh sponsored or you know under insurance."	10	Add payment classification		